A Book of Lemmas

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Introduction

"I am too fond of reading books to care to write them."

— Oscar Wilde

This book is meant as a collection of interesting facts and problems in Olympiad Euclidean geometry. I use this file as a personal reference as well as for preparing workshops and lectures for high school students.

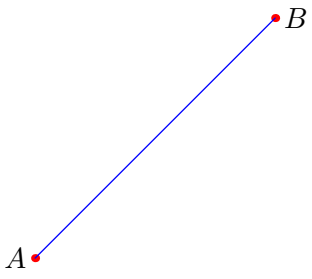
I tried giving credit and links to problems wherever I could. If you have any suggestions or comments – please email me. All the diagrams are drawn by me in **Asymptote: The Vector Graphics Language**.

I would also like to thank my high school teachers for introducing me to the world of geometry – Dominik Burek, Jacek Dymel and Tomasz Kobos.

Notes on notation

In this book I've included the following notation:

- $(P_1P_2P_3\dots)$ means the circumcircle of $P_1, P_2, P_3, \dots$
- $P_1 - P_2 - P_3 - \dots$ means line $P_1, P_2, P_3, \dots$
  (hence those points are collinear)



Recommendations

If you're just getting started with olympiad geometry, I highly recommend:

- *Euclidean Geometry in Mathematical Olympiads*, Evan Chen [Che16a]
  Widely considered the modern gold standard for competition geometry.
- OLYmpiad GEOmetry Articles in aops (+ books + problem sets)
  Amazing list of resources.
- Yufei Zhao's Math Olympiad training handouts
- Evan Chen's Olympiad Articles
  In polish:
- *Planimetria dla Olimpijczykow* by Piotr Ambroszczyk and Mieszko Komisarczyk [Pio23]

Problem collections:

- *Geometry At Its Best*, Eric Shen [She20]
  Nice collection of problems.

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Part I

Orthocenter lemmas

Chapter 1

Orthocenter configurations

Let us establish the standard notation for the configurations discussed in this chapter. Consider a triangle $\triangle ABC$.

- Γ is the circumcircle of $\triangle ABC$.
- O is the circumcenter and R is the circumradius.
- H is the orthocenter, G is the centroid.
- D, E, F are the feet of the altitudes from vertices A, B, C, respectively. $\triangle DEF$ is the *orthic triangle*.
- M is the midpoint of side BC.
- A', B', C' are the *antipodes* of A, B, C with respect to Γ (i.e., A' is the point on Γ such that AA' is a diameter).
- denote $\angle BAC = \alpha, \angle ABC = \beta, \angle ACB = \gamma$
- H_A, H_B, H_C the reflections of point H over the sides BC, AC, AB respectively.
- R_A, R_B, R_C the midpoints of AH, BH, CH

1.1 General properties

We claim some well-known properties:

- $(AEHF)$ is cyclic with diameter AH
- $(BFEC)$ is cyclic with center at M
- $\angle BHC = 180^o - \alpha$

Lemma 1: Reflections of the orthocenter

The reflections H_A, H_B, H_C of the orthocenter H across sides BC, AC, AB lie on the circumcircle (ABC).

Proof. Let's see that for H_A. Equality $\angle BHC = 180^o - \alpha$ implies $\angle BH_AC = 180^o - \alpha$ so $(BACH_A)$ is concyclic. □

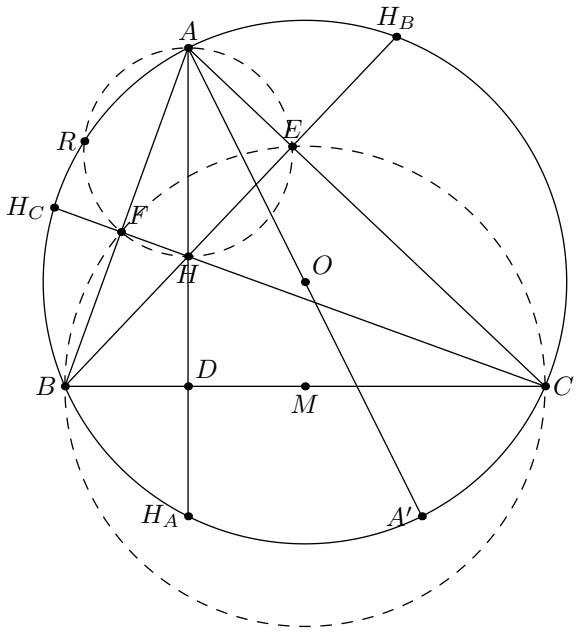


Figure 1.1: *Configuration*

Lemma 2: Points O and H are isogonal conjugates

Points H and O are isogonal conjugates; that is, $\angle BAH = \angle OAC = 90^\circ - \beta$.

Proof. This is a relatively simple angle chasing calculation. □

Lemma 3: Orthic Triangle Incenter

Point H is the incenter of $\triangle DEF$.

Proof. Again this is an angle chasing proof requiring you to show $\angle FDH = \angle HDE$. □

Lemma 4: Orthocentric System

l The set of four points $\{A, B, C, H\}$ forms an *orthocentric system*. This means any one of the points is the orthocenter of the triangle formed by the other three.

Proof. This is the direct consequence of the perpendicular lines $AH \perp BC$, $AB \perp CH$, $AC \perp BH$. □

Lemma 5: Three Tangents Lemma

In $\triangle ABC$ with orthic triangle $\triangle DEF$ and midpoint M of BC:

(a) The line lines ME and MF are tangent to the circumcircle of $\triangle AEF$.

(b) The line through A parallel to BC is also tangent to (AEF).

Proof. Since $BCEF$ is cyclic with center M, $ME = MF = MB = MC$. In isosceles triangle $\triangle MEC$, $\angle MEC = \angle MCE = \gamma$. Since $AEHF$ is cyclic with diameter AH, the center of (AEF) is R_A (the midpoint of AH).

To show tangent-chord angle condition for ME:

$$\angle MEH = 90^\circ - \angle MEC = 90^\circ - \gamma = \angle EAH.$$

By the converse of the tangent-chord theorem, ME is tangent to (AEF) at E. By symmetry, MF is tangent at F.

For the line ℓ through A parallel to BC, $\ell \perp AD$. Since AD is a diameter-line of (AEF), ℓ is perpendicular to the radius at A, and hence tangent to (AEF). □

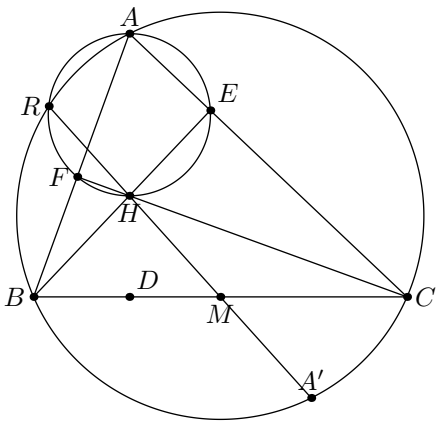


Figure 1.2: *Lemma 6*

Lemma 6: Antipode & Orthocenter Lemma

Let R be the intersection of $(AEHF)$ with (ABC), $R \neq A$.

- $R - H - M - A'$
- $BC \parallel H_A A'$
- M is the center of $(HH_A A')$

Proof. Segments AA' and AH are the diameters of circles (ABC), $(AEHF)$ respectively. Then $\angle ARA' = 90^o = \angle ARH$ so points R, H, A' are collinear.

As H_A is the reflection of H over BC we have $HM = H_A M$. Notice also that $H_A M = A'M$ because line OM is the perpenticular bisector of segment $H_A A'$. Combining this with the fact that $HH_A A'$ is a right triangle, we have that points H, M, A' lie on one line.

□

Lemma 7

Let EF intersect (ABC) at K and L, K on the other side of AC than B. Further, let K' be the reflection of K over AC. Then the following holds:

- $AK = \sqrt{AH \cdot AD}$
- $(AEK'L)$

- $(AHK'C)$
- $C - K' - L$
- $D - K' - E$
- A is the center of $(KK'L)$

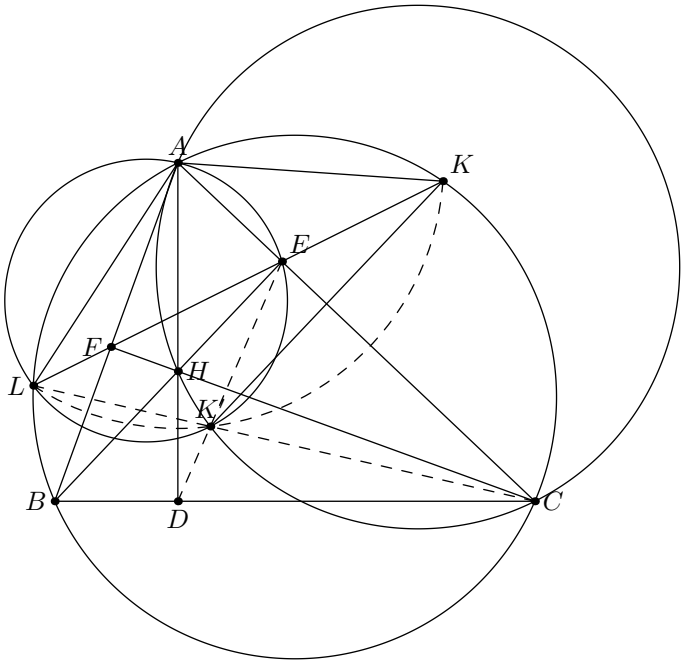


Figure 1.3: *Lemma 7.*

Proof. Consider an inversion $I(A, \sqrt{AH \cdot AD})$. Since $AD \perp BC$ and $AEHF$ is cyclic, $AE \cdot AC = AH \cdot AD = AH \cdot AD$. Thus, under this inversion:

$$H \to D, \quad E \to C, \quad F \to B.$$

$$(ABC) \to EF$$

Since $K, L \in (ABC) \cap EF$, points K and L are fixed points of the inversion! Hence $\sqrt{AH \cdot AD} = AL = AK = AK'$.

Furthermore, $\angle ALK = \angle AKL = \angle AK'E$ so $(AEK'L)$ and by the inversion $L - K' - C$. Again by angle chasing:

$\angle AK'C = \angle AEK' = 180^o - \angle ALK' = 180^o - \angle ALC = \angle AKC = \angle AHC$

so $(AHK'C)$ and by inversion $D - K' - E$. □

1.2 Pedal triangle

Let P be a point inside triangle ABC and points P_A, P_B, P_C be the foots of P onto sides BC, AC, AB respectively.

Then $\triangle P_A P_B P_C$ is the *Pedal trangle* of P with respect to ABC.

- The orthic triangle $\triangle DEF$ is the pedal triangle of H.
- The medial triangle $\triangle M_A M_B M_C$ is the pedal triangle of O.

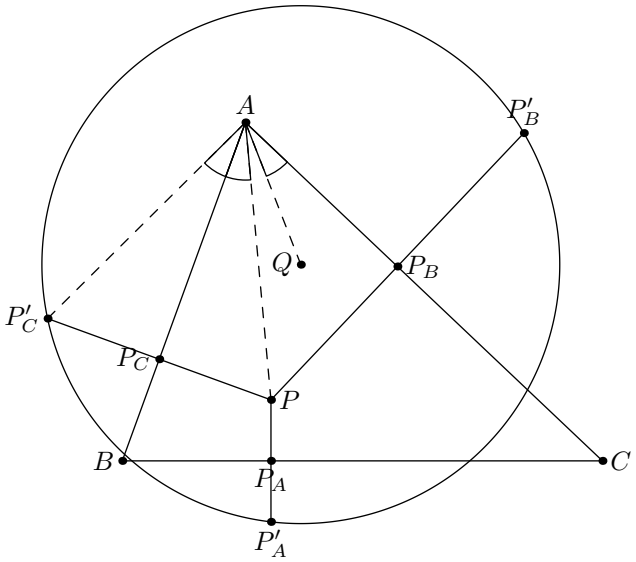


Figure 1.4: *Lemma 8*

Lemma 8

Let P'_A, P'_B, P'_C be the reflections of P about points P_A, P_B, P_C respectively. Then circumcenter Q of $(P'_A P'_B P'_C)$ is the isogonal conjugate of P.

Proof. See [i3421] Theorem 1.2.2. □

Lemma 9: Six Point Circle

Let P and Q be points in ABC and N be the midpoint of PQ. Then let $P_A P_B P_C$ be the pedal triangle of P, and let $Q_A Q_B Q_C$ be the pedal triangle of Q. Then $P_A, P_B, P_C, Q_A, Q_B, Q_C$ all lie on a circle with center N iff P and Q are isogonal conjugates.

Proof. Great proof can be found in [Ron20]. □

1.3 Problems

Problem 1. Iran TST 2011 P1

In acute triangle ABC, let E and F be the feet of the altitudes from B and C, respectively. Let M, K, L denote the midpoints of BC, ME, MF, respectively. Line KL intersects the line through A parallel to BC at T. Prove that $TA = TM$.

Problem 2. USA TSTST 2016 P2

Let ABC be a scalene triangle with orthocenter H and circumcenter O. Denote by M, N the midpoints of $\overline{AH}$, $\overline{BC}$. Suppose the circle γ with diameter $\overline{AH}$ meets the circumcircle of ABC at $G \neq A$, and meets line AN at a point $Q \neq A$. The tangent to γ at G meets line OM at P. Show that the circumcircles of $\triangle GNQ$ and $\triangle MBC$ intersect at a point T on $\overline{PN}$.

Problem 3. USA TSTST 2012

In scalene triangle ABC, let the feet of the perpendiculars from A to BC, B to CA, C to AB be A_1, B_1, C_1, respectively. Denote by A_2 the intersection of lines BC and B_1C_1. Define B_2 and C_2 analogously. Let D, E, F be the respective midpoints of sides BC, CA, AB. Show that the perpendiculars from D to AA_2, E to BB_2 and F to CC_2 are concurrent.

Problem 4. Iran TST 2018

Let ABC be a triangle with $A \neq 90^o$, and let BE and CF be its altitudes. The bisector of A intersects EF and BC at M and N respectively. Let P be a point such that $MP \perp EF$ and $NP \perp BC$. Prove that line AP bisects BC.

Problem 5. IMO SL 2008 G4

In an acute triangle ABC segments BE and CF are altitudes. Two circles passing through the point A and F and tangent to the line BC at the points P and Q so that B lies between C and Q. Prove that lines PE and QF intersect on the circumcircle of triangle AEF.

Problem 6. IMO SL 2017 G3

Let O be the circumcenter of an acute triangle ABC. Line OA intersects the altitudes of ABC through B and C at P and Q, respectively. The altitudes meet at H. Prove that the circumcenter of triangle PQH lies on a median of triangle ABC.

1.4 Reading

[Ron20] — Victor Rong, *A Few Configurations* (Sections 1 & 3)
Homothety relations with the nine-point circle, and pedal triangle theory for general isogonal conjugates.

[i3421] — *"American" Olympiad Triangle Geometry Configurations* (Section 1.1)
Introduction to the orthocenter.

Chapter 2

Nine-point circle

It is what you read when you don't have to that determines what you will be when you can't help it.
— Oscar Wilde

Now let's look at a particularly useful circle - with many stunning properties.

For the sake of this section, let M_A, M_B, M_C denote the midpoints of sides BC, AC, AB. Furthermore, let O_A be the circumcenter of (BHC) and R_A, R_B, R_C be the midpoints of segments AH, BH, CH respectively.

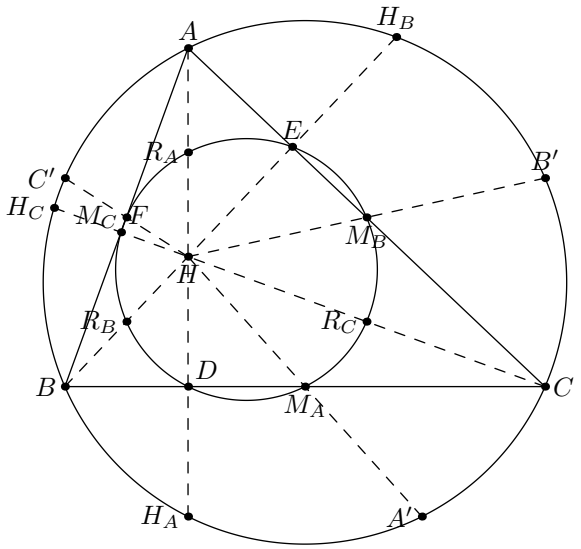


Figure 2.1: *Nine-point circle*

Lemma 10: Nine-point circle - n_9

Points D, E, F, M_A, M_B, M_C, R_A, R_B, R_C all lie on one circle with center N_9 called the *Nine-point circle* of ABC.

Furthermore,

- N_9 is the midpoint of HO
- r_9 the radius of the nine-point circle is equal to $\frac{1}{2}R$
- N_9 lies on R_AM_A

Proof 1 (Via Homothety). Recall from orthocenter properties that the reflection of H across side midpoint M_A is the antipode A' (*Lemma 1*). Thus, M_A is the midpoint of HA'. Similarly, D is the midpoint of HH_A.

Consider a homothety $\mathrm{h}(H, \frac{1}{2})$ centered at H with scale factor $\frac{1}{2}$. Then it maps:

- $(ABC) \to n_9$ with radius $r_9 = \frac{1}{2}R$.
- $O \to N_9$, the midpoint of HO.
- $A \to R_A$ (since R_A is the midpoint of AH).
- $A' \to M_A$ (since M_A is the midpoint of HA').
- $H_A \to D$ (since D is the midpoint of HH_A).

Applying the same homothety to vertices B, C, their antipodes, and orthocenter reflections shows that all nine points $D, E, F, M_A, M_B, M_C, R_A, R_B, R_C$ lie on n_9. Since AA' is a diameter of (ABC), its image under $\mathrm{h}(H, \frac{1}{2})$ is R_AM_A, which must be a diameter of n_9 passing through its center N_9. □

Proof 2 (Via Isogonal Conjugates). Recall that H and O are isogonal conjugates in $\triangle ABC$ (*Lemma 2*). The orthic triangle $\triangle DEF$ is the pedal triangle of H, while the medial triangle $\triangle M_AM_BM_C$ is the pedal triangle of O.

By the Six-Point Pedal Circle Theorem (*Lemma 9*), we have $(DEFM_AM_BM_C)$.

Proving that R_A, R_B, R_C also lie on that circle can proceed as above. □

Lemma 11: Euler line

Let R denote the radius of (ABC) and r the radius if the incircle.

- $H - G - N_9 - O$ [*Euler line*]
- $HG = 2 \cdot GO$
- $OI^2 = R(R - 2r)$

Proof. Consider a homothety $\mathrm{h}(G, -\frac{1}{2})$ centered at the centroid G with scale factor $-\frac{1}{2}$. This homothety maps $\triangle ABC$ to its medial triangle $\triangle M_AM_BM_C$. Under this mapping:

- $(ABC) \to n_9$
- orthocenter H of $\triangle ABC \to$ the orthocenter of $\triangle M_AM_BM_C$, which is the circumcenter O of $\triangle ABC$.

Thus, $\mathrm{h}(G, -\frac{1}{2})$ maps $H \mapsto O$ and $O \mapsto N_9$. This implies that G, H, O, N_9 are collinear, with vectors $\vec{GO} = -\frac{1}{2}\vec{GH}$, or $HG = 2 \cdot GO$.

For dot three, see [Ron20] Lemma 4.

□

Lemma 12: Connecting n_9 and (BHC)

- O_A is the reflection of O about line BC
- $AH = 2 \cdot OM_A$
- $A - N_9 - O_A$

- $AHOO_A$ is a parallelogram

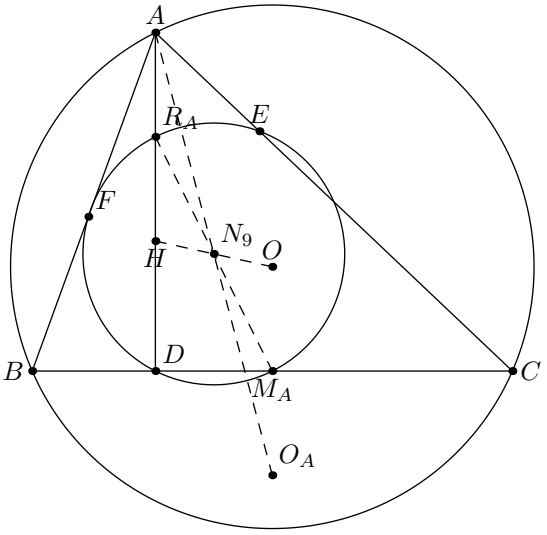


Figure 2.2: *Lemma 12*

Proof. Clearly from *Lemma 1* point H_A is the reflection of H about line BC and hence triangles BHC and BH_AC are reflections of each other BC. Notice that O is the circumcenter of BH_AC and O_A, from definition, is the circumcentre of BHC. Therefore, they must also be reflections.

Angle $\angle M_ADR_A = 90^o$ implies that R_AM_A is a diameter of n_9 and so, as N_9 is its circumcenter, $R_A - N_9 - M_A$. Then we have $R_AH = OM_A$ from parallelogram R_AHM_AO.

Then, we see that R_A is the midpoint of AH and M_A is a midpoint of OO_A, and so $AHOO_A$ is a parallelogram as well. Which implies $A - N_9 - O_A$. □

Lemma 13

Nine-point circles of AHB, BHC, CHA, and ABC are the same.

Lemma 14

Consider an inversion I with center A and radius $\sqrt{\frac{1}{2}AB \cdot AC}$ combined with a reflection about the bisector of $\angle BAC$. Then it transforms the nine-point circle of ABC into the circumcircle of $\triangle BOC$.

2.1 Feuerbach theorem

Now let's take a look at a beutiful but spectacularly useless theorem.

Lemma 15: Feuerbach theorem

The incircle and the *Nine-point circle* are internally tangent at the *Feuerbach point* of the triangle.

The *Nine-point circle* is also externally tangent to the three excircles.

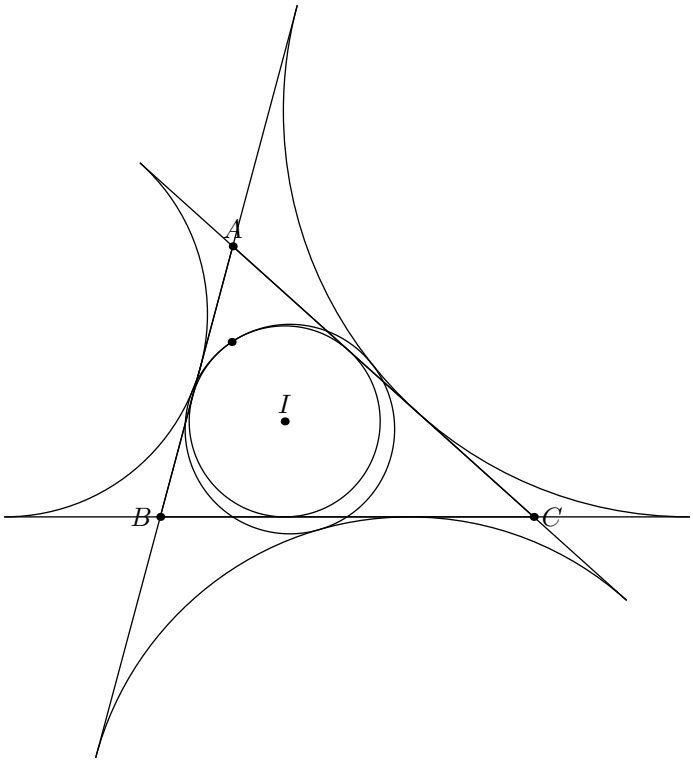


Figure 2.3: *Feuerbach theorem*

I won't prove it - because it is quite hard and its rather useless in mathematical olympiads. But it's a nice fact to know. Its also a nice excuse to make this diagram.

Lemma 16

The following inequality holds: $2r \leq R$.

Proof. We've seen that n_9 is two times smaller than (ABC). Now, we can add to that that the incircle fits inside n_9. Hence, the inequality holds.

□

Correction, I know exactly 1 problem which uses this theorem.

Problem 7. Let $\triangle ABC$ with circumcentre O, incenter I and $\Omega = (BOC)$ and ω be the circle tangent to AB, AC at E, F and to Ω at X.

Show that EF is the perpendicular bisector of AI.

Solution. Let M be the midpoint of EF. From symmetry we already know that $AI \perp EF$ - we just need to prove that M is the midpoint of AI.

Use inversion I with center A and radius $\sqrt{\frac{1}{2}bc}$ and reflection about the angle bisector, then (see the properties of the inversion in *Chapter 17*):

$$I(\Omega) = n_9$$

then $I(\omega)$ is tangent to AB, AC and externally to n_9 so from *Feuerbach theorem* it must be the A-excircle. Let the A-excircle be tangent to AB, AC at K, L respectively and have circumcentre I_A.

$$I(E) = L, I(F) = K$$

so $I(M) = I_A$ as $(AKLI_A)$ and $I(E - M - F) = (AKLI_A)$.
Then we have:

$$AM \cdot AI_A = \frac{1}{2} b \cdot c$$

Moreover from *Lemma 97* we have $AI \cdot AI_A = b \cdot c$. So

$$AM = \frac{b \cdot c}{2 \cdot AI_A} = \frac{1}{2} AI$$

and we are done. □

2.2 Poncelet points

The concept of the Nine-Point Circle generalizes naturally to four arbitrary points.

Lemma 17: Poncelet Point

Let A, B, C, P be 4 points not in an orthocentric system. The nine-point circles of $\triangle ABC, \triangle BCP, \triangle CAP$, and $\triangle ABP$ all meet at one point P', the *Poncelet Point* of the system $ABCP$.

Lemma 18

The Poncelet Point P' of $\{A, B, C, P\}$ lies on the circumcircle of the pedal triangle of P with respect to $\triangle ABC$.

Remark. When $P = I$ (the incenter of $\triangle ABC$), its pedal triangle is the intouch triangle $\triangle D_1E_1F_1$. The Poncelet Point of $\{A, B, C, I\}$ is precisely the *Feuerbach point*!

2.3 Best problem

My favourite problem in the whole world.
It deserves its own section.
It uses lots of things from *Chapter 18* but whatever.

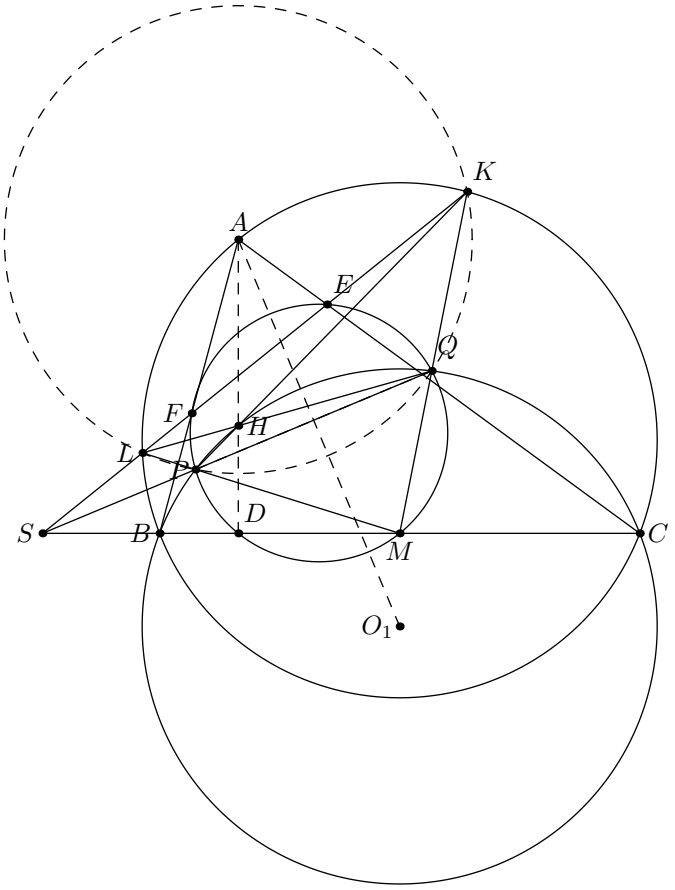
Problem 8. Let ABC be a triangle with orthocenter H, let D, E, F be the foot of altitudes and M the midpoint of BC. Circumcircles of (DEM) and (BHC) intersect at P, Q where P is at the same side of AD as B. Prove that EF, PH, MQ and (ABC) have a common point.

Proof. $(DMEF)$ is obviously the n_9 of $\triangle ABC$. Let $S = BC \cap PQ$ and we have:

$$n_9 \cap (BHC) = PQ$$
$$(ABC) \cap (BHC) = BC$$

and, as S lies on the *orthic axis* because of the 3 power axis theorem, from *Lemma 20* we have $S - E - F$.

Now, denote by N the circumcenter of n_9 and O_1 the circumcenter of (BHC) - and invoke *Lemma 12* to get $A - N - O_1$. Then, as $O_1P = O_1Q$, AO_1 must be the perpendicular bisector of PQ.



Let $K = MQ \cap EF$, $L = MP \cap EF$, then (because $ME = MF$) $\angle LFM = \angle EQM = \angle MPF = \angle MEK$ and

$$MP \cdot ML = MF^2 = ME^2 = MQ \cdot MK$$

so $(PQKL)$.

Notice that SM is the third diagonal of $(PQKL)$ and $(PQMD)$, then using *Lemma 134*, D must be the *Miquel point* of $(PQKL)$ and we know $O(PQKL) \in DH$.

As noted earlier, AO_1 is the perpendicular bisector of PQ - then $DH \cap AO_1 = A$ must be the $O(PQKL)$.

From PowPoint, $SL \cdot SK = SP \cdot SQ = SB \cdot SC$ so $(BCKL)$. Further, from *Lemma 2*, $AO \perp EF$, but because of $AK = AL$, AO must be the perpendicular bisector of KL, so it must be $O = O(BCKL)$ and hence $(ABCKL)$.

Let $H' = PK \cap QL$, again from *Lemma 134* H' must lie on AD and $AH' \cdot AD = AK^2$ because H' and D are inverses wrt $(PQKL)$.

Now comes the boring part:

$$AM^2 - AK^2 = Pow_{(PQKL)}(M) = MP \cdot ML = MF^2 = MB^2$$

$$AH' \cdot AD = AK^2 = AM^2 - MB^2 = Pow_{(BCEF)}(A) = AF \cdot AB = AH \cdot AD$$

and $H = H'$ so we are done. □

For this last part, we could have used *Lemma 7*, but we are staying faightful to our original solution.

2.4 Worst problem

The most insane problem I've seen in my life.

Problem 9. GQMO 2020 P8

Let ABC be an acute scalene triangle, with the feet of A, B, C onto BC, CA, AB being D, E, F respectively. Let W be a point inside ABC whose reflections over BC, CA, AB are W_a, W_b, W_c respectively. Finally, let N and I be the circumcenter and the incenter of $W_aW_bW_c$ respectively. Prove that, if N coincides with the nine-point center of DEF, the line WI is parallel to the Euler line of ABC.

2.5 Other problems

Problem 10. USA TSTST 2016 P2

Let ABC be a scalene triangle with orthocenter H and circumcenter O. Denote by M, N the midpoints of $\overline{AH}$, $\overline{BC}$. Suppose the circle γ with diameter $\overline{AH}$ meets the circumcircle of ABC at $G \neq A$, and meets line AN at a point $Q \neq A$. The tangent to γ at G meets line OM at P. Show that the circumcircles of $\triangle GNQ$ and $\triangle MBC$ intersect at a point T on $\overline{PN}$

Problem 11. Let ABC be an acute-angled triangle with orthocenter H. Let M, N, K, L, P be the midpoints of segments AB, AC, CH, BH, BC, respectively, and let Q be the foot of the altitude from vertex A. The lines PM and NL intersect at point X, and the lines KM and NQ intersect at point Y. Prove that the line XY is parallel to BC.

Solution. Note that KL is the midsegment connecting the midpoints of BH and CH in triangle $\triangle HBC$, so $KL \parallel BC$. Similarly, in $\triangle ABC$, MN is the midsegment connecting the midpoints of AB and AC, so $MN \parallel BC$.

By the properties of the Nine-Point Circle, quadrilateral $KLMN$ is cyclic. Since $MN \parallel KL$, $KLMN$ is an isosceles trapezoid. The perpendicular bisector of MN is therefore also the perpendicular bisector of KL, as well as PQ (since M, N, P, Q lie on the same nine-point circle).

Thus, the construction of points X and Y is completely symmetric with respect to this perpendicular bisector. By symmetry, the line XY must be perpendicular to this axis of symmetry, which implies $XY \parallel BC$.

□

Problem 12. In triangle ABC, let I be the incenter. Let M be the midpoint of side BC, and let S be the midpoint of arc BAC (the arc BC containing vertex A) on the circumcircle of $\triangle ABC$. Prove that $\angle IMB = \angle ISA$.

Solution. Let I_B and I_C denote the B-excenter and C-excenter of triangle ABC, respectively.

Recall that the circumcircle of $\triangle ABC$ is the Nine-Point Circle of the excenter triangle $\triangle I_AI_BI_C$ (*Lemma 98*). Therefore, the midpoint S of segment I_BI_C lies on the circumcircle (ABC).

Since quadrilateral I_CBCI_B is cyclic, triangles $\triangle I_CII_B$ and $\triangle BIC$ are similar. The points S and M are the midpoints of the corresponding sides I_BI_C and BC in these similar triangles. Hence, the angle formed at the midpoint M in $\triangle BIC$ equals the angle formed at S in $\triangle I_CII_B$, which yields $\angle IMB = \angle ISA$.

□

Problem 13. Let ABC be an acute-angled triangle. Points M and N are the reflections of vertex A across vertices B and C, respectively. Line k passes through B, is perpendicular to AB, and intersects line AC at K. Line l passes through C, is perpendicular to AC, and intersects line AB at L. Lines k and l intersect at P. Prove that the five points K, N, M, L, P are concyclic.

Solution. Let K', L', P' be the midpoints of segments AK, AL, AP, respectively.

In triangle $\triangle AKL$, B and C are the feet of the altitudes from L and K onto sides AK and AL, so $P = k \cap l$ is the orthocenter of $\triangle AKL$.

The midpoints K', L', P' of the segments connecting the vertices of $\triangle AKL$ to its orthocenter P, along with the altitude feet B and C, all lie on the Nine-Point Circle of $\triangle AKL$.

Applying a dilation $\mathrm{h}(A, 2)$ centered at A with scale factor 2 maps:

- $K' \mapsto K$, $L' \mapsto L$, $P' \mapsto P$,
- $B \mapsto M$ (since B is the midpoint of AM),
- $C \mapsto N$ (since C is the midpoint of AN).

Since homothety preserves concyclicity, the points K, N, M, L, P lie on the image circle under this dilation, completing the proof. □

Problem 14. Let AK, BL, CM be the altitudes of an acute triangle ABC, and let H be its orthocenter. Let $S = BL \cap KM$, let P be the midpoint of segment AH, and let $T = LP \cap AM$. Prove that the line TS is perpendicular to BC.

Solution. Set $\gamma = \angle ACB$. Then $\angle TLH = 90^\circ - \angle TLA = 90^\circ - \angle PAL = \gamma$ and $\angle BMK = \gamma$ from $(AMKC)$.

Hence, we get that $(TLSM)$ is cyclic.

Since $(ALHM)$ is cyclic with diameter AH, inscribed angle equality gives $\angle MAS = \angle MLS = \angle MAH$. Thus, ray AS coincides with ray AH, which means T, S, H are collinear and $TS \parallel AH$. Since $AH \perp BC$, it follows directly that $TS \perp BC$.

□

Problem 15. USAMO 2015 P2

Quadrilateral $APBQ$ is inscribed in circle ω with $\angle P = \angle Q = 90^\circ$ and $AP = AQ < BP$. Let X be a variable point on segment $\overline{PQ}$. Line AX meets ω again at S (other than A). Point T lies on arc AQB of ω such that $\overline{XT}$ is perpendicular to $\overline{AX}$. Let M denote the midpoint of chord $\overline{ST}$. As X varies on segment $\overline{PQ}$, show that M moves along a circle.

Proof. Let $K = PQ \cap ST$, and let Y, Z be the orthogonal projections of points S, A onto lines AT and ST, respectively. Then quadrilaterals $STYX$ and $XYZM$ are cyclic.

Using power of a point:

$$KP \cdot KQ = KT \cdot KS = KX \cdot KY = KZ \cdot KM.$$

This implies that quadrilateral $PQZM$ is cyclic. Let O' be the circumcenter of $(PQZM)$, which is the intersection of the perpendicular bisector of PQ and the perpendicular bisector of MZ.

We will show that the perpendicular bisector of MZ bisects the segment AO. Indeed, since $AZ \parallel OM$, quadrilateral $AOMZ$ is an isosceles trapezoid. Its midline is the perpendicular bisector of ZM, which bisects AO.

Hence, point M lies on a fixed circle centered at the midpoint of AO with radius $O'P$, independent of the choice of X.

□

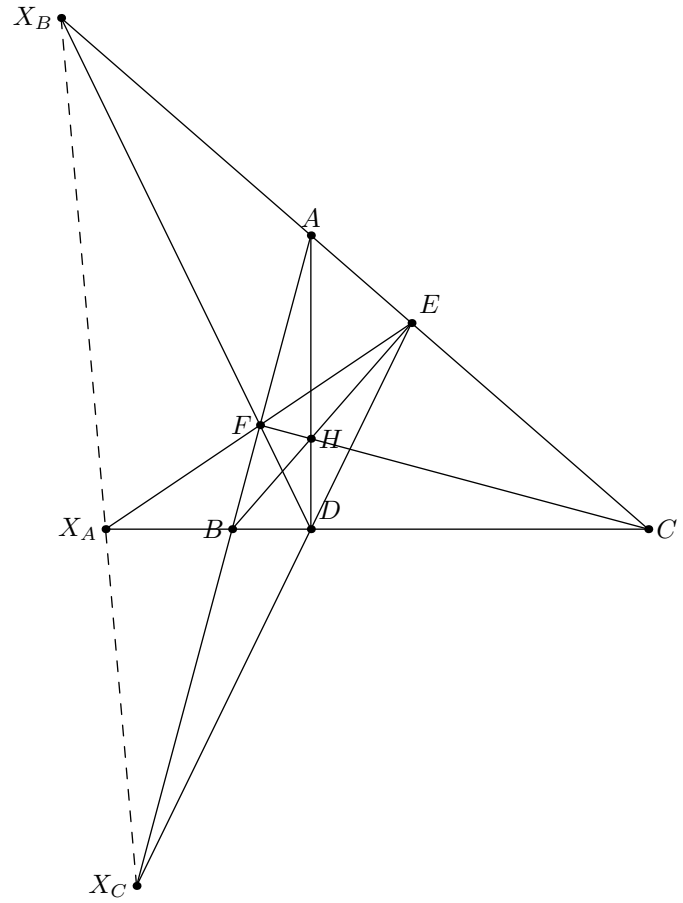
2.6 Reading

[i3421] — *"American" Olympiad Triangle Geometry Configurations* (Sections 1.2)

Includes a more detailed discussion on Poncelet Points.

Chapter 3

Orthic axis



Lemma 19: Ex-point

Let $X_A = EF \cap BC$ - the *A-Ex Point* of $\triangle ABC$.

- $(X_A, D; B, C) = -1$

Lemma 20: Orthic axis

Points X_A, X_B, X_C lie on a line, called the *orthic axis* - power axis of (ABC) and n_9. Therefore it is perpenticular to the *Euler line* of $\triangle ABC$.

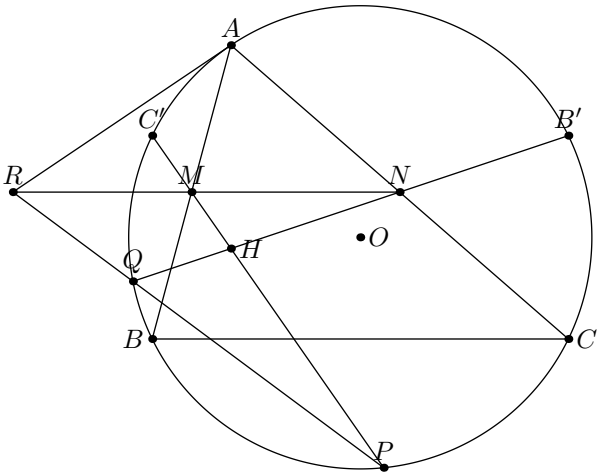


Figure 3.1: *Problem 16.*

Proof. The Nine-Point Circle n_9 passes through altitude feet D, E, F and side midpoints M_A, M_B, M_C. Power of point X_A with respect to n_9 is $X_AE \cdot X_AF$, while power with respect to (ABC) is $X_AB \cdot X_AC$. Since $BCEF$ is cyclic, $X_AE \cdot X_AF = X_AB \cdot X_AC$. Thus X_A lies on the radical axis of (ABC) and n_9. By symmetry, X_B and X_C also lie on this radical axis.

Since the centers of (ABC) and n_9 are O and N_9 (which lie on the Euler line HO), the radical axis is perpendicular to HO. □

Lemma 21

Tangent to (ABC) at A cuts line M_BM_C at Y. Then Y lies on the orthic axis of ABC.

Problem 16. USA TSTST 2011/4

Acute triangle ABC is inscribed in circle ω. Let H and O denote its orthocenter and circumcenter, respectively. Let M and N be the midpoints of sides AB and AC, respectively. Rays MH and NH meet ω at P and Q, respectively. Lines MN and PQ meet at R. Prove that $OA \perp RA$.

Solution. Let B', C' be the antipodes of B, C wrt (ABC). Then from *Lemma 6*, $P-H-M-C'$ and $Q-H-N-B'$. Furthermore, $MN \parallel B'C'$ as it is the midline in $\triangle HB'C'$.

Then:

$$\angle QPC' = \angle QB'C' = \angle QNM$$

so $(PQMN)$.

Then

$$Pow_{(ABC)}(R) = RQ \cdot RP = RM \cdot RN = Pow_{n_9}(R)$$

so it is the intersection of the A-midline of $\triangle ABC$ with the orthic axis. Then, from *Lemma 21*, RO is tangent to (ABC) and we are done.

□

Now one of my favourite problems:

Problem 17. Let ABC be an acute trangle, H its orthocenter and Ω its circumcircle. Circles with diameters AH, BH, CH intersect Ω at A_1, B_1, C_1, different from vertices of the triangle, respectively. Tangent to Ω at A cuts B_1C_1 at A_2, define B_2, C_2 similarly. Prove that A_2, B_2, C_2 are collinear.

Solution. Let A', B', C' be antipodes of A, B, C wrt Ω and M_A, M_B, M_C be the midpoints of BC, AC, AB respectively.

By *Lemma 6* we have B_1-H-M_B-B', C_1-H-M_C-C' and $HM_B = M_BB'$, $HM_C = M_CC'$. Hence $B'C' \parallel M_BM_C$ and we get $(B_1M_CM_BC_1)$.

Now consider 3 circles Ω, $(B_1M_CM_BC_1)$ and n_9:

$$(B_1M_CM_BC_1) \cap \Omega = B_1C_1$$
$$(B_1M_CM_BC_1) \cap n_9 = M_BM_C$$

so by 3 axis $B_1C_1 \cap M_BM_C \in \Omega \cap n_9 =$ *orthic axis*

Combining that with *Lemma 21* we get $B_1C_1 \cap M_BM_C = A_2 \in$ *orthic axis* Similarly we show B_2, $C_2 \in$ *orthic axis* which finishes the proof. □

Problem 18. Vietnam TST 2003/2

Given a triangle ABC. Let O be the circumcenter of this triangle ABC. Let H, K, L be the feet of the altitudes of triangle ABC from the vertices A, B, C, respectively. Denote by A_0, B_0, C_0 the midpoints of these altitudes AH, BK, CL, respectively. The incircle of triangle ABC has center I and touches the sides BC, CA, AB at the points D, E, F, respectively. Prove that the four lines A_0D, B_0E, C_0F and OI are concurrent. (When the point O concides with I, we consider the line OI as an arbitrary line passing through O.)

Solution. Let N be the second intersection of A_0D with the incircle. Combining *Lemma 95 and 113* we have that DN is the polar of R, defined in the latter lemma, and R lies on the orthic axis of ABC.

Analogusly, the poles of lines B_0E and C_0F also lie on the orthic axis and by *Lemma ?* those three polars coincide.

Let ω_A be the circle with center R and radius RD. and define ω_B and ω_C analogusly. Then, DN is also the power axis of ω_A and the incircle so the point where lines A_0D, B_0E, C_0F concurr lies on the power axis of ω_A, ω_B and ω_C (note that it's a line, not a point as the centers of those circles are collinear).

Notice that points I and O also lie on this power axis as ω_A is orthogonal to both the incircle and the circumcircle, and hence we're done. □

Lemma 22

H is the orthocenter of $\triangle AX_AM$.

3.1 Humpty and Queue points

Lemma 23: Queue point

Let $R_A = HM \cap (ABC)$ - the *A-Queue Point* of $\triangle ABC$. Recall the properties of R_A from *Lemma 6*.

- $X_A - R_A - A$
- $R_A \in (AEHF)$
- $R_A - H - M - A'$
- R_A is the *Miquel point* of the $(BFEC)$

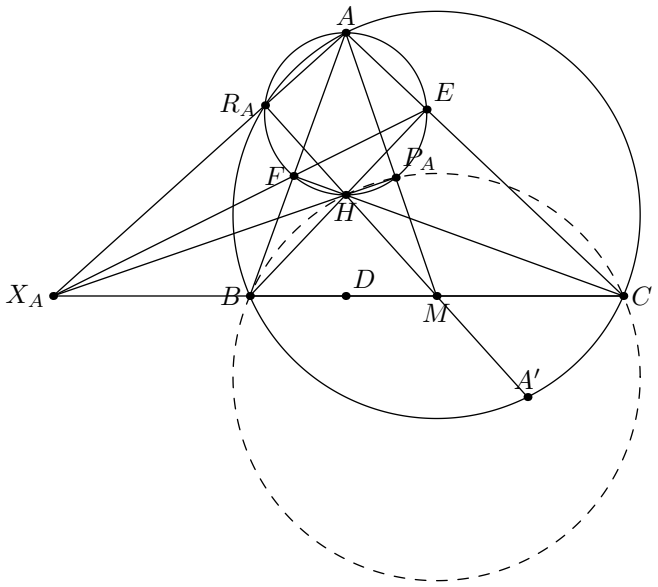


Figure 3.2: *Lemma 24*

- R_A is the center of spiral similarity from FE to BC

Lemma 24: HM/Humpty point

Let P_A be the foot of H onto AM - call it the *A-HM Point* or the *A-Humpty Point* of $\triangle ABC$.

It has the following properties:

- $P_A \in (AEHF)$
- $X_A - H - P_A$
- (BHP_AC)
- P_A lies on the A-Apollonius Circle, i.e. $\frac{AB}{AC} = \frac{P_AB}{P_AC}$

Problem 19. USA TSTST 2017

Let ABC be a triangle with circumcircle Γ, circumcenter O, and orthocenter H. Assume that $AB \neq AC$ and that $\angle A \neq 90^\circ$. Let M and N be the midpoints of sides AB and AC, respectively, and let E and F be the feet of the altitudes from B and C in $\triangle ABC$, respectively. Let P be the intersection of line MN with the tangent line to Γ at A. Let Q be the intersection point, other than A, of Γ with the circumcircle of $\triangle AEF$. Let R be the intersection of lines AQ and EF. Prove that $PR \perp OH$.

Solution. By *Lemma 21*, $P \in$ *orthic axis* of $\triangle ABC$.

By *Lemma 24*, $R = X_A \in$ *orthic axis* of $\triangle ABC$.

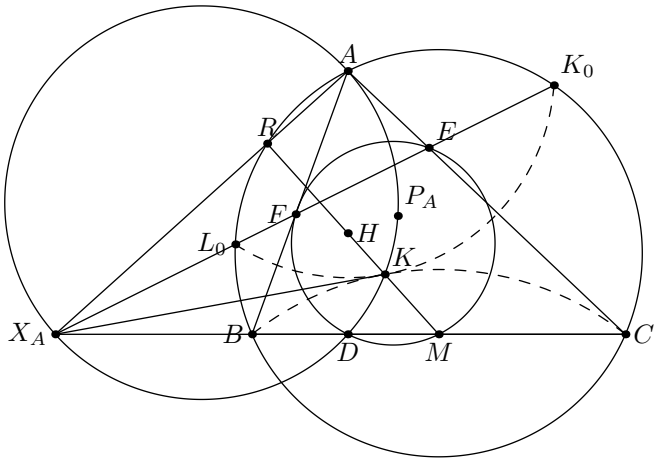
Hence $PR =$ *orthic axis* $\perp OH$ by *Lemma 20* and we are done. □

Problem 20. RMM 2013 P3

Let $ABCD$ be a quadrilateral inscribed in a circle ω. The lines AB and CD meet at P, the lines AD and BC meet at Q, and the diagonals AC and BD meet at R. Let M be the midpoint of the segment PQ, and let K be the common point of the segment MR and the circle ω. Prove that the circumcircle of the triangle KPQ and ω are tangent to one another.

Using 3.4, this problem can be restated as:

Problem 21. Let D be the foot of the A-altitude and H be the orthocenter of $\triangle ABC$. Let γ be the circle centered at A with radius $\sqrt{AH \cdot AD}$. Let M be the midpoint of $\overline{BC}$, and let $\overline{MH} \cap \gamma = K$. Show that (KBC) is tangent to γ.



Now my solution to Problem 21:

Solution. Let K_0, L_0 be the intersections of EF with (ABC).

Considering an inversion wrt γ:

$$H \longleftrightarrow D, B \longleftrightarrow F, C \longleftrightarrow E, P_A \longleftrightarrow M, R \longleftrightarrow X_A$$

so it takes (ABC) into line EF and points K_0, L_0 remain unchanged by the inversion so they lie on γ.

Under inversion the line $R-H-K-M$ becomes (X_ADKP_AA) and we have $90^o = \angle X_ADA = \angle X_AKA$ and hence X_AK is tangent to γ.

Therefore,

$$X_AK^2 = X_AL_0 \cdot X_AK_0 = X_AB \cdot X_AC$$

which ends the proof.

□

Lemma 25

Let ω_B be the circle through A tangent to BC at B and ω_C be the circle through A tangent to BC at C. Then ω_B and ω_C intersect again at P_A. Therefore $\angle P_ABC = BAP_A$ and $P_AAC = P_ACB$.

Now inspired by the following problem:

Problem 22. Polish MO 2018 P5

An acute triangle ABC in which $AB < AC$ is given. Points E and F are feet of its heights from B and C, respectively. The line tangent in point A to the circle escribed on ABC crosses BC at P. The line parallel to BC that goes through point A crosses EF at Q. Prove PQ is perpendicular to the median from A of triangle ABC.

Solution. (A), (AEFH), (BFEC) 3 axis theorem

□

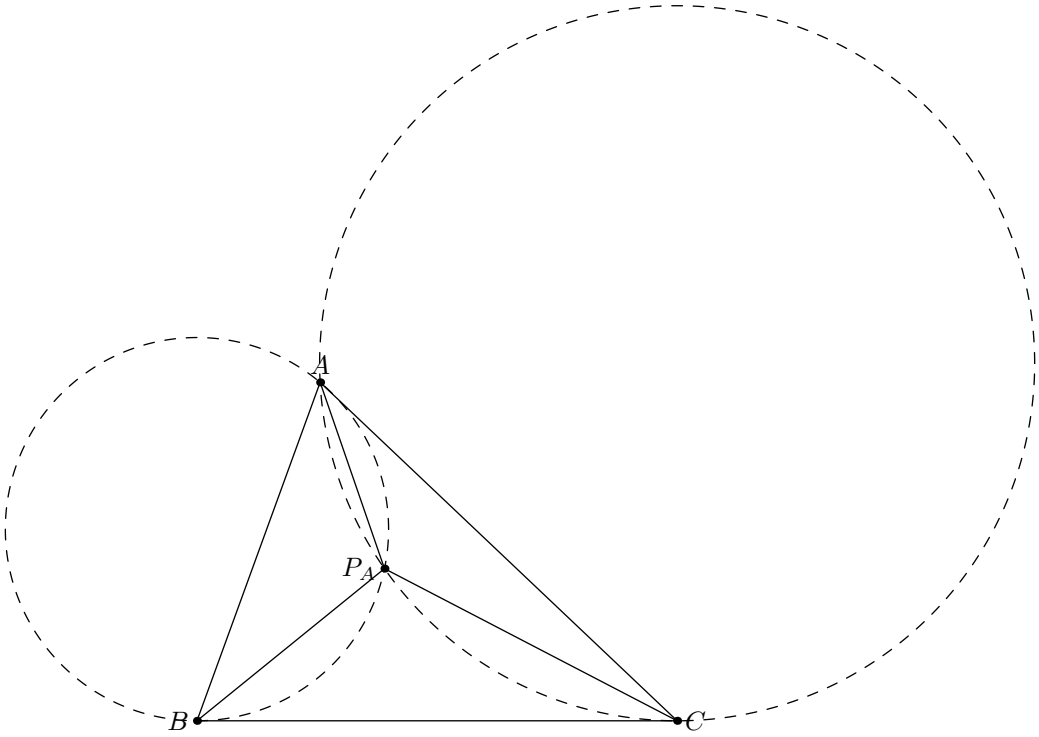


Figure 3.3: *Lemma 25.*

we can formulate the following lemma:

Lemma 26

Let $P = BC\cap$ tangent to (ABC) at A, $Q = EF\cap$ parallel to BC through A, $R =$ midpoint of AH and O_1 be the circumcenter of (BHC). Then

- $PQ \perp AM$
- $P - Q - R$
- PQ bisects segment AP_A
- R is the orthocenter of $\triangle AMP$
- PP_A is tangent (BHP_AC)
- (PRP_AMO_1)

Proof. We've already proven above that $PQ \perp AM$ and that PQ is the radical axis of (A) and $(BFEC)$.

Recalling *Lemma 5* we have $\angle REM = 90^o$ and

$$Pow_{(A)}(R) = AR^2 = RE^2 = RM^2 - ME^2 = Pow_{(BFEC)}(R)$$

and we have $R \in PQ$. □

with which we'll defeat the following problem

Problem 23. ELMO 2018 P4

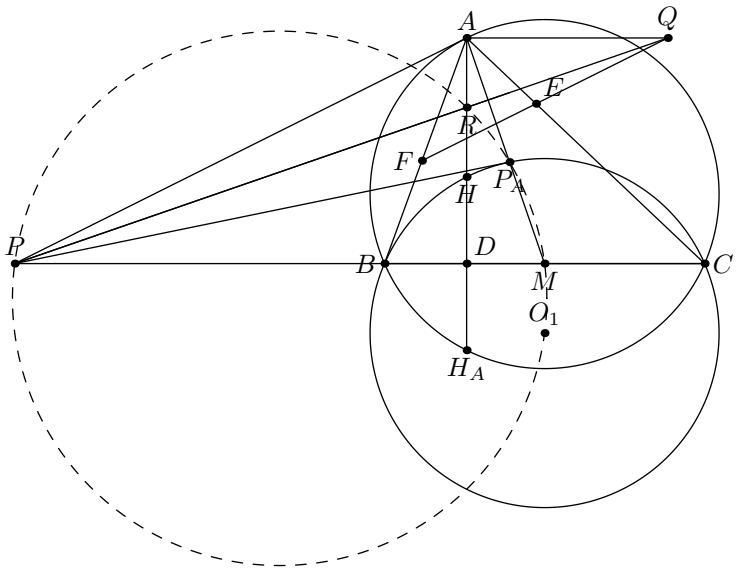


Figure 3.4: *Lemma 26*

Let ABC be a scalene triangle with orthocenter H and circumcenter O. Let P be the midpoint of $\overline{AH}$ and let T be on line BC with $\angle TAO = 90°$. Let X be the foot of the altitude from O onto line PT. Prove that the midpoint of $\overline{PX}$ lies on the nine-point circle of $\triangle ABC$.

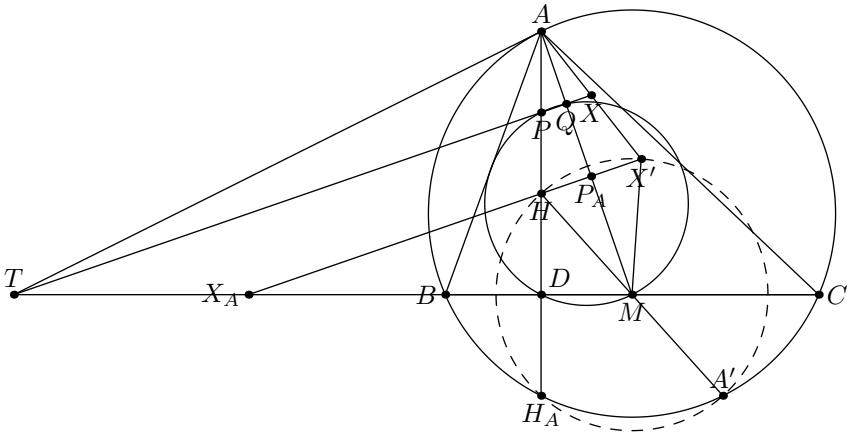


Figure 3.5: *Problem 23*

Solution. Recalling *Lemma 26*, we have $PT \perp AM$. Let $Q = PT \cap AM$, A' be the antipode of A wrt to (ABC) and P_A - the Humpty point of $\triangle ABC$.

Now consider a homothety with center A and ratio 2:

$$P \longrightarrow H, Q \longrightarrow P_A, O \longrightarrow A'$$

and let X' be a point such that $X \longrightarrow X'$.

Relation $\angle OXP = 90^o$ is transformed into $\angle A'X'H = 90^o$ and combining it with *Lemma 6* we have $(HH_AA'X')$ with has center at M. Hence $MH = MX'$ and P_A is the midpoint of HX' and, reversing the homothey, we have Q is the midpoint of PX.

Now $\angle PQM = 90^o = \angle PDM$ and $Q \in n_9$. □

Lemma 27: Humpty reflections

Let P_1, P_2 be the reflection of P_A over M, BC respectively and R be the midpoint of AH. Then:

- $A - P_A - M - P_1$
- $P_1 \in (ABC)$
- AP_2 is the A-symmedian
- $X_A - H_A - P_2$
- $(R_A, H_A; B, C) = -1$
- $R_A - D - P_2$
- $(R_A R P_A D)$ is the *nine- point circle* of $\triangle AMX_A$

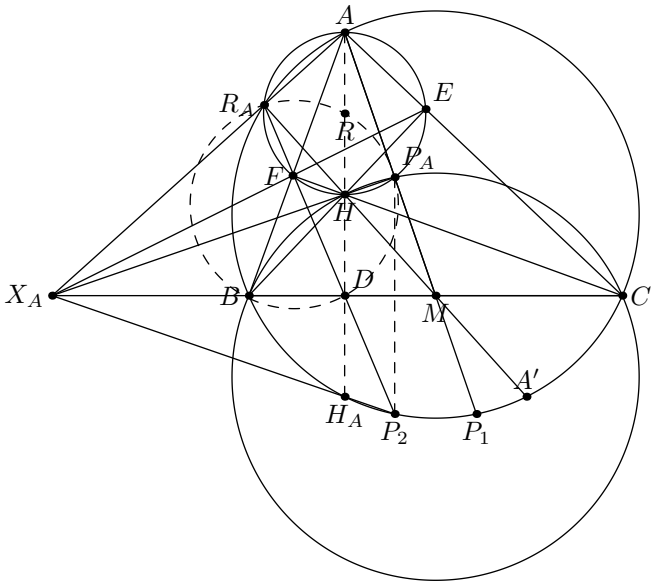


Figure 3.6: *Lemma 27*

Problem 24. ELMO SL 2013 G3

In $\triangle ABC$, a point D lies on line BC. The circumcircle of ABD meets AC at F (other than A), and the circumcircle of ADC meets AB at E (other than A). Prove that as D varies, the circumcircle of AEF always passes through a fixed point other than A, and that this point lies on the median from A to BC.

3.2 Why points

Let A_1 be the intersection of (ABC) and line parallel to BC through A, and G be the centroid.

Lemma 28

It holds that $D - G - A_1$.

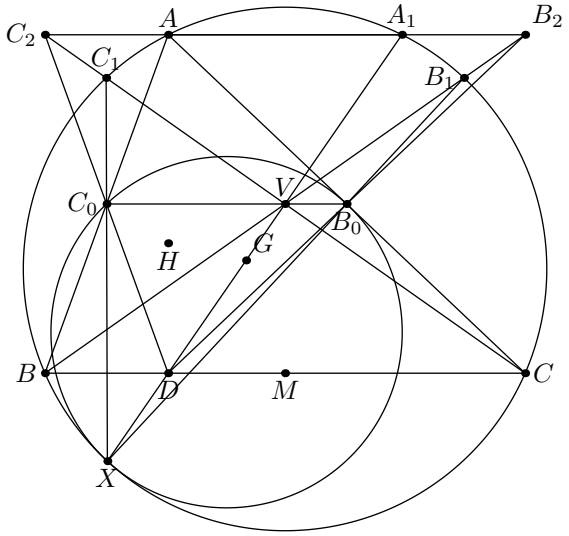


Figure 3.7: *Problem 25*

Proof. Homothety with ratio $-\frac{1}{2}$ and center G takes (ABC) to n_9. □

Before we dive deeper into what the *Why point* is let's see the motivation for it.

Problem 25. IMO SL 2011 G4

Let ABC be an acute triangle with circumcircle Ω. Let B_0 be the midpoint of AC and let C_0 be the midpoint of AB. Let D be the foot of the altitude from A and let G be the centroid of the triangle ABC. Let ω be a circle through B_0 and C_0 that is tangent to the circle Ω at a point $X \neq A$. Prove that the points D, G and X are collinear.

Proof. Let A_1 be the intersection of (ABC) and line parallel to BC through A, then from *Lemma 28*, we can instead prove $X - D - A_1$ and forget about G.

Let B_1, C_1 be the intersections of (ABC) with XC_0, XB_0, respectively and let $V = BB_1 \cap CC_1$. Then from the first Pascal: $B_0 - V - C_0$.

Moreover, let $B_2 = BB_1 \cap AA_1$, $C_2 = CC_1 \cap AA_1$.

Denote $D' = B_0B_2 \cap C_0C_2$, then $D' \in BC$ because $\triangle BCV$, $\triangle B_2VC_2$ are reflections about B_0C_0, and what follows so are $\triangle ABC$, $\triangle DB_2C_2$.

From the other Pascal:

$$XA_1 \cap BC \in B_0B_2 \to XA_1 \cap BC = D$$

which finishes the proof.

□

We can also note that $V \in XA_1$ though it's not necessary for the proof. It follows from the fact that reflection about V transforms $\triangle DB_2C_2$ into $\triangle A_1BC$.



Figure 3.8: Pascals for *Problem 25*

Let T be the point where the tangent to (ABC) at A intersects line BC and A' be the antipode to A in (ABC).

Lemma 29: Why point

Point $Y_A = A_1D \cap (ABC)$ is called the *why point* in $\triangle ABC$ and satisfys:

- (DMP_1Y_A)
- $X_A - Y_A - P_1$
- $T - Y_A - A'$
- $(R_AHDY_AX_A)$

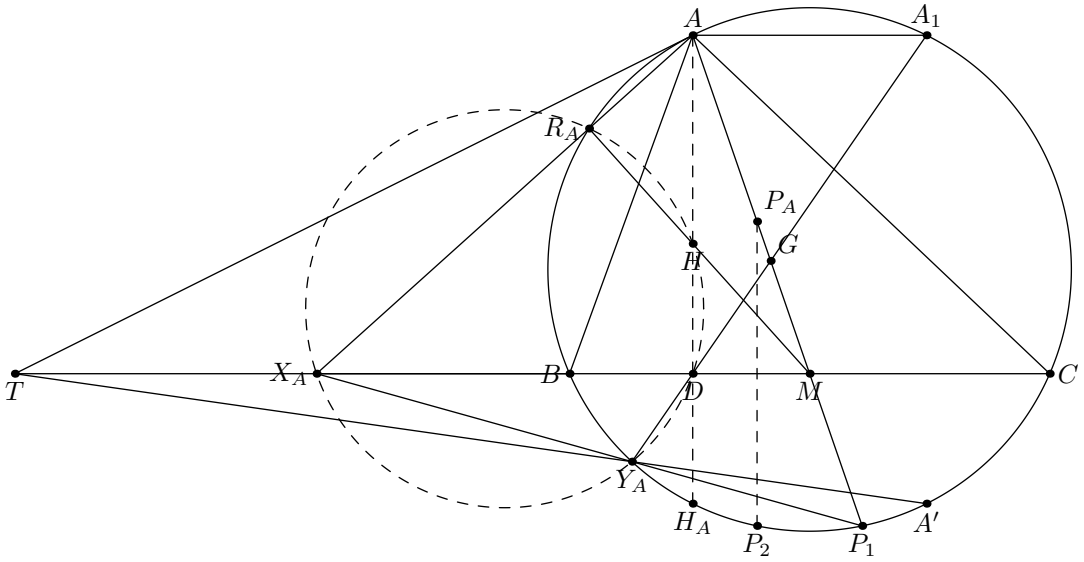


Figure 3.9: *Lemma 29*

3.3 Problems

Problem 26. EGMO 2016 P4

Two circles ω_1 and ω_2, of equal radius intersect at different points X_1 and X_2. Consider a circle ω externally tangent to ω_1 at T_1 and internally tangent to ω_2 at point T_2. Prove that lines X_1T_1 and X_2T_2 intersect at a point lying on ω.

Problem 27. USA TST 2008

Let ABC be a triangle with G as its centroid. Let P be a variable point on segment BC. Points Q and R lie on sides AC and AB respectively, such that $PQ \parallel AB$ and $PR \parallel AC$. Prove that, as P varies along segment BC, the circumcircle of triangle AQR passes through a fixed point X such that $\angle BAG = \angle CAX$.

3.4 Reading

[i3421] — *"American" Olympiad Triangle Geometry Configurations* (Sections 1.4)

[Ana17] — ***A Special Point on the Median*, Anant Mudgal, Gunmay Handa**

[Pau]— ***On Two Special Points in Triangle*, Kapil Pause**

The A-Humpty point; Preliminaries

The A-Humpty point; Inversion

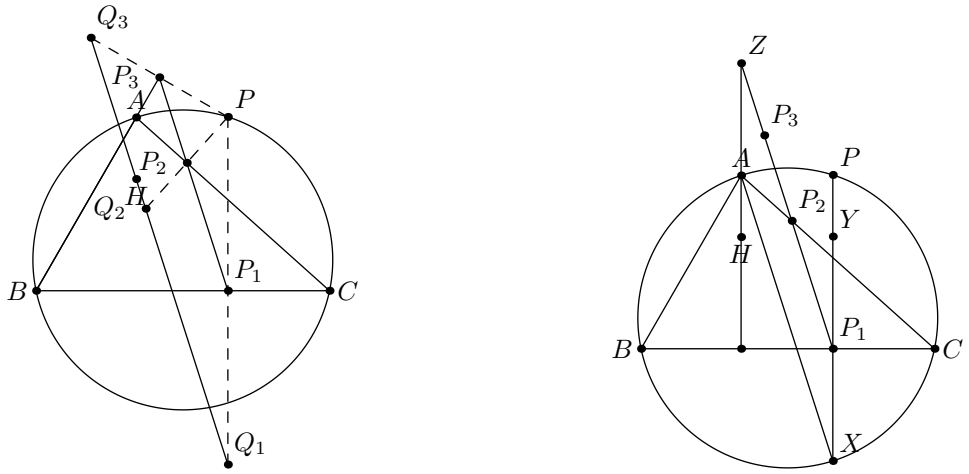
The Ex-points and the Queue-points - Part One

The Ex-points and the Queue-points - Part Two

Chapter 4

Simson & Steiner

4.1 Conection between Simson and Steiner lines



(a) *Lemma 30& 31* (b) *Lemma 32*

Figure 4.1: *Simson & Steiner*

Lemma 30: Simson line

Let P be a point and P_1, P_2, P_3 be the feet of P onto sides BC, AC, AB respectively. Then $P_1 - P_2 - P_3$ iff P lies on (ABC).

Lemma 31: Steiner line

Let P be a point on (ABC) and Q_1, Q_2, Q_3 be the reflections of P over sides BC, AC, AB respectively. Then $Q_1 - Q_2 - Q_3 - H$.

Lemma 32: Properties of the Simson line

Let $X = PP_1 \cap (ABC)$, $Y = X \curvearrowright_{ref} P_1$, $Z =$ simson line of $P \cap AH$

- $AX \parallel$ simson of P
- Y is the orthocenter of $\triangle PBC$

- $APYH$ is a parallelogram
- PP_1HZ is a parallelogram

Lemma 33: Simson Line Bisection Theorem

- the *Simson line* of P bisects the segment HP
- *Simson line* and *Steiner line* of P are connected by a homothety with origin P and ratio 2
- the midpoint of HP lies on the nine-point circle of $\triangle ABC$

Lemma 34

The Simson lines of point P and its antipode wrt (ABC) - Q are perpendicular and meet on the nine-point circle.

Lemma 35: Perpendicular Simson Lines

Let P and Q be diametrically opposite points on the circumcircle Γ. Their Simson lines ℓ_P and ℓ_Q are perpendicular to each other, and their point of intersection lies on the Nine-Point Circle n_9.

Proof. It is a classical result that the Simson line of any point $P \in \Gamma$ bisects the segment HP. Let X be the midpoint of HP and Y be the midpoint of HQ. Since PQ is a diameter of Γ, the midpoint of PQ is O. In $\triangle HPQ$, segment XY connects side midpoints, so $XY \parallel PQ$ and $XY = \frac{1}{2}PQ = R$.

Since N_9 is the midpoint of HO, $N_9X = \frac{1}{2}OP = \frac{1}{2}R$ and $N_9Y = \frac{1}{2}OQ = \frac{1}{2}R$. Thus X and Y lie on n_9. Angle chasing confirms that the Simson lines ℓ_P and ℓ_Q pass through X and Y at right angles, making their intersection a point on n_9. □

4.2 Anti-Steiner point

Lemma 36: Anti-Steiner point

Let ℓ be a line passing through H and ℓ_A, ℓ_B, ℓ_C be the reflections of ℓ over sides BC, AC, AB respectively. Then lines ℓ_A, ℓ_B, ℓ_C coincide at a point lying on (ABC) - P called the *Anti-Steiner point*.

Notice that this is a converse of *Lemma 31*.

Lemma 37

Let Q be a point on ℓ such that AP and AQ are isogonal conjugates wrt $\angle BAC$. Then $AQ \perp l$.

The next lemma is inspired by *Problem 28*.

Lemma 38: Further properties of the Anti-Steiner point

Let XY be a line passing through H where X, $Y \in (ABC)$. Further, let Z be such a point on (ABC) that line YZ is parallel to the Simson line of X with respect to ABC. Then Z is the *Anti-Steiner point* of XY.

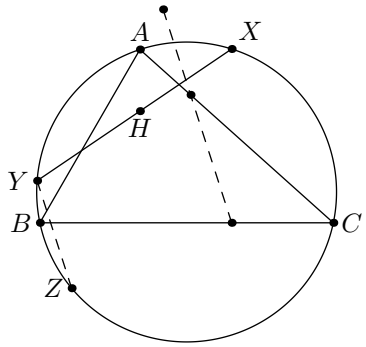


Figure 4.2: *Lemma 38*

Proof. Denote by X_A, Y_A the reflections of points X, Y over BC. Then from *Lemma 31* points X_A, H_A, Y_A are collinear.

Let $Z' = X_AY_A \cap (ABC)$ and denote $\angle Z'YX = \theta$, then $\angle XH_AY_A = \angle XH_AZ' = 180^o - \theta$. As X_AY_A is a reflection of XY over BC, we have $180^o - \theta = \angle XH_AY_A = \angle X_AHY$. Then $\angle XHX_A = \theta$ and, therefore, lines X_AH and $Z'Y$ are parallel.

Line HX_A is, obviously, the Steiner line of X so it is parallel to the Simson line of X. Then the Simson line is parallel to $Z'Y$ so we have $Z = Z'$ and we are done because Z' was the Anti-Steiner point by definition.

□

Problem 28. China TST 2011 - Quiz 3 - D2 - P1

Let H be the orthocenter of an acute trangle ABC with circumcircle Γ. Let P be a point on the arc BC (not containing A) of Γ, and let M be a point on the arc CA (not containing B) of Γ such that H lies on the segment PM. Let K be another point on Γ such that KM is parallel to the Simson line of P with respect to triangle ABC. Let Q be another point on Γ such that $PQ \parallel BC$. Segments BC and KQ intersect at a point J. Prove that $\triangle KJM$ is an isosceles triangle.

Solution. Let $X = PM \cap BC$ and H_A be a reflection of H over BC. From *Lemma 38* K is the Anti-Steiner point of PM so points K, X, H_A are collinear.

It follows that $\angle KMP = \angle KQP = \angle KJX$ so quadrilateral $(KMJX)$ is cyclic. Then

$$\angle MKJ = \angle MXJ = \angle H_AXJ = \angle BXK = 180^o - \angle KXJ = \angle KMJ$$

so triangle KJM is isosceles. □

4.3 Further configurations

Lemma 39

Let $P = M_AH_C \cap AB$,$Q = M_AH_B \cap AC$. Then points P,Q,H are collinear and $PQ \perp AM_A$.

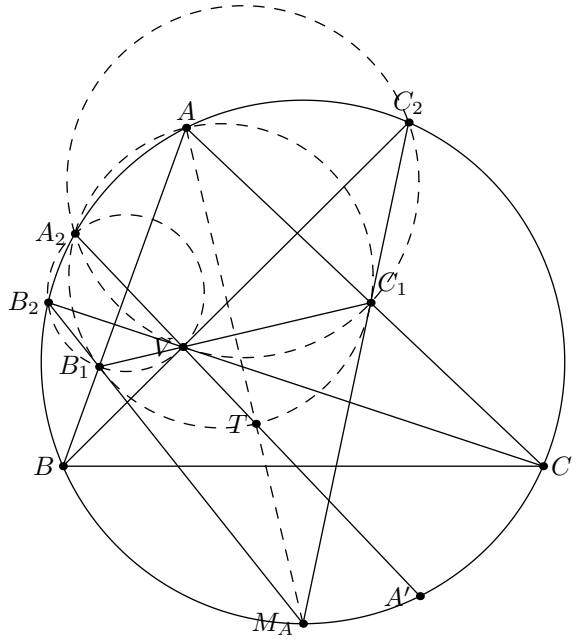


Figure 4.3: *Lemma 40*

Problem 29. IMO SL 2005 G5

Let ABC be an acute-angled triangle with $AB \neq AC$, let H be its orthocenter and M the midpoint of BC. Points D on AB and E on AC are such that $AE = AD$ and D,H,E are collinear. Prove that HM is orthogonal to the common chord of the circumcircles of triangles ABC and ADE.

The solution to *Problem 29* can be generealized into the following lemma:

Lemma 40: Generalization of the previous lemma

Let ABC be a triangle, B_1, C_1 points on sides AB, AC respectively such that $AB_1 = AC_1$, M_A the midpoint of arc BC not containing A and A' the antipode of A. Further, let B_2, C_2 be the second intersections of lines M_AB_1, M_AC_1 with circle (ABC) and $V = BC_2 \cap CB_2$. Denote by A_2, T be the intersections of line $A'V$ with circle (ABC) and segment AM_A, respectively.

Then:

- $B_1 - V - C_1$
- $(A_2VB_1B_2)$,$(A_2VC_1C_2)$
- points $(AA_2B_1TC_1)$ all lie on one circle

Problem 30. 74 Polish MO first stage P10

A selene acute triangle ABC is inscribed in circle Ω. Points M, N are the midpoints of arc BC containing and not containing A, respectively. Perpenticular bisector of AN intersects segments AB, AC at X and Y, respectively. Denote by Z the common point of Ω and (NXY), $Z \neq N$. Show that line MZ passes through the orthocenter of ABC - H.

Solution. Let P, Q be the intersections of line through H perpendicular to AN with AB, AC and H_B, H_C be the intersections of altitudes from B, C with (ABC). Moreover, let $\angle BAC = 2\alpha$,

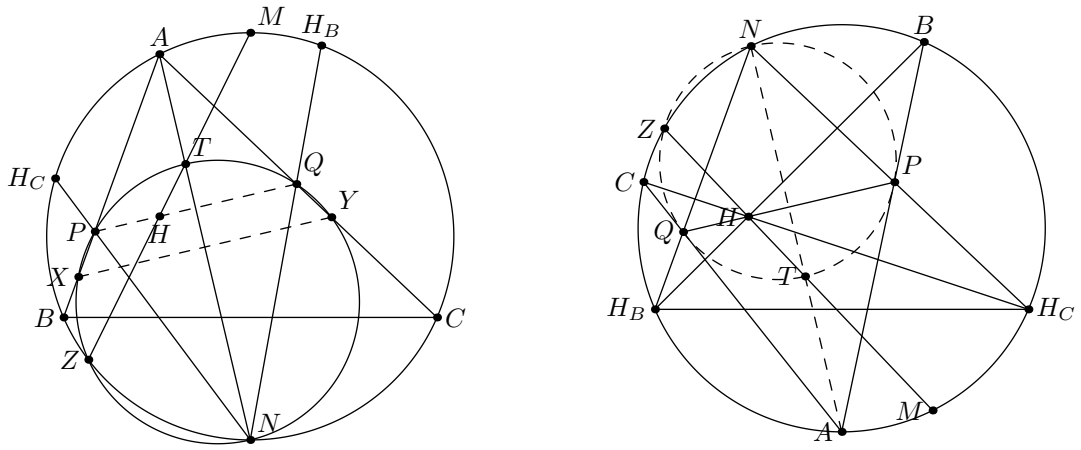


Figure 4.4: *Problem 30.*

From *Lemma 39*, $N-P-H_C$, $N-Q-H_B$. Now the angle chasing: (notice that $XA = XN$)

$$\angle XNQ = \angle XNA + \angle ANQ = \angle XAN + \angle ABH_B = \alpha + (90^o - 2\alpha) = 90^o - \alpha$$

$$\angle XPQ = 180^o - \angle APQ = 180^o - (90^o - \alpha) = 90^o + \alpha$$

so $X \in (PQN)$, similarly $Y \in (PQN)$ and we have $(PQXYNZ)$.

Now let's use *Lemma 40* in the following way (look figure), from which we get $Z - H - M$ which finishes the proof.

□

4.4 Reading

As a fun aside, I encountered Steiner line again during my *Proof assistants* course in Cambridge– you can check it out here.

Part II

Circles and midpoints

Chapter 5

Spiral Similiarly

Lemma 41: Spiral similarity

Let circles o_1, o_2 intersect at A and S, and let two lines through S intersect o_1, o_2 at B, X and C, Y respectively.

Then $\triangle ABC$ is *spiraly similar* to $\triangle AXY$, meaning that they are similar triangles, orientated the same way and they share a common vertex.

$\triangle ABC \backsimeq \triangle AXY$

We then say that the spiral similarity at A sends $B \to X$ and $C \to Y$.

Lemma 42: Spiral pair

Spiral similarities come in pairs - if a spiral similarity at A sends $BC \to XY$ then there also exists a spiral similarity at A that sends $BX \to CY$.

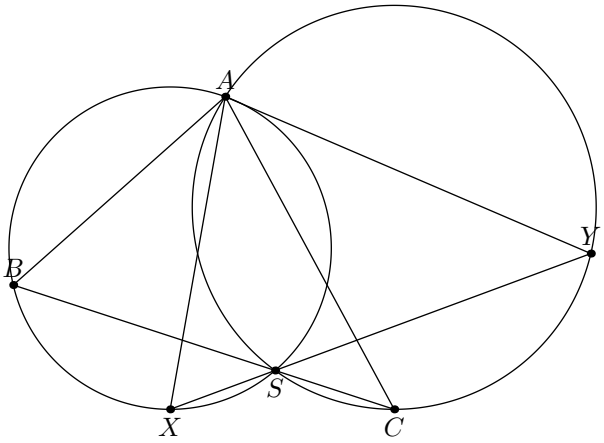


Figure 5.1: *Spiral*

Problem 31. Circles o_1, o_2 with centers O_1, O_2 respectively intersect at points A, B. Line through B cuts o_1, o_2 at C, D. Tangent to o_1, o_2 at C, D respectively cut at E. Line AE intersects $o = (AO_1O_2)$ at $F \neq E$. Prove that the length of segment EF is equal to the length of the diameter of o.

Solution. From tangencies:

$$\angle ECA = 180^o - \angle CBA = \angle DBA = 180^o - \angle EDA$$

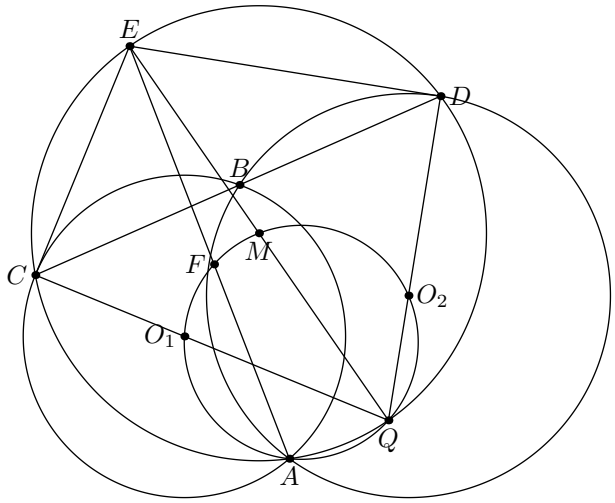


Figure 5.2: *Problem 31*

so $(ECAD)$. Further

$$\angle CO_1A = 2 \cdot \angle CBA = 2 \cdot (180^o - \angle ABD) = \angle AO_2D$$

and $CO_1 = O_1A$, $AO_2 = O_2D$ so $\triangle CO_1A \backsim \triangle DO_2A$.

Let $Q = CO_1 \cap DO_2$, then from *Lemma 41* we have $Q \in (AO_1O_2) = o$ and $Q \in (ECAD)$.

Denote $M = EQ \cap o$. Considering $90^o = \angle ECQ = \angle EAQ$ we obtain that FQ is the diameter of o. Again from spiral similarity we get $\triangle CO_1A \backsim \triangle EMA$, hence $\angle MEA = \angle EAM = \angle FQM$ and $EF = FQ$ which finishes the solution. □

Lemma 43

Let BC be the shortest side of $\triangle ABC$ and D, E be points on AB, AC respectively such that $BD = BC = CE$.

Moreover, let M_{BC} be the midpoint of arc BC containing A. Then $(AM_{BC}DE)$ and $M_{BC}C = M_{BC}E$.

Proof. Obviously, $M_{BC}B = M_{BC}C$ and $\angle ABM_{BC} = \angle ACM_{BC}$ so $\triangle ABM_{BC} \backsim \triangle ACM_{BC}$.

Then $M_{BC}C = M_{BC}E$ and

$$\angle DM_{BC}E = \angle DM_{BC}B + \angle BM_{BC}C - \angle EM_{BC}C = \angle BM_{BC}C = \angle BAC$$

□

Problem 32. Let BC be the shortest side of $\triangle ABC$ and C_0, B_0 be points on AB, AC respectively such that $BC_0 = BC = CB_0$.

Furthermore, let $F = BB_0 \cap CC_0$. Show that incenter I, orthocenter H of $\triangle B_0C_0F$ and M_{BC} midpoint of arc BC containing A lie on one line.

Proof. From *Lemma 43*, $(AM_{BC}B_0C_0)$.

Angle bisectors BI, CI are the perpendicular bisectors of lines CC_0, BB_0 respectively because of the equal segment condition. This means that I is the orthocenter of $\triangle BCF$.

Let $X = CI \cap B_0H$ and $Y = BI \cap C_0H$. Then by angle chasing:

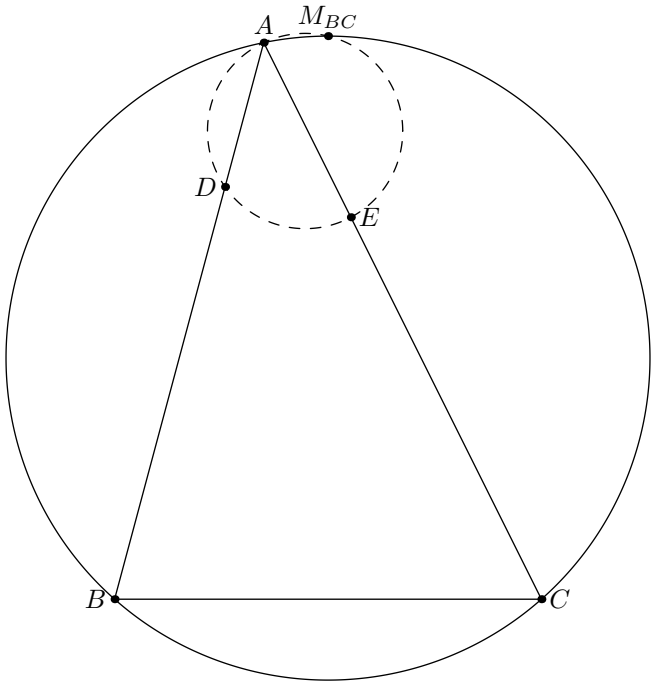


Figure 5.3: *Lemma 43*

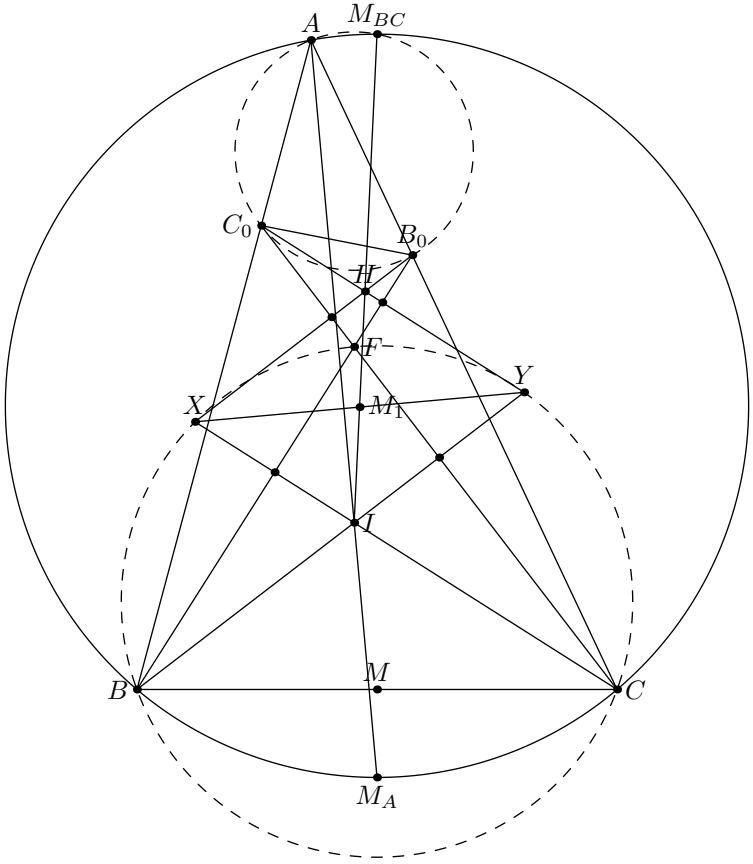


Figure 5.4: *Problem 32*

$$\angle XBB_0 = \angle XB_0B = \angle B_0BY = \angle C_0CX = \angle CC_0Y = \angle YCC_0$$

from which we can deduce that $(BCYFX)$ and we have that $\triangle IBC \backsim \triangle IXY$.

Now, denote M the midpoint of BC and M_1 the midpoint of XY. Quadrilateral $XIYH$ is a rombus so $H - M_1 - I$ and $\angle HIY = \angle M_1IY = \angle MIC = \angle M_{BC}IY$ where the last equality holds because $M_{BC}I$ is the I-symmedian is $\triangle BIC$ by *Lemma 59.* Then $M_{BC} - H - I$.

□

5.1 Problems

Problem 33. Let $ABCD$ be a quadrilateral. Let diagonals AC and BD meet at P. Let O_1 and O_2 be the circumcenters of $\triangle APD$ and $\triangle BPC$. Let M, N, and O be the midpoints of AC, BD, and O_1O_2. Show that O is the circumcenter of $\triangle MPN$.

Problem 34. IMO SL 2002 G4

Circles S_1 and S_2 intersect at points P and Q. Distinct points A_1 and B_1 (not at P or Q) are selected on S_1. The lines A_1P and B_1P meet S_2 again at A_2 and B_2 respectively, and the lines A_1B_1 and A_2B_2 meet at C. Prove that, as A_1 and B_1 vary, the circumcenters of triangles A_1A_2C all lie on one fixed circle.

Problem 35. Iran TST 2010 P5

Circles ω_1 and ω_2 intersect at points P and K. Line XY is tangent to ω_1 at X and ω_2 at Y. Let lines YP and XP meet ω_1 and ω_2, respectively, again at B and C, respectively, and let lines BX and CY meet at A. Prove that if the circumcircles of ABC and AXY meet again at Q, then $\angle QXA = \angle QKP$.

5.2 Reading

[Zha10]— Yufei Zhao, *Three Lemmas in Geometry* (Section 2: Center of Spiral Similarity)

[Bac] — Jafet Baca, *On a Special Center of Spiral Similarity*

Chapter 6

Length-based lemmas

As much as I regret to admit it, you should know all those things.

But I don't like it very much. I've included the most important theorems for completness but I don't think I want to dive deep into it and include all the proofs.

There are great resources at the end written by people who liked it more than me.

That being said, remember that length-bashing and all trigonometry should be your last resort. Always try to make as many synthetic observations as you can first!

Lemma 44: Angle bisector theorem

Let the angle bisector of $\angle BAC$ intesect BC at D. Then:

$$\frac{AB}{AC} = \frac{BD}{DC}$$

This is a special case of a more general and powerful theorem:

Lemma 45: The Ratio Lemma

Let D be a point on the side BC of a triangle $\triangle ABC$.

$$\frac{BD}{CD} = \frac{BA}{CA} \cdot \frac{\sin(\angle BAD)}{\sin(\angle CAD)}$$

Now some definitions.

Cevian is a segment connecting one of the verticies of a triangle to the opposite side of that triangle (or the extension of that side).

Lemma 46: Isogonal Cevians and Isogonal Conjugates

Two cevians AP and AQ passing through vertex A of $\triangle ABC$ are said to be *isogonal cevians* (or *isogonal lines*) if they are reflections of each other across the interior angle bisector of $\angle BAC$; that is, $\angle BAP = \angle CAQ$.

If P is any point in the plane not on the side lines of $\triangle ABC$, the isogonals of AP, BP, CP with respect to $\angle A, \angle B, \angle C$ are concurrent at a unique point P^*, called the *isogonal conjugate* of P.

Lemma 47: Steiner's Theorem for Isogonal Cevians

Let AP and AQ be isogonal conjugate cevians of $\triangle ABC$ that intersect the side BC at points P and Q. Then the following ratio holds:

$$\frac{BP}{CP} \cdot \frac{BQ}{CQ} = \left(\frac{AB}{AC}\right)^2$$

Lemma 48: Isotomic Cevians and Isotomic Conjugates

Two points D and D' on side BC of $\triangle ABC$ are *isotomic points* if they are symmetric with respect to the midpoint M of BC (i.e., $BD = CD'$). The cevians AD and AD' are called *isotomic cevians*.

If cevians AD, BE, CF concur at a point P, then their isotomic cevians AD', BE', CF' also concur at a point P^t, called the *isotomic conjugate* of P.

6.1 Triangles

Length lemmas inside the triangle.

Lemma 49: Ceva's Theorem

Let D, E, F be points on the sides BC, CA, AB of a triangle $\triangle ABC$. The cevians AD, BE, CF are concurrent if and only if:

- **(Length Form)** $\dfrac{BD}{DC} \cdot \dfrac{CE}{EA} \cdot \dfrac{AF}{FB} = 1$
- **(Trigonometric Form)** $\dfrac{\sin(\angle BAD)}{\sin(\angle CAD)} \cdot \dfrac{\sin(\angle CBE)}{\sin(\angle ABE)} \cdot \dfrac{\sin(\angle ACF)}{\sin(\angle BCF)} = 1$

Lemma 50: Menelaus' Theorem

Let a line intersect the sides (or their extensions) BC, CA, AB of a triangle $\triangle ABC$ at points D, E, F respectively. The points D, E, F are collinear if and only if:

$$\frac{BD}{DC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = 1$$

Lemma 51: Carnot's Theorem

Let D, E, F be points on the sides BC, CA, AB of a triangle $\triangle ABC$. The perpendiculars to the sides at points D, E, F are concurrent if and only if:

$$(BD^2 - DC^2) + (CE^2 - EA^2) + (AF^2 - FB^2) = 0$$

Lemma 52: Cevian Nest Theorem

Let $A_1B_1C_1$ be a triangle, $A_2B_2C_2$ a triangle inscribed in $A_1B_1C_1$ ($A_2 \in B_1C_1, B_2 \in C_1A_1, C_2 \in A_1B_1$), and $A_3B_3C_3$ a triangle inscribed in $A_2B_2C_2$ ($A_3 \in B_2C_2, B_3 \in C_2A_2, C_3 \in A_2B_2$).

If any two of the following statements hold, then so does the third:

1. The cevians A_2A_3, B_2B_3, C_2C_3 concur at a point P in $\triangle A_2B_2C_2$.
2. The cevians A_3A_1, B_3B_1, C_3C_1 concur at a point Q in $\triangle A_3B_3C_3$.
3. The cevians A_1A_2, B_1B_2, C_1C_2 concur at a point R in $\triangle A_1B_1C_1$.

When these statements hold, the three triangles $A_1B_1C_1, A_2B_2C_2, A_3B_3C_3$ are said to form a *cevian nest*.

6.2 Projective and Hexagon Theorems

Let $\triangle ABC$ and $\triangle A'B'C'$ be two triangles in the plane.

1. The two triangles are *perspective from a point* (or *radially perspective*) if the lines joining corresponding vertices—AA', BB', CC'—are concurrent at a point O. The point O is called the *center of perspective* or *perspective point*.
2. The two triangles are *perspective from a line* (or *axially perspective*) if the intersection points of corresponding side lines—$P = BC \cap B'C'$, $Q = CA \cap C'A'$, $R = AB \cap A'B'$—are collinear along a line ℓ. The line ℓ is called the *axis of perspective* or *perspective line*.

Lemma 53: Desargues' Theorem

Two triangles $\triangle ABC$ and $\triangle A'B'C'$ are perspective from a point if and only if they are perspective from a line.

That is, the lines AA', BB', CC' concur at a point O if and only if the intersection points $P = BC \cap B'C'$, $Q = CA \cap C'A'$, $R = AB \cap A'B'$ are collinear.

Lemma 54: Pappus's Hexagon Theorem

Let A, B, C be three distinct points lying on line ℓ_1, and let D, E, F be three distinct points lying on line ℓ_2. Then the three intersection points of opposite sides:

$$P = AB \cap DE, \qquad Q = BC \cap EF, \qquad R = CD \cap FA$$

are collinear.

Remark. Pappus's Theorem is a famous degenerate case of Pascal's Theorem. While Pascal's Theorem applies to six points lying on a smooth conic (such as a circle or ellipse), Pappus's Theorem corresponds to the special case where the conic degenerates into a pair of intersecting straight lines $\ell_1 \cup \ell_2$. We will explore Pascal's Theorem further in *Chapter 9*.

Lemma 55: Brianchon's Theorem

Let $ABCDEF$ be a hexagon circumscribed about a conic (or circle). Then the three main diagonals:

$$AD, \qquad BE, \qquad CF$$

are concurrent at a single point (called the *Brianchon point*).

Remark (Projective Duality)**.** Brianchon's Theorem is the exact projective dual of Pascal's Theorem: points correspond to lines, lines correspond to points, inscribed polygons correspond to circumscribed polygons, and collinearity of points corresponds to concurrency of lines.

Lemma 56

6.3 Other

Lemma 57: Ptolemy's Theorem

A quadrilateral $ABCD$ is cyclic if and only if the sum of the products of the lengths of its opposite sides equals the product of the lengths of its diagonals:

$$AB \cdot CD + BC \cdot DA = AC \cdot BD$$

Lemma 58

Suppose $2 \cdot BC = AB + AC$. Then $AI \perp IO$.

Problem 36 (Náboj Math 2023)**.** Let I and O be respectively the incenter and the circumcenter of triangle ABC with the properties $AB = 495$, $AC = 977$ and $\angle AIO = 90^o$. Determine the length of BC.

Well there's nothing in this problem that the lemma doesn't already tell you but it is included here because it was solved by me on this competition.

Problem 37 (HMMT 2013)**.** Let triangle ABC satisfy $2BC = AB + AC$ and have incenter I and circumcircle ω. Let D be the intersection of AI and ω (with A, D distinct). Prove that I is the midpoint of AD.

6.4 Reading

I've attached some more resources on trigonometry and bashing.

[Che16a] — Evan Chen, *Euclidean Geometry in Mathematical Olympiads* Chapter 5: Lengths and Ratios

[Ron21d] — Victor Rong, *Ratio Chasing*
Concurrency/collinearity tests, isogonal lines, the Cevian Nest Theorem, and ratio applications to olympiad problems.

[Ron20] — **Victor Rong, *A Few Configurations* (Section 3: Isogonal Conjugates)**

[Son23] — **Alex Song, *Angle Geometry***

[mir22] — **mira74, *Lengthbashing***

[mir20] — **mira74, *Ratio lemma***

Chapter 7

Symedian

In a triangle ABC, the A-Symmedian is a line isogonal to the A-median.

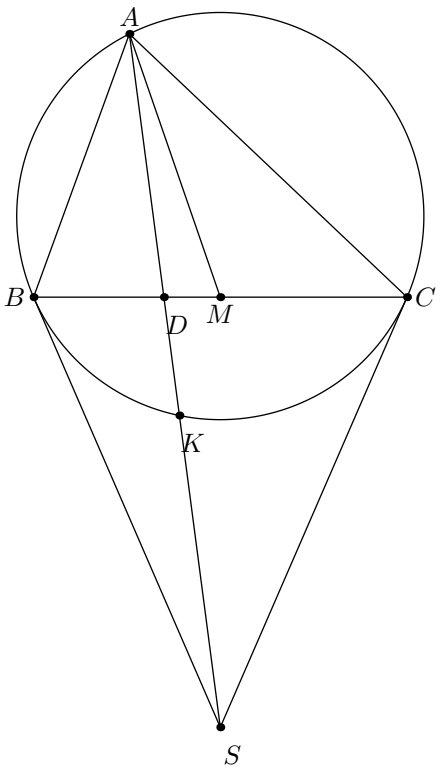


Figure 7.1: *Lemma 59.*

Lemma 59: Symmedian theorem

Let tangents to (ABC) at B and C meet at S. Then AS is the A-symmedian.

Proof. For 3 nice proofs see: [Zha10] – Section 3. □

Lemma 60: Symmedian harmonic

Let the A-symmedian intersect (ABC) again at K.

Then the quadrilateral $ABKC$ is harmonic i.e. $(A, K; B, C) = -1$.

Lemma 61: Symedian segment ratio

Let $D = AS \cap BC$. Then

$$\frac{BD}{DC} = \frac{AB^2}{AC^2}$$

Notice that this is a special case of *Lemma 47*.

Lemma 62: Symmedian on a circle construction

Let M_A, M_{BC} be the midpoints of the arc not containing and containing A respectively. Moreover let $D_1 = AM_A \cap BC$, $S_1 = AM_{BC} \cap BC$ and $K = M_{BC}D_1 \cap (ABC)$, $K \neq M_{BC}$.

Then AK is the A-symmedian.

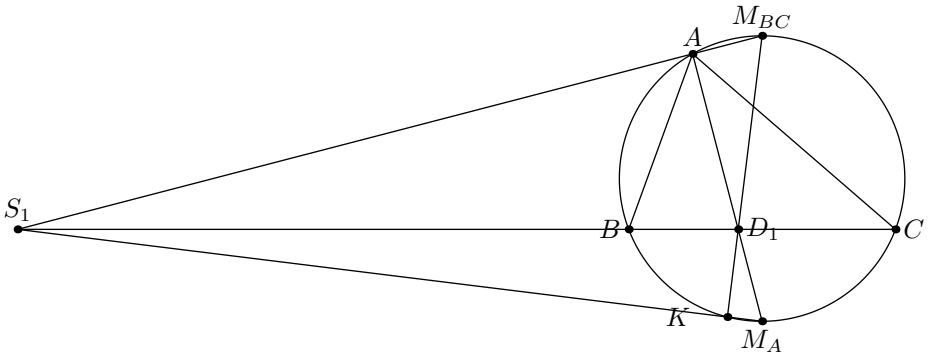


Figure 7.2: *Lemma 62*

Proof. From *Lemma 121*, we have $-1 = (S_1, D_1; B, C) \underset{M_{BC}}{=} (A, K; B, C)$ so by *Lemma 60* we are done. □

Lemma 63: Four symmedians

Four symmedians

In quadrilateral $ABKC$, we also have

- KA is K-symmedian in $\triangle KBC$
- CB is C-symmedian in $\triangle ACK$
- BC is B-symmedian in $\triangle ABK$

Lemma 64

Combining *Lemmas 61 and 63*, we can formulate the following relation

$$AB \cdot CK = AC \cdot BK$$

Lemma 65

Let P be a point inside $\triangle ABC$ nad K, L be the feet of P onto AB, AC respectively. Then $P \in A$-symmedian

$$\Leftrightarrow \frac{PK}{PL} = \frac{AB}{AC}$$

.

Proof. Let K', L' be the feet of M onto AB, AC respectively. Then:

$$\triangle AKP \sim AL'M \Leftrightarrow \frac{PK}{AP} = \frac{ML'}{AM}$$

$$\triangle ALP \sim AK'M \Leftrightarrow \frac{PL}{AP} = \frac{MK'}{AM}$$

So:

$$\frac{PK}{PL} = \frac{ML'}{MK'} = \frac{BE}{CF} = \frac{AB}{AC}$$

where E, F are the feet of altitude from B, C respectively. □

Lemma 66: Antiparallel lines

Let the *anti-parallel* lines with respect to vertex A and side BC be such lines that if we denote their intersections with sides AB, AC, X and Y respecively, then $(BCXY)$ is cyclic.

In other words, triangles $\triangle ABC \simeq \triangle AYX$ are similiar.

We will denote $BC \asymp XY$.

Lemma 67: Symmedian midline duality

A cevian from A bisects the antiparallels from A to BC iff it is the A-Symedian.

Lemma 68

Let D, E,F be the altitudes from A, B, C respectively and M_B, M_C be midpoints of sides AC, AB respectively. Then:

- $DE \cap MM_C \in A$-symmedian
- $DF \cap MM_B \in A$-symmedian

Proof. We'll prove only the first dot, the second one follows analogusly.

Let $X = DE \cap MM_C$ and mark $\angle BAC = \alpha$, $\angle ABC = \beta$, $\angle ACB = \gamma$.

By angle chasing:

$$AC \parallel MM_C \longrightarrow \angle M_C MB = \gamma$$

$$ED \asymp AB \longrightarrow \angle EDC = \alpha$$

so $\angle DXM = \beta$ and $(BDXM_C)$.

Point M_C is the center of (ADB) so $BM_C = M_CD$ and $\angle M_CDB = \angle M_CBD = \beta$. Further, $\angle M_CDX = \gamma$ and, using $(BDXM_C)$, $\angle M_CBX = \gamma$.

Now let $X' = BX \cap AC$. Segment MM_C is a midline so X is the midpoint of BX' and AX is a median in $\triangle ABX'$ which is a similar to $\triangle ABC$ with a reflection so AX is its symmedian. □

7.1 Symmedian point

Lemma 69: Symmedian point

The three symmedians of $\triangle ABC$ concide at a point K called the *Symmedian point* or *Lemoine point*.

Notice that it is the isogonal conjugate of the centroid.

Lemma 70: First Lemoine circle

Let line parallel to BC through K intersect AB, AC at A_1, A_2. Define B_1, B_2, C_1, C_2 analogously.

Then $(A_1A_2B_1B_2C_1C_2)$ is called the *First Lemoine circle*.

Lemma 71: Second Lemoine circle

Let antiparallel to BC through K intersect AB, AC at A_1, A_2. Define B_1, B_2, C_1, C_2 analogously.

Then $(A_1A_2B_1B_2C_1C_2)$ is called the *Second Lemoine circle*.

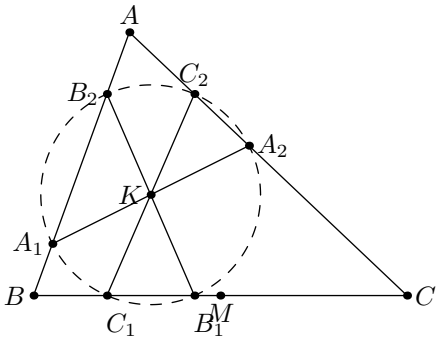


Figure 7.3: *Lemma 71*

Lemma 72: Schwatt Line

The midpoint of the A-altitude, the midpoint of BC, and the symmedian point of ABC are concurrent.

Lemma 73: Lemoine's Pedal Triangle Theorem

The symmedian point K is the only point in a ABC that is the centroid of its pedal triangle.

7.2 Dumpty point

Let H_A be the foot of altitude from A.

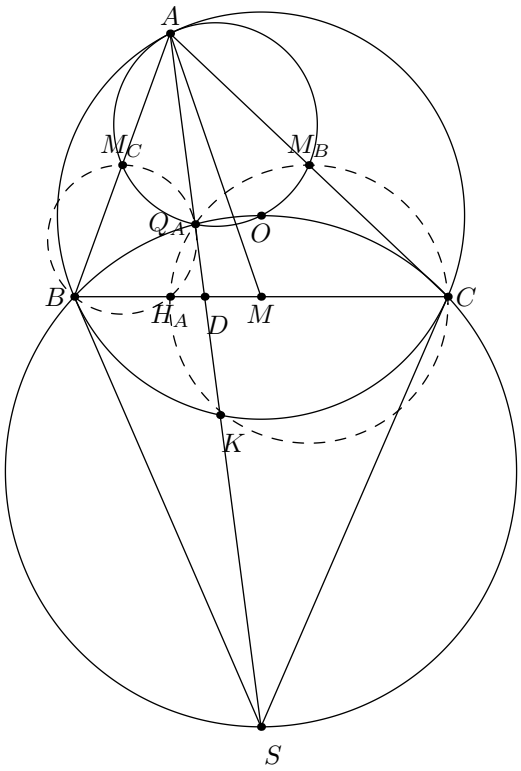


Figure 7.4: *Lemma 74*

Lemma 74: Dumpty point

Let Q_A be the foot of O onto the A-symmedian. Then it is the *A-Dumpty Point* of $\triangle ABC$.

- (BQ_AOCS)
- $(AM_BOQ_AM_C)$
- Q_A is the midpoint of AK
- $(BH_AQ_AM_C)$, $(CH_AQ_AM_B)$
- Q_A is the center of spiral similarity taking AB to AC
  therefore $\angle Q_ABA = Q_AAC$ and $\angle Q_AAB = \angle Q_ACA$

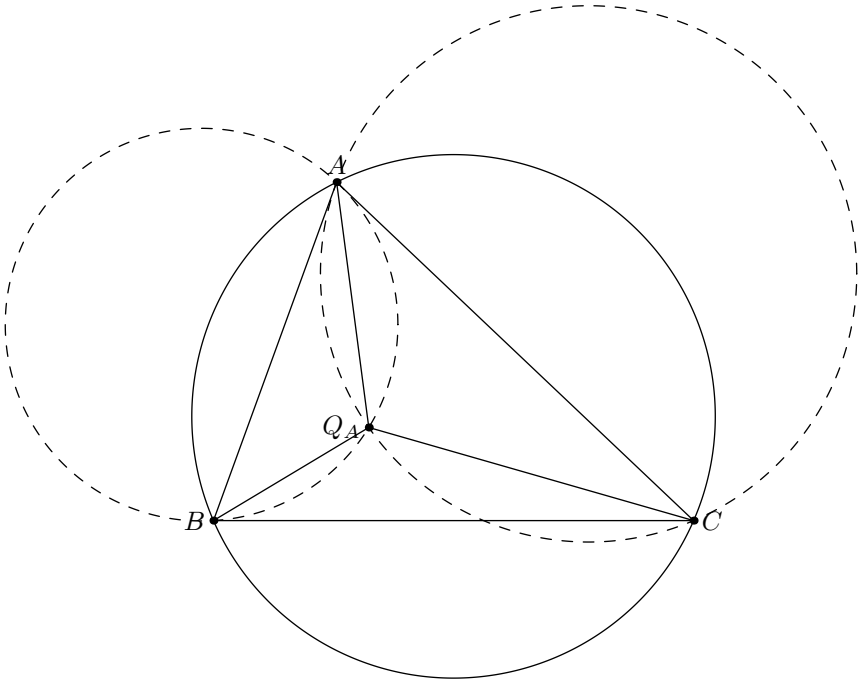


Figure 7.5: *Lemma 76*

Lemma 75

The A-Dumpty point is the isogonal conjugate of the A-Humpty point.

Lemma 76: Q_A inside triangle

Let ω_B denote the circle tangent to AC at A passing through B and ω_C the circle tangent to AB at A passing through C.

Then Q_A is the second intersection of ω_B and ω_C.

$\triangle AQ_AB \backsim \triangle CQ_AA$

Problem 38. Altitudes AA_1, BB_1, CC_1 of triangle $\triangle ABC$ intersect at H. Perpendiculars of B_1C_1, A_1C_1 meet CA, CB at P, Q, respectively. Show that line through C perpendicular to A_1B_1 bisects segment PQ.

Solution. By the magic of angle chasing:

$$\angle HQC = \angle HA_1C_1 = \angle C_1CA = \angle HCA$$

$$\angle HPC = \angle HB_1C_1 = \angle C_1CB = \angle HCB$$

so $\triangle HPC \backsim \triangle HCQ$ so H is the *Humpty point* of $\triangle CPQ$ by *Lemma 76*.

Then CH is the symmedian and so it is isogonal to CM where M is the midpoint of PQ. But we also know from *Lemma 2* that CH is isogonal to CO so by *Lemma 56*, line $C - O - M$ in perpendicular to A_1B_1.

□

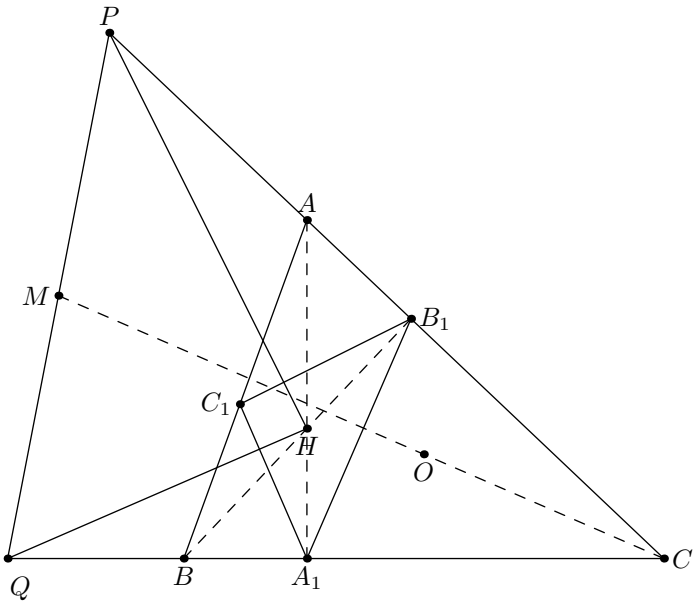


Figure 7.6: *Problem 38.*

Problem 39. ELMO 2014

Let ABC be a triangle with circumcenter O and orthocenter H. Let ω_1 and ω_2 denote the circumcircles of triangles BOC and BHC, respectively. Suppose the circle with diameter AO intersects ω_1 again at M, and line AM intersects ω_1 again at X. Similarly, suppose the circle with diameter AH intersects ω_2 again at N, and line AN intersects ω_2 again at Y . Prove that lines MN and XY are parallel.

Solution. In [Pau]. □

This problem is basically a collection of lots of things we've seen before - we just need to notice them.

Problem 40. Vietnam TST 2001

In the plane, two circles intersect at A and B, and a common tangent intersects the circles at P and Q. Let the tangents at P and Q to the circumcircle of triangle APQ intersect at S, and let H be the reflection of B across the line PQ. Prove that the points A, S, and H are collinear.

Solution. First we notice that from *Lemma 80*, AB is the median of $\triangle APQ$.

Then we can make that claim stronger by noticing that from *Lemma 25*, B is the Humpty point of $\triangle ABC$ and from *Lemma 27*, AH is a symmedian.

Then, as S also lies on the symmedian from *Lemma 59*, we get $A - H - S$ as desired.

□

Very famous problem for symmedians, there are 11 solutions on AOPS and I think most are worth reading:

Problem 41. USAMO 2008 P2

Let ABC be an acute, scalene triangle, and let M, N, and P be the midpoints of $\overline{BC}$, $\overline{CA}$, and $\overline{AB}$, respectively. Let the perpendicular bisectors of $\overline{AB}$ and $\overline{AC}$ intersect ray AM in points D and E respectively, and let lines BD and CE intersect in point F, inside of triangle ABC. Prove that points A, N, F, and P all lie on one circle.

Problem 42. Let $\triangle ABC$ be an acute triangle where $AB < AC$, D is the midpoint of BC and P is a point on segment AD. Further, let E, F be the intersections of (CDP), (BDP) with AC, AB respectively. Angle bisector of $\angle EPF$ intersects EF at Q. Prove that the tangent to (AQP) at A is perpenticular to BC.

Solution.

- Power of the point

$$AE \cdot AC = AP \cdot AD = AF \cdot AB$$

which means $(BFEC)$ and have $\angle AEF = \angle ABC = \angle FPA$ so $(AEPF)$

- Symmedian

From *Lemma 67* we have that AD is the A-symmedian in $\triangle AEF$ and from *Lemmas 64 and 44* we have:

$$\frac{EQ}{FQ} = \frac{EP}{FP} = \frac{AE}{AF}$$

so AQ is the angle bisector in $\angle EAF$

- Angle chasing

Let A' be the foot of A onto BC and denote $\angle BAC = 2\alpha$, $\angle ABC = 2\beta$. Now:

$$\angle QPA = \angle APF - \angle QPF = 2\beta - \frac{180^o - 2\alpha}{2} = 2\beta + \alpha - 90^o =$$

$$\alpha - (90^o - 2\beta) = \angle BAQ - \angle BAA' = \angle A'AQ$$

which concludes the proof.

□

7.3 Reading

[Pau] — **Kapil Pause, *On Two Special Points in Triangle***

[Zha10] — **Yufei Zhao, *Three Lemmas in Geometry* (Section 3: Symmedian)**

[Sar21] — **Srijon Sarkar, *Midpoint of Symmedian Chord***

Chapter 8

Circle lemmas

This chapter explores several fundamental lemmas related to circles in Euclidean geometry. These lemmas are powerful tools for solving a wide range of geometry problems.

8.1 Power of a Point

I don't believe you've made it this far into this book without ever hearing of POP. So this section is really only included for completeness. You can check out some more detailed resources in *Section 8.8*.

Let Ω be a circle with center O and radius R. Then, the *power of a point* P with respect to Ω is defined as the value $\text{Pow}_\Omega(P) = OP^2 - R^2$.

It is positive if P is outside Ω, negative if P is inside Ω, and exactly 0 if P lies on the circle.

Theorem 77: Power of a Point Theorem

Let A, B, C, D be four points, no three collinear, and $P = AB \cap CD$.

Then:

$$PA \cdot PB = PC \cdot PD$$

iff $(ABCD)$ lie on a circle.

In particular, we have the degenerate tangent case $A = B$, where PA is tangent to the circle, giving us $PA^2 = PC \cdot PD$.

Lemma 78: Radical Axis

Let ω_1, ω_2 be circles with circumcenters O_1, O_2 respectively. The radical axis of ω_1 and ω_2 is the set of points P safisfying:

$$Pow_{\omega_1}(P) = Pow_{\omega_2}(P)$$

Then this set forms a line perpenticular to O_1O_2.

If the circles intersect, their radical axis is the line passing through their intersection points. If they are tangent, it is their common tangent.

Here, we can also have the degenerate case where one or two of the circles is a point (a circle with radius 0). The radical axis of two points is the perpendicular bisector of the segment between them. The radical axis of a point and a circle is the line bisecting tangents from the point to the circle.

If two circles are non-intersecting, the midpoints of their internal and external common tangents always lie on their radical axis.

Lemma 79: Radical center

Let ω_1, ω_2, ω_3 be three circles and take the radical axes of every pair of circles. Then those three lines coincide at a single point (the *radical center*), unless the centers of the three circles are collinear, in which case the radical axes are pairwise parallel.

Lemma 80: Common tangent bisector

Let ω_1 and ω_2 be two circles intersecting at X and Y. Let the external tangent to the circles closer to Y touch ω_1 at A and ω_2 at B. Then XY bisects segment AB.

Proof. Let $M = XY \cap AB$. Since M lies on the line XY (the common chord, and thus the radical axis of the two circles), it has equal power to both circles. Therefore, $MA^2 = \text{Pow}_{\omega_1}(M) = \text{Pow}_{\omega_2}(M) = MB^2$, meaning $MA = MB$. □

Lemma 81: Orthogonal circles

Let two circles ω_1, ω_2 with centers O_1, O_2 intersect at points X and Y.

Then, we say that ω_1 and ω_2 are *orthogonal* if $\angle O_1XO_2 = 90^o = \angle O_1YO_2$.

This implies that an inversion around ω_1 preserves ω_2 in place and vice versa.

8.2 Tangent circles

Lemma 82: Shooting lemma

Also known as *Death Star lemma*.

Let Ω be a circle with a chord AB and let ω be a circle tangent internally to AB and Ω at points X, Y respectively.

Denote by M the midpoint of arc AB not containing Y - then $Y - X - M$.

Furthermore, let P, Q be the points where tangents from M touch ω - then $P, Q \in$ circle with center M and radius MA, denote it γ, and γ and ω are orthogonal.

If you think about it, it kind of looks like the Death Star.

Solution. Consider the inversion around γ. Then A and B stay in place, so we must have:

$$\Omega \leftrightarrow AB$$

So, as ω is tangent to both, it also must say in place and points X, Y get interchanged, $X \leftrightarrow Y$. This means that $Y - X - M$, and ω and γ are orthogonal.

Then $MP^2 = MX \cdot MY = MA^2$ and $P, Q \in \gamma$. □

Problem 43. Let ω be a circle internally tangent to circle Ω. Point I is the circumcenter of ω, chord AB of Ω is tanget to ω. Let M, N are the midpoints of arc AB of Ω containing and not containing the point of tangency to ω, respectively. Tangents from N to ω meet it at C, D, where $ABCD$ is a concave quadrilateral. Let $X = AD \cap BC$. Show that $X - I - M$.

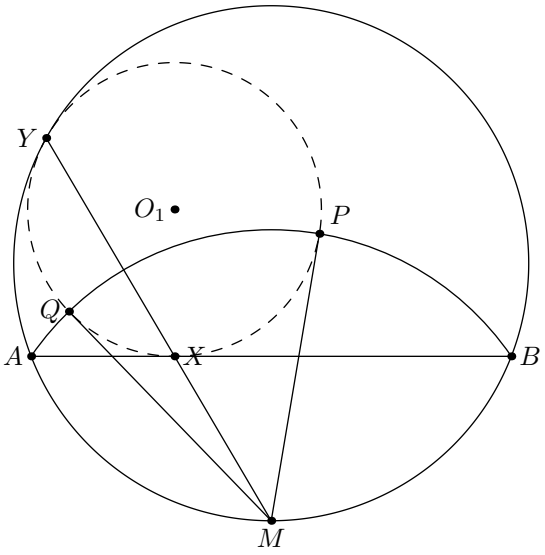


Figure 8.1: *Lemma 82*

Proof. From *Lemma 82*, it's clear that $(ABCD)$, denote if γ. Let $Y = AB \cap CD$, $Z = AC \cap BD$.

Then wrt γ:

- polar of $M = AB$ (because $MAN = MBN = 90^o$)
- polar of $I = CD$ (because γ and ω are orthogonal)
- polar of $X = YZ$ (from *Lemma 127*)

Then, as those lines meet at Y, from *Lemma 126* we have $M - I - X$. □

Now a terribly fun problem that doesn't require almost any geo knowledge.

Problem 44. Segment AB is a chord of circle Γ. Different circles ω_1, ω_2 are internally tangent to Γ at points P and Q, respectively, and to the segment AB at a common point. Chords AB and PQ meet at D. Let C be the midpoint of arc AB of circle Γ that contains point P. Prove that line CD passes through the center of ω_1.

Solution. Let X be the point of tangency of ω_1 and ω_2, M be the antipode of C wrt Γ and S be the intersection of tangents to Γ at P and Q.

By *Lemma 79* for ω_1, ω_2, Γ we get $S - A - B$.

By *Lemma 82*, we have $P - X - M$ and $C - X - Q$.

Now, let's define ω_S to be the circle with center S and radius SP and ω_M to be the circle with center M and radius MA. We'll prove that the desired line is the radical axis of those two circles.

Consider:

$$\omega_S \cap \Gamma = PQ$$

$$\omega_M \cap \Gamma = AB$$

so again by *Lemma 79* $D = PQ \cap AB \in \omega_S \cap \omega_M$.

Furthermore, let r_1 be the radius of ω_1. Then $Pow_{\omega_S}(O_1) = r_1^2$ as $\angle O_1PS = 90^o$ and $Pow_{\omega_M}(O_1) = r_1^2$ as from *Lemma 82* circles ω_1 and ω_M are orthogonal. Hence $O_1 \in \omega_S \cap \omega_M$.

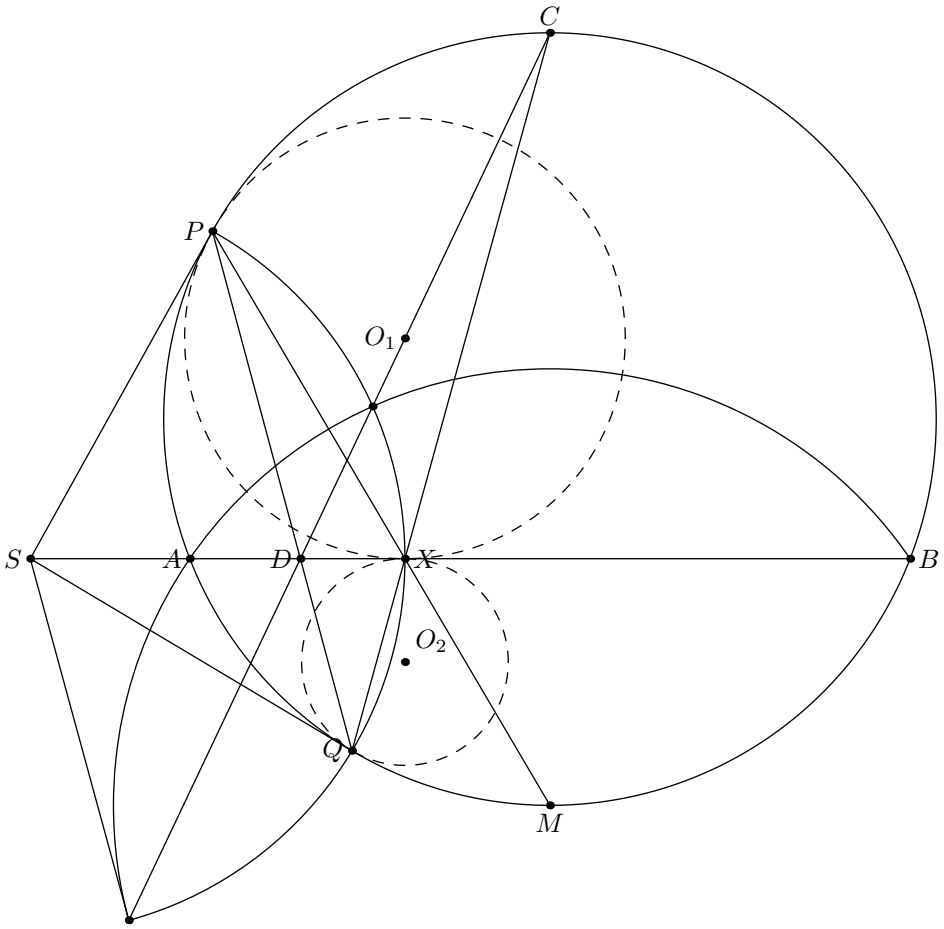


Figure 8.2: *Problem 44*

Finally, $Pow_{\omega_M}(C) = CA^2$ and $Pow_{\omega_S}(C) = CX \cdot CQ = CA^2$ as again from *Lemma 82* circle with center C and radius CA and circle ω_2 are orthogonal. Therefore $C \in \omega_S \cap \omega_M$ and we have that $D - O_1 - C$ as they all lie on the radical axis of ω_S and ω_M.

□

8.3 Curvilinear incircles

Curvilinear incircles—circles tangent to two sides of a triangle and the circumcircle. We will see a special case–*the mixtlinear incircle* later on in *Cabpter 12*.

Lemma 83: Curvilinear - Sawayama Thebault

Let I be the incenter of $\triangle ABC$, D be a point on BC and let ω_1, ω_2 be circles tangent to AD, internally to (ABC) and DC, BD respectively. Furthermore, let ω_1 be tangent to DC, AD at X_1, Y_1 respectively, and let ω_2 be tangent to BD, AD at X_2, Y_2 respectively.

Then:

- $X_1 - Y_1 - I$
- $X_2 - Y_2 - I$
- $O_1 - I - O_2$

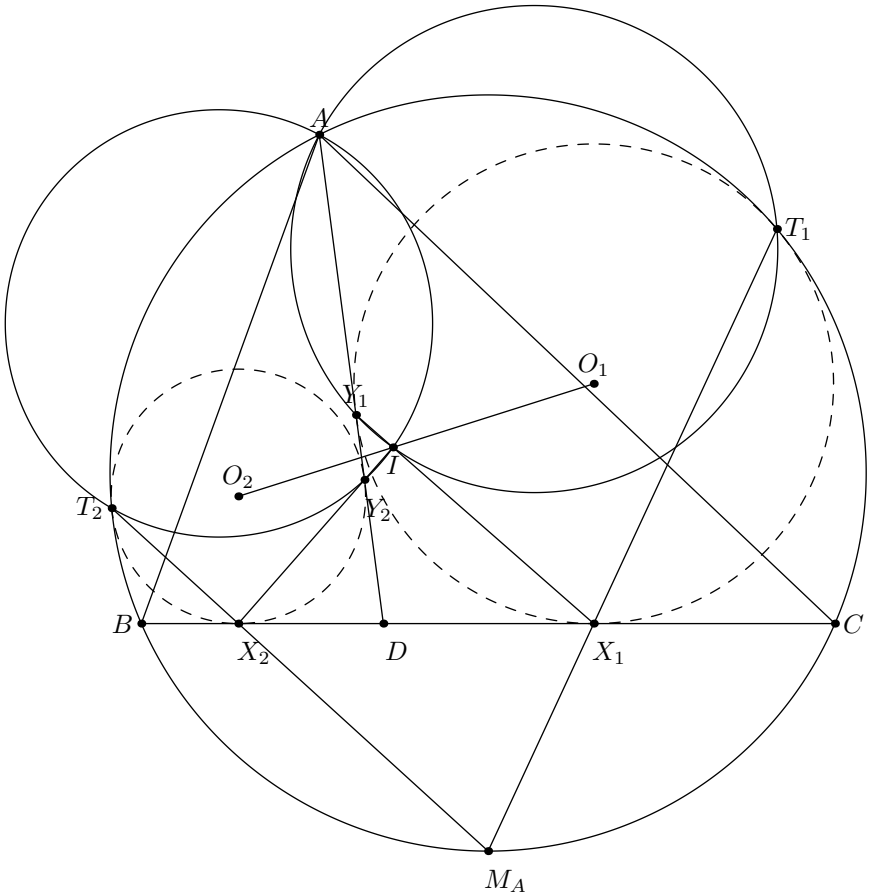


Figure 8.3: *Lemma 83 & 84*

Lemma 84: Curvilinear - Configuration 2

Furthermore, let T_1, T_2 be points of tangency of ω_1, ω_2 to (ABC) and M_A be the midpoint of arc BC not containing A.

Then:

- $T_1 - X_1 - M_A$
- $T_2 - X_2 - M_A$
- (AY_1IT_1)
- (AY_2IT_2)

8.4 Monge theorem

Let $\omega_1, \omega_2, \omega_3$ be three circles with centers O_1, O_2, O_3.

For any two, noncoaxial circles ω_i and ω_j, we define two *centers of similitude* as follows:

- *Positive Center of Similitude (S_{ij}^+):*

  The center of homothety taking ω_i to ω_j with a positive ratio. This is the intersection of the external common tangents.

- *Negative Center of Similitude (S_{ij}^-):*

  The center of homothety taking ω_i to ω_j with a negative ratio. This is the intersection of the internal common tangents.

Lemma 85: Monge's Theorem

For any noncoaxial circles ω_i, ω_j, ω_k:

1. $(+++)$ *Configuration:* The three positive centers $S_{ij}^+, S_{jk}^+, S_{ki}^+$ are collinear.
2. $(+--)$ *Configuration:* One positive center and two negative centers (e.g., $S_{ij}^+, S_{jk}^-, S_{ki}^-$) are collinear.

Problem 45. Given is an acute-angled triangle ABC inscribed in the circle Ω. Point D is the midpoint of minor arc BC of circle Ω. Circle ω, centered at D, is tangent to segment BC at E. Tangents to ω passing through point A intersect line BC at K, L, and points B, K, L, C lie in this order on line BC. Circle γ_1 is tangent to segments AL and BL and to Ω at point M. Similarly, circle γ_2 is tangent to segments AK and CK and to Ω at point N. Lines KN and LM intersect at P. Prove that $\angle KAP = \angle EAL$.

Solution. Let's rename points: $E \to M$, $D \to M_A$, $M \to T_1$ and $N \to T_2$.

Let S be the interesection of tangents to (ABC) at points B and C. Now:

$$\angle SBC = \angle BAC = 2 \cdot \angle M_AAC = 2 \cdot \angle M_ABC$$

and similarly, $\angle BCS = 2 \cdot \angle BCM_A$. Therefore, M_A is the incenter of $\triangle BCS$ so ω is tangent to sides BS and CS.

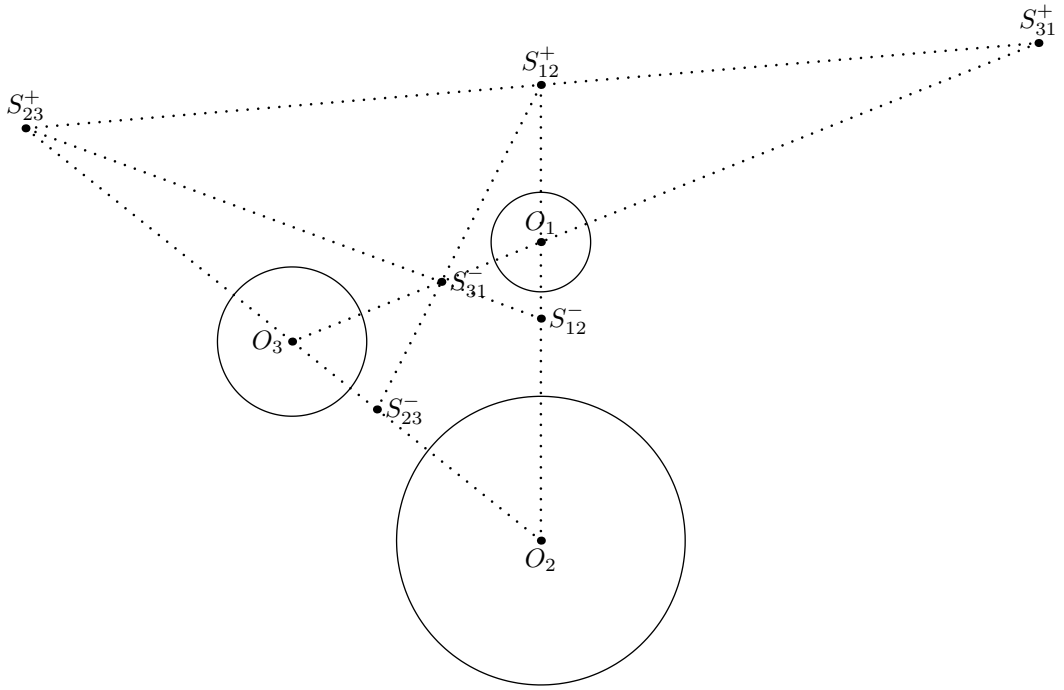


Figure 8.4: Monge's Theorem

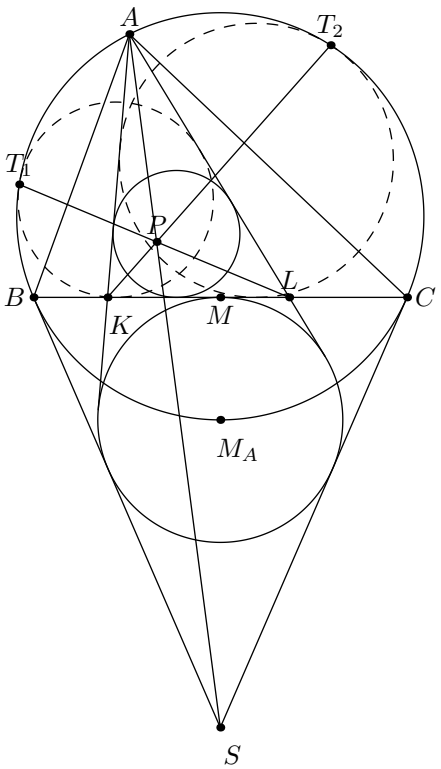


Figure 8.5: *Problem 45.*

Let γ be the incircle of $\triangle AKL$. We have:

$$S^+_{\gamma_1\gamma} = L$$

$$S^+_{\gamma_1\Omega} = T_1$$

so we have $S^+_{\gamma\Omega} \in LT_1$.
Similarly,

$$S^+_{\gamma_2\gamma} = K$$

$$S^+_{\gamma_2\Omega} = T_2$$

and $S^+_{\gamma\Omega} \in KT_2$. Hence $S^+_{\gamma\Omega} = P$.
Now consider:

$$S^+_{\gamma\omega} = A$$

$$S^+_{\omega\Omega} = S$$

so $A - P - S$.

By *Lemma 59* we also have that AS and AM are isogonal conjugates in $\triangle ABC$. To finish the proof notice that tangents from A to ω AK and AL are symmetric wrt the angle bisector AM_A. Therefore lines AS and AM are also isogonal conjugates wrt $\triangle AKL$.

□

Problem 46. USA TSTST 2017 P5

Let ABC be a triangle with incenter I. Let D be a point on side BC and let ω_B and ω_C be the incircles of $\triangle ABD$ and $\triangle ACD$, respectively. Suppose that ω_B and ω_C are tangent to segment BC at points E and F, respectively. Let P be the intersection of segment AD with the line joining the centers of ω_B and ω_C. Let X be the intersection point of lines BI and CP and let Y be the intersection point of lines CI and BP. Prove that lines EX and FY meet on the incircle of $\triangle ABC$.

Proof. make pretty diagram □

8.5 Butterfly

Lemma 86: The Butterfly Theorem

Let M be the midpoint of a chord PQ of a circle Γ. Let AB and CD be two other chords of Γ passing through M. Let X be the intersection of the chord AC with PQ, and let Y be the intersection of the chord BD with PQ. Then, M is the midpoint of the segment XY; that is, $XM = MY$.

Problem 47. Polish MO 2022

Let ABC be a triangle satisfying $AB < AC$. Let M be the midpoint of BC. A point P lies on the segment AB with $AP > PB$. A point Q lies on the segment AC with $\angle MPA = \angle AQM$. The perpendicular bisectors of BC and PQ intersect at S. Prove that $\angle BAC + \angle QSP = \angle QMP$.

8.6 Poncelet's Priorism

8.7 Problems

Problem 48. Let ABC be a non-right triangle with orthocenter H. Let M and N be points on sides AB and AC, respectively. Prove that the common chord of the circles with diameters CM and BN passes through H.

Problem 49. Two circles ω_1 and ω_2 are given. One of their common external tangents touches ω_1 at A, and the other common external tangent touches ω_2 at D. The segment AD intersects ω_1 and ω_2 again at B and C, respectively. Prove that $AB = CD$.

Problem 50. Let C be a point on a circle with diameter AB, and let D be the midpoint of arc AC not containing B. Let E be the projection of D onto the line BC, and let F be the intersection of the line AE with the circle. Prove that the line BF bisects the segment DE.

Problem 51. Let the incircle of triangle ABC touch the sides BC, CA, and AB at D, E, and F, respectively. Let Y_1, Y_2, Z_1, Z_2, and M be the midpoints of segments BF, BD, CE, CD, and BC, respectively. Let X be the intersection of the lines Y_1Y_2 and Z_1Z_2. Prove that the line MX is perpendicular to BC.

Problem 52. Let $ABCD$ be a cyclic quadrilateral with sides AB and CD not parallel. Let M be the midpoint of CD. Let P be a point inside quadrilateral $ABCD$ such that $PA = PB = CM$. Prove that the lines AB, CD, and the perpendicular bisector of segment MP are concurrent.

Problem 53. Let $ABCD$ be a tangential quadrilateral (a quadrilateral in which a circle can be inscribed) such that $BC = 2AB$. The perpendicular bisector of BC and the angle bisector of $\angle DCB$ intersect at point X. Prove that the lines AX and BD are perpendicular.

Problem 54. Let ABC be an acute triangle with $AB < AC$. Let E and F denote the feet of the altitudes from B and C, respectively. The tangent to the circumcircle of triangle ABC at point A intersects the line BC at point P. The line parallel to BC passing through A intersects EF at point Q. Prove that PQ is perpendicular to the median of triangle ABC from A.

Problem 55. Let circles ω_1 and ω_2, with centers O_1 and O_2 respectively, intersect at points P and Q. Their common tangent, located closer to point P, touches the circles at points A and B, respectively. Let the line perpendicular to segment BP passing through A intersect segment O_1O_2 at C. Prove that $\angle APC = 90^\circ$.

Problem 56. Points A and B lie on a circle ω, and let W be the midpoint of the minor arc AB. Choose a point C on the major arc AB. The tangent to ω at point C intersects the tangents at A and B at points X and Y, respectively. Segments WX and WY intersect AB at points N and M, respectively. Prove that the length of segment NM does not depend on the choice of point C.

Problem 57. 2000 IMO P1

Two circles G_1 and G_2 intersect at two points M and N. Let AB be the line tangent to these circles at A and B, respectively, so that M lies closer to AB than N. Let CD be the line parallel to AB and passing through the point M, with C on G_1 and D on G_2. Lines AC and BD meet at E; lines AN and CD meet at P; lines BN and CD meet at Q. Show that $EP = EQ$.

Problem 58. 1995 IMO P1

Let A, B, C, D be four distinct points on a line, in that order. The circles with diameters AC and BD intersect at X and Y. The line XY meets BC at Z. Let P be a point on the line XY other than Z. The line CP intersects the circle with diameter AC at C and M, and the line BP intersects the circle with diameter BD at B and N. Prove that the lines AM, DN, XY are concurrent.

Problem 59. Given is triangle ABC ($AB > BC$) inscribed in the circle Ω. On the sides AB and BC, points M and N are chosen, respectively, so that $AM = CN$. Lines MN and AC intersect at point K. Let P be the center of the inscribed circle of triangle AMK, and Q the center of the K-excircle of the triangle CNK. Prove that the midpoint of the arc AC, containing point B, of the circle Ω is equidistant from the points P and Q.

Problem 60. Let circles ω_1 and ω_2, with centres in O_1, O_2, respectively, intersect at two distinct points P and Q. Their common tangent, closer to point P, touches the circles at A, B respectively. Let the perpendicular from A to the line BP meet O_1O_2 at C. Prove that $\angle APC = 90°$.

Problem 61. Let ABC be a triangle with circumcircle Ω and incenter I, and let M be the midpoint of BC. The points D, E, F are selected on sides BC, CA, AB respectively, such that $ID \perp BC$, $IE \perp AI$, and $IF \perp AI$. Suppose that the circumcircle of $\triangle AEF$ intersects Ω at a point X other than A. Prove that lines XD and AM meet on Ω.

Problem 62. Let O be the circumcenter and H the orthocenter of an acute-angled triangle ABC such that $BC > CA$. Let F be the foot of the altitude CH of triangle ABC. The perpendicular to the line OF at the point F intersects the line AC at P. Prove that $\angle FHP = \angle BAC$.

Problem 63. USA TST 2010

Let ABC be a triangle. Point M and N lie on sides AC and BC respectively such that $MN||AB$. Points P and Q lie on sides AB and CB respectively such that $PQ||AC$. The incircle of triangle CMN touches segment AC at E. The incircle of triangle BPQ touches segment AB at F. Line EN and AB meet at R, and lines FQ and AC meet at S. Given that $AE = AF$, prove that the incenter of triangle AEF lies on the incircle of triangle ARS.

8.8 Reading

- [Zhu21] **Power of a Point** by Albert Zhu.
  Including the "Forgotten Coaxiality Lemma" and "Black Magic" combining PoP with barycentric coordinates.
- [Zha11] **Power of a Point** by Yufei Zhao.
  Introducing the Radical Axis Theorem, Radical Centers, and their applications to Olympiad problems.
- [Ron21b] **Incircles** by Victor Rong.
  Covers Monge's Theorem, Sawayama-Thébault, and includes a collection of great problems.
- [Zha08] **Circles** by Yufei Zhao.
  Including Death Star/Shooting lemma configurations and projective geometry quickies.
- [Zha07] **Lemmas in Euclidean Geometry** by Yufei Zhao.
  Includes discussion on curvilinear incircles.

Chapter 9

Pascal

You might think: Why does Pascal theorem deserves to have its own chapter?
Because I won a silver medal of it.

9.1 Definition

Lemma 87: Pascal's Theorem

Consider 6 points X_1, X_2, X_3, X_4, X_5, X_6 and define:

- $X_7 = X_1X_5 \cap X_2X_4$
- $X_8 = X_1X_6 \cap X_3X_4$
- $X_9 = X_2X_6 \cap X_3X_5$

Then, points X_7, X_8, X_9 are collinear if and only if points X_1, X_2, X_3, X_4, X_5, X_6 lie on a conic.

Also known as the *hexagrammum mysticum theorem*, Latin for mystical hexagram.

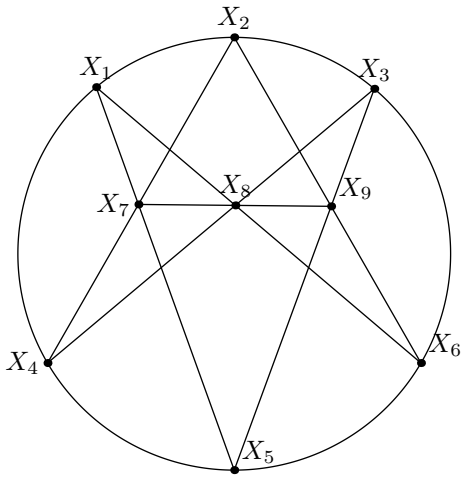


Figure 9.1: The Pascal theorem template

9.2 *Hexagrammum Mysticum*

Lemma 88: Number of different lines given by reordering points

If six unordered points are given on a conic section, they can be connected into a hexagon in 60 different ways, resulting in 60 different instances of Pascal's theorem and 60 different Pascal lines. This configuration of 60 lines is called the *Hexagrammum Mysticum*.

Proof. calculation □

Forgive me but I will not include a drawing of *Hexagrammum Mysticum* here. It terrifies me a little bit. Also, drawing that would take me hours.

9.3 Pascal theorem and me

Problem 64. EGMO 2022 P6

Let $ABCD$ be a cyclic quadrilateral with circumcentre O. Let the internal angle bisectors at A and B meet at X, the internal angle bisectors at B and C meet at Y , the internal angle bisectors at C and D meet at Z, and the internal angle bisectors at D and A meet at W. Further, let AC and BD meet at P. Suppose that the points X, Y, Z, W, O and P are distinct. Prove that O, X, Y, Z and W lie on the same circle if and only if P, X, Y, Z and W lie on the same circle.

What's the the first thing that should come to your mind when you see six points on a circle? Pascal's Theorem (it's reverse to be percise).

That was indeed my first thought. My second thought, however, was that perhaps we can generalise the problem statment claiming that P, O, X, Y, Z and W all lie on one conic. Easy to see that then the problem's statement must be true. That conic would also allow us to apply Pascal.

Okay but we have only 4 points on the circumcircle now. How do we get more? Let's extend the angle bisectors to the midpoints of the corresponding arcs. There are 8 points here now. But notice something - point O is the intersection of segments joining the arcs!

Now the the configuration looks somewhat Pascal-y. But there it's any way to include the primary definition of O now, it there?

1. What if the definition of O as the circumcentre doesn't matter? But then why should the definitions of the arc midpoints matter? Let's forget about what and consider 4 arbitrary points on the circle $(ABCD)$ and let O the intersection of the new quadrilateral's diagonals then.
2. If we're going to be using only Pascal anyway, why should the circle $(ABCD)$ matter? It could in fact be any conic.

That's how we arrive at the following lemma. See that calling it the generalisation of the problem is an understatement.

Lemma 89

Let A_1, A_2, A_3, A_4, A_5, A_6, A_7, A_8 be 8 points on a conic. Define as follows:

- $B_1 = A_1A_2 \cap A_3A_4$
- $B_2 = A_3A_4 \cap A_5A_6$

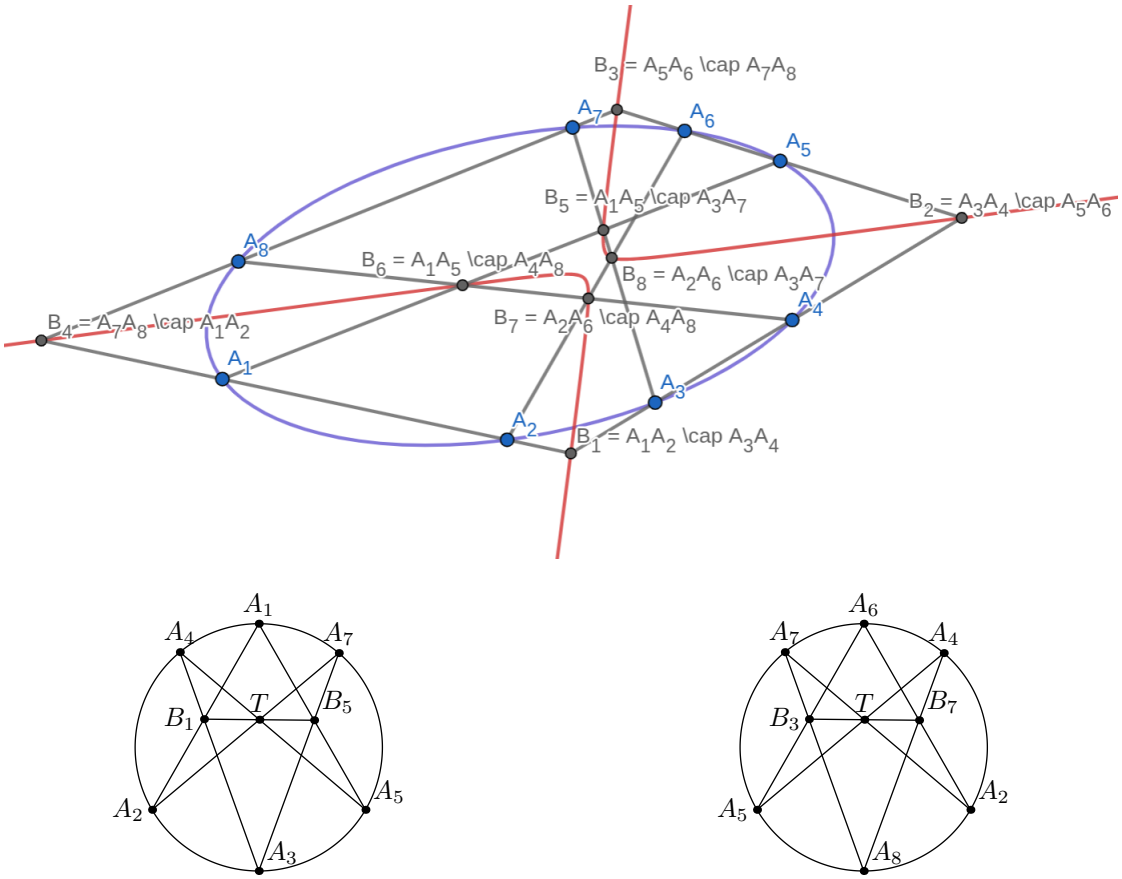


Figure 9.2: First pair of Pascals

- $B_3 = A_5A_6 \cap A_7A_8$
- $B_4 = A_7A_8 \cap A_1A_2$
- $B_5 = A_1A_5 \cap A_3A_7$
- $B_7 = A_2A_6 \cap A_4A_8$

Then points B_1, B_2, B_3, B_4, B_5, B_7 lie on one conic.

Also, you can see in the diagram below that points B_6 and B_8 lie on that conic as well but I didn't figure it out during the competition.

. . . . . .

Solution 1. First, let's prove the lemma.

Define $T = A_2A_7 \cap A_4A_5$. Then using the following Pascals (see *Figure 9.2*), we obtain $B_1 - T - B_5$ and $B_3 - T - B_7$.

Define $V = A_2A_3 \cap A_5A_8$. Then from the second pair of Pascals (see *Figure 9.3*), we get $B_5 - V - B_4$ and $B_2 - V - B_7$.

Finally, define $R = A_3A_4 \cap A_7A_8$. Notice that $A_3A_4 = B_1B_2$ and $A_7A_8 = B_3B_4$. So, equivalently, $R = B_1B_2 \cap B_3B_4$.

Now consider this third pair of Pascals (see *Figure 9.4*). From the first one, we get $T - R - V$. In the second one, we will use the converse of Pascal theorem and consider 6 points $B_3, B_5, B_2, B_1, B_7, B_4$. We deduce that their appropriate intersections correspond to points T, R, V from the collinearlities we established earlier. Then, as $T - R - V$, we get that those 6 points lie on one conic, which we wanted to prove.



Figure 9.3: Second pair of Pascals



Figure 9.4: Third pair of Pascals

Note – we can get that B_6 and B_8 lie on this conic just by renaming points.

□

Now we can finish *Problem 64* by defining – let M, N be the midpoints of arc BD containing and not containing A, respectively, and let K, L be the midpoints of arc AC containing and not containing B, respectively. Set $A \to A_1$, $B \to A_3$, $C \to A_5$, $D \to A_7$ and $N \to A_2$, $L \to A_4$, $M \to A_6$, $K \to A_8$.

Then *Lemma 89*, gives us that $X \to B_1$, $Y \to B_2$, $Z \to B_3$, $W \to B_4$, $P \to B_5$, $O \to B_7$ lie on one conic, which finishes the proof.

. . . . . .

I suppose that it isn't the only set of Pascals that solves the problem. Nethertheless, finding those sets is long and sad.

For the record, I do consider the solution to be an overkill. It is, however, the third official solution even though it doesn't mention that O doesn't need to be the circumcenter and $ABCD$ doesn't really need to be a circle at all for the proof to hold.

This problem can also be solved in a Pascal-free way.

But i won't do it.

You can read about it in the official solution.

Solution 2. Official Solutions □

Problem 65. Vietnam TST 2016 P3

Let ABC be triangle with circumcircle (O) of fixed BC, $AB \neq AC$ and BC not a diameter. Let I be the incenter of the triangle ABC and $D = AI \cap BC, E = BI \cap CA, F = CI \cap AB$. The circle passing through D and tangent to OA cuts for second time (O) at G $(G \neq A)$. GE, GF cut (O) also at M, N respectively. Let $H = BM \cap CN$. Prove that AH goes through a fixed point.

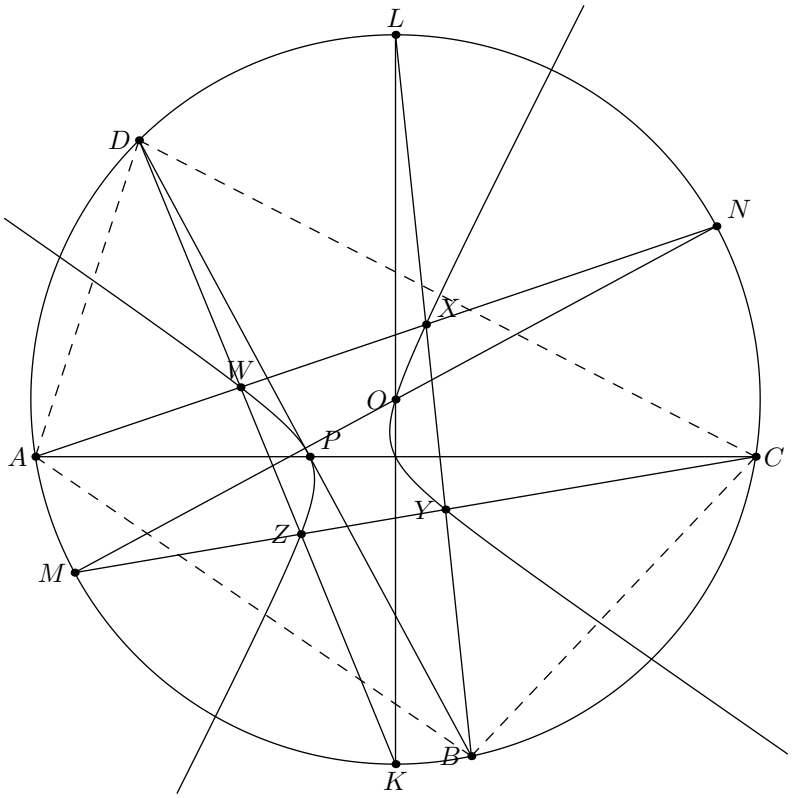


Figure 9.5: *Problem 64.*

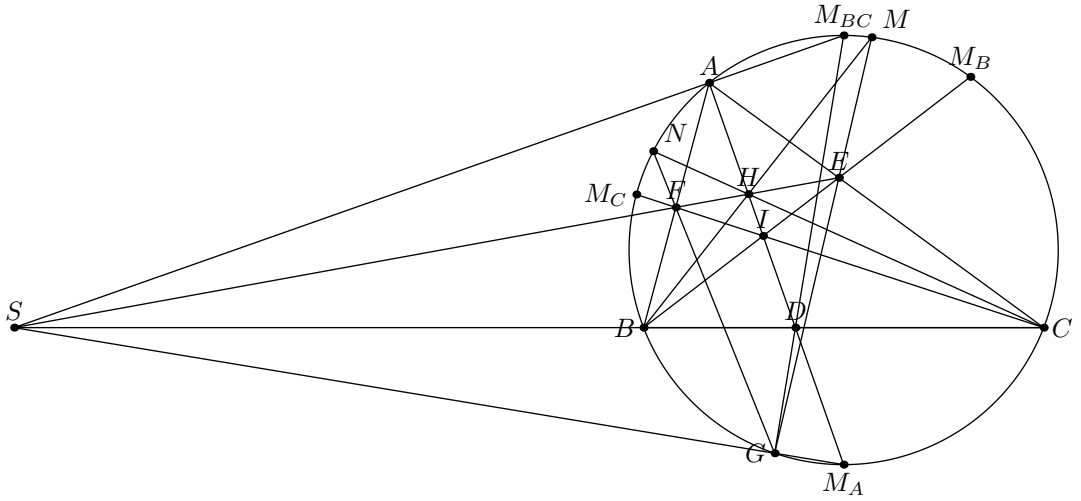


Figure 9.6: *Problem 65.*

Solution. Let points M_A, M_B, M_C be the intersections of lines AI, BI, CI with the circumcircle, respectively. Furthermore, let M_{BC} be the midpoint of arc BC containing A and H_A be the intersection of the altitude from A with the circumcircle.

As O and the orthocenter are isogonal conjugates,

$$\angle AGD = \angle OAM_A = \angle M_AAH_A = \angle M_{BC}H_AA = \angle M_{BC}GA$$

hence we have $G - D - M_{BC}$.

Let $S = BC \cap EF$. As $(B, C; S, D) = -1$, from the properties of cross-ratio we have $S - A - M_{BC}$ as well as $S - G - M_A$, as D is the orthocenter of M_ASM_{BC}.

Now consider the following 3 Pascals:

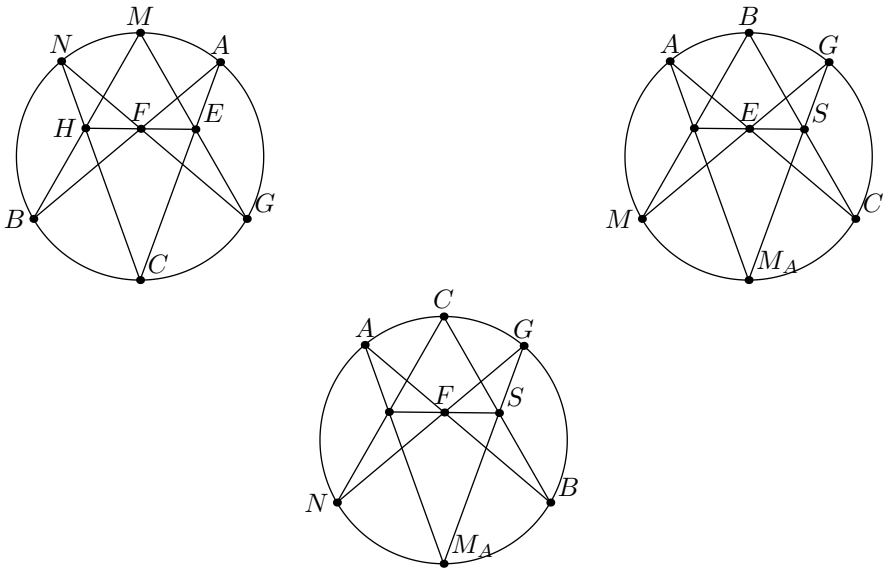


Figure 9.7: A figure with two subfigures

Therefore, we have:

$$S - F - H - E$$

$$AM_A \cap BM \in SE$$

$$AM_A \cap CN \in SE$$

and $H \in AM_A$ and M_A is the desired fixed point.

□

9.4 Reading

[Ron21c] — Victor Rong, *Projective Geometry*

[Qui17] — CJ Quines, *Pascal's Theorem*

Part III

Incircle lemmas

Chapter 10

Incircle configurations

Let the incircle of ABC touch sides BC, AC and AB at points D, E and F, respectively. Denote by M the midpoint of BC and by D_1, E_1, F_1 the touch points of A, B, C-excircles with sides BC, AC, AB respectively.

Furthermore, let M_A be the midpoint of arc (BC) not containing A and M_{BC} its antipode.

Let $\angle BAC = 2 \cdot \alpha$, $\angle ABC = 2 \cdot \beta$, $\angle BCA = 2 \cdot \gamma$.

We claim some well-known properties:

- $\angle BIC = 90^o + \alpha$
- $A - I - I_A$
- $BD = \frac{AB+BC-AC}{2}$

Lemma 90: Trident lemma

Also knows as *trillium theorem* or *Incenter-excenter lemma*.

Points B, I, C I_A lie on a circle with circumcenter M_A.

Lemma 91: The ABC-Lemma

Also called *Iran lemma*.

In triangle ABC the A-touch chord, B-bisector and C-midline coincide.

Namely, lines EF, BI, MN have a common point X, where N is the midpoint of AC. Furthermore, $\angle BXC = 90^o$.

It gets its second name from this problem:

Problem 66. Iran TST 2009 P19

In triangle ABC, D, E and F are the points of tangency of incircle with the center of I to BC, CA and AB respectively. Let M be the foot of the perpendicular from D to EF. P is on DM such that $DP = MP$. If H is the orthocenter of BIC, prove that PH bisects EF.

Lemma 92: Gergonne Point

Lines AD, BE, CF coincide in point called *Gergonne Point*.

Furthermore, it is also the symmedian point of $\triangle DEF$.

Lemma 93: Median-Incircle Concurrency

The A-Median, touchchord EF and line DI intersect at one point.

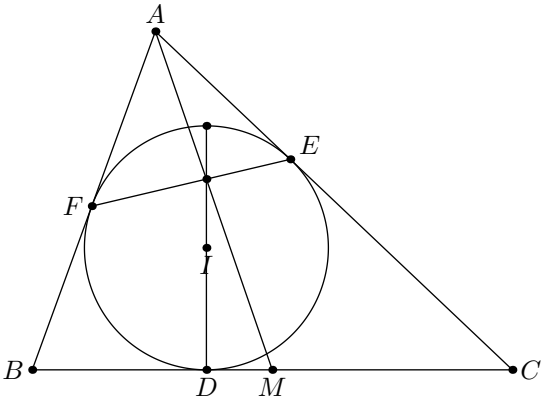


Figure 10.1: *Lemma 93*

Lemma 94

Let ω be the circle with center M and radius MD. Then M_BM_C is the radical axis of ω and (BIC).

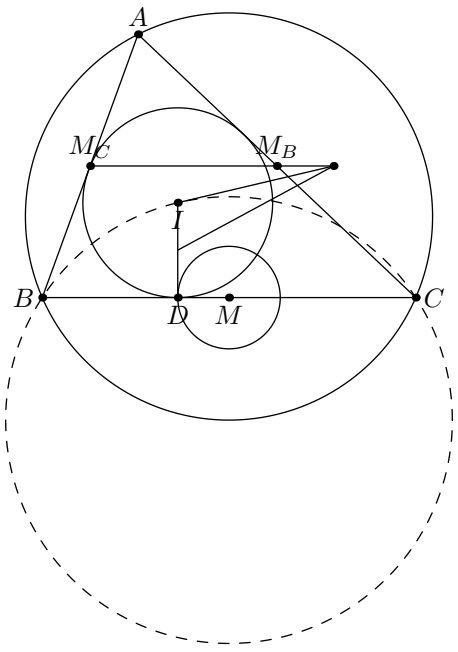


Figure 10.2: *Problem 137*

Problem 67. CAMO 2020/3

Let ABC be a triangle with incircle ω, and let touch BC, CA, AB at D, E, F, respectively. Point M is the midpoint of EF, and T is the point on ω such that DT is a diameter. Line MT meets the line through A parallel to BC at P and ω again at Q. Lines DF and DE intersect line AP at X and Y respectively. Prove that the circumcircles of $\triangle APQ$ and $\triangle DXY$ are tangent.

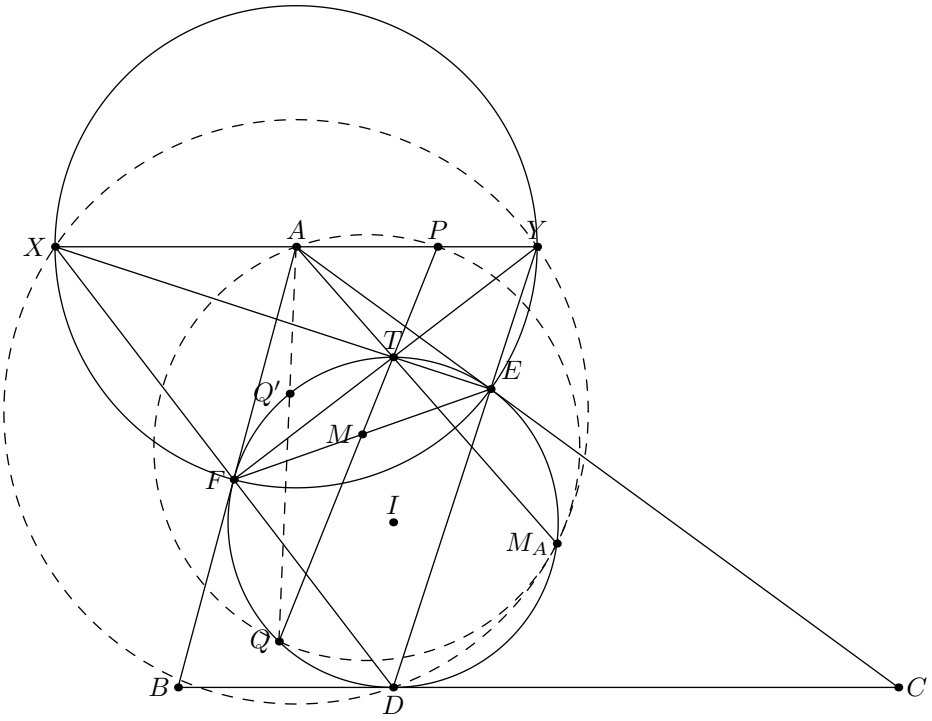


Figure 10.3: *Problem 67*

Solution. As a starter - angle chasing. Notice:

$$\angle AFX = \angle BFD = \angle BDF = \angle AXF$$

which means that $AX = AF$. Analogusly, we get that $AY = AF$ and hence A is the circumcenter of $(XYEF)$ which we will call γ.

Which also means that E, F are the foot of altitudes in $\triangle DXY$. Combining it with $(DEFT)$ and $DT \perp XY$, we get that T is the orthocenter of $\triangle DXY$, therefore $X - T - E$ and $Y - T - F$.

Now, let's rotate our diagram to get the following configuration:

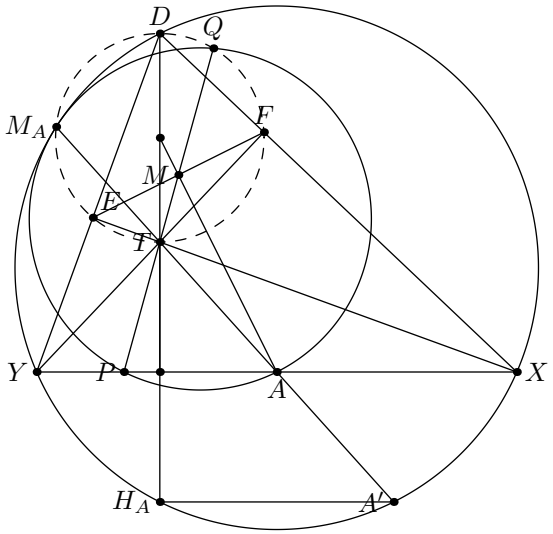


Figure 10.4: *Problem 67* – Part 2

Let M_A be the second intersection of AT with (DEF), A' the antipode of D wrt (DXY) and H_A the reflection of T over XY. Then M_A lies on (DXY) and $M_A - T - A - A'$, as a consequence of *Lemma 6*. Chasing angles we get:

$$\angle M_AQP = \angle M_AQT = \angle M_ADT = \angle M_ADH_A = \angle M_AA'H_A = \angle M_AAP$$

so (M_AQAP).

Now, let's go back to the first diagram. Notice that from *Lemma 59* that AM_A is a symmedian in $\triangle TEF$. Then, as TQ is the median, $\angle ETM_A = \angle FTQ$ and arcs FQ and M_AE are equal. So AQ and AM_A are isogonal conjugates in $\triangle ABC$.

Finally, consider an inversion around $(XYEF)$. Let $Q' = AQ \cap (DEF)$. It sends:

$$T \leftrightarrow M_A$$

$$Q \leftrightarrow Q'$$

$$(APM_AQ) \leftrightarrow Q'T$$

$$(DM_AYX) \leftrightarrow (TXY)$$

To prove (APM_AQ) and (DM_AYX) are tangent at M_A, it is sufficient to prove that (TXY) is tanget to $Q'T$. By angle chasing:

$$\angle XTQ' = \angle XEF \text{ (as } Q'T \parallel EF) = \angle XYF$$

which finishes the proof.

□

10.1 Reading

[Che16b] — Evan Chen, *The Incenter/Excenter Lemma*

Chapter 11

Excircle

This chapter needs a long introduction.

Not because it is an especially important chapter.

But because we have this figure on the right and LaTeXwon't stop screaming at me.

There are many things that can be said about excircles.

They are, for instance, very big.

Very round.

It's hard to draw them in a pdf.

Because the entire diagram scales up.

You should think hard before drawing one.

Because once you add it to the diagram.

There is no going back.

The smaller your diagram, the more likely some lines will overlap and you will have a very confusing diagram.

That's why vector graphics is great.

Because I can just change coordinates if something looks bad.

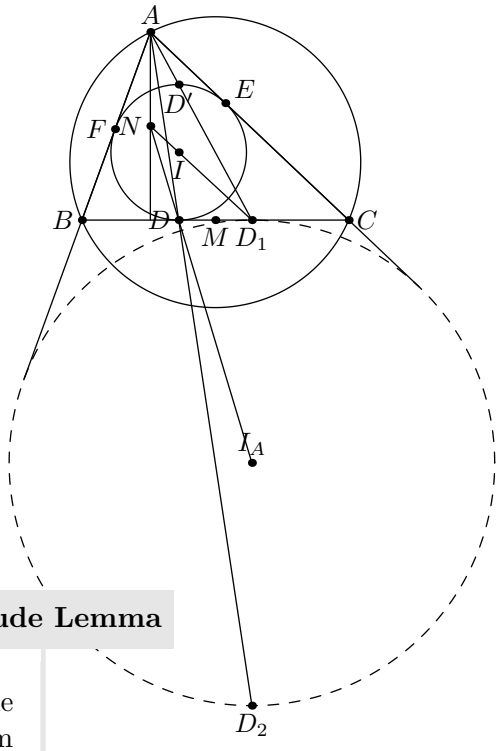


Figure 11.1: *Big Picture I*

Lemma 95: The Big Picture I - Midpoint of Altitude Lemma

Let D' be the antipode of D wrt the incircle, D_2 the antipode of D_1 wrt the excircle and N the midpoint of altitude from A.

- $BD = CD_1$
- $A - D' - D_1$
- $N - I - D_1$
- $N - D - I_A$
- $A - D - D_2$
- $IM \parallel AD_1$
- $AD \parallel I_AM$

This is the first of the Big Pictures we will see in this chapter. They are meant to be configurations you can easily

recognise in a problem and be able to recall any of the properties.

Lemma 96: Nagel Point

Lines AD_1, BE_1, CF_1 coincide at the *Nagel point*.

Furthermore, it is the isotomic conjugate of the *Gergonne Point*.

Lemma 97

$AI \cdot AI_A = AB \cdot AC$, so under $\sqrt{bc}$ inversion I goes to I_A.

Lemma 98: The Big Picture II - Excentral-Orthic Duality

Let M_{BC}, M_{AC},M_{AB} be antipodes wrt (ABC) of points M_A, M_B, M_C respectively.

- I is the orthocenter of $\triangle I_A I_B I_C$
- points M_{BC}, M_{AC},M_{AB} are the midpoints of segemnts $I_B I_C$, $I_A I_C$, $I_A I_B$ respectively
- (ABC) is the nine-point circle of $\triangle I_A I_B I_C$
- lines $I_A D_1$, $I_B E_1$, $I_C F_1$ coincide at the *Bevan point* - B_{ABC} of $\triangle ABC$
- O is the midpoint of IB_{ABC} which is the Euler line of $\triangle I_A I_B I_C$

Lemma 99: The Big Picture III

Let the B-excircle touch sides BC, AB at points B_1, B_2 and the C-excircle touch sides BC, AC at points C_1, C_2, respectively.

Furthermore, let $X = C_1C_2 \cap I_B I_C$ and $Y = B_1B_2 \cap I_B I_C$. Then:

- $C_1 - F_1 - Y$
- $B_1 - E_1 - X$
- $(B_1 Y X C_1)$
- $(X C_2 Y E_1)$
- $(X B_2 Y F_1)$

Proof. Let $X' = B_1E_1 \cap C_1C_2$ and $Y = C_1F_1 \cap B_1B_2$.

We have $C_1F_1 \perp I_C B \parallel B_1B_2$ so $C_1Y' \perp B_1B_2$. Similarly, $B_1X' \perp C_1C_2$ and so $(C_1X'Y'B_1)$, let's call it Ω.

Now consider $\omega_1 = (F_1Y'B_2)$ and see:

$$Pow_{\omega_1}(I_C) = I_C F_1^2 = I_C C_1^2 = Pow_{\Omega}(I_C)$$

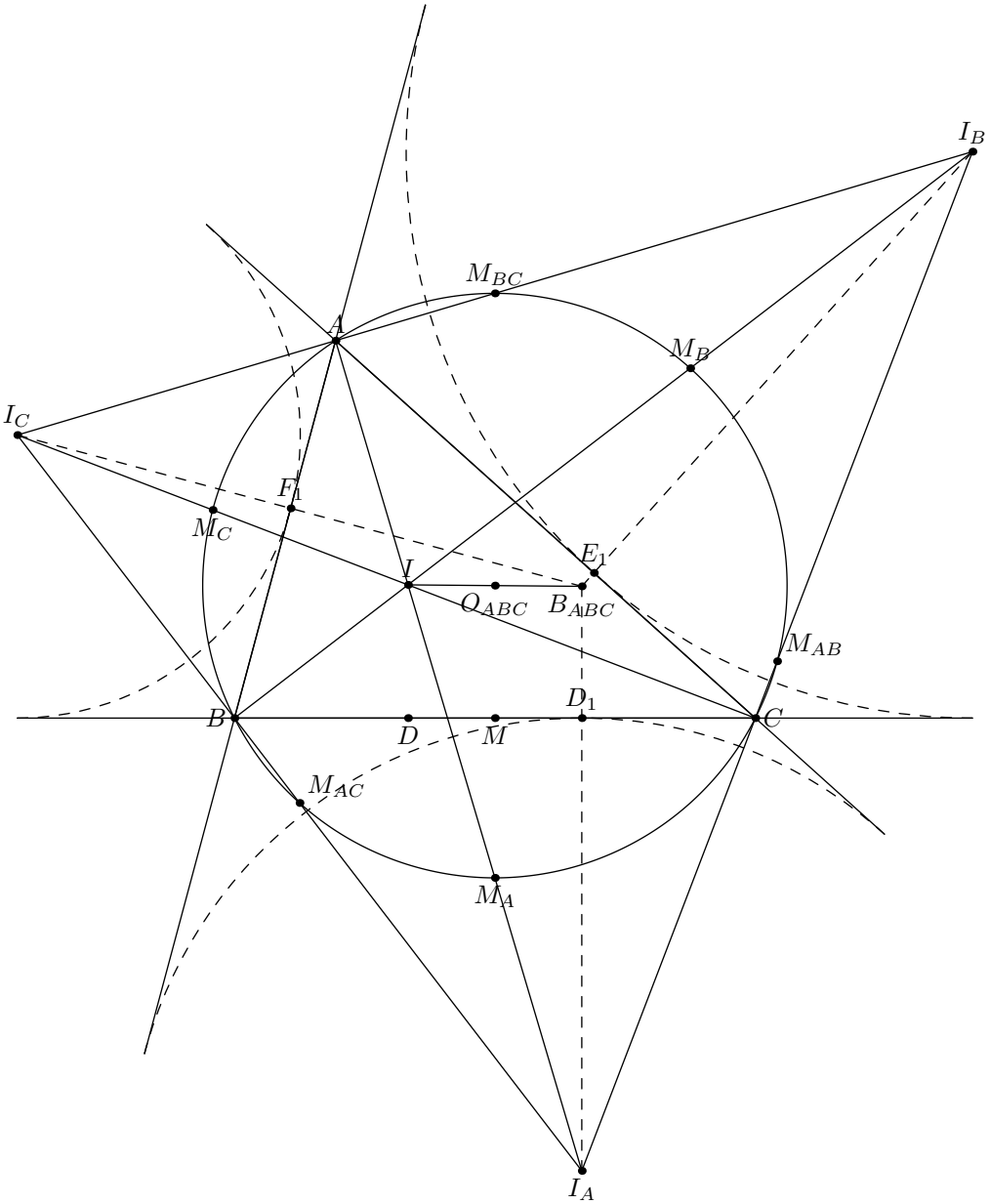


Figure 11.2: *Big Picture II*

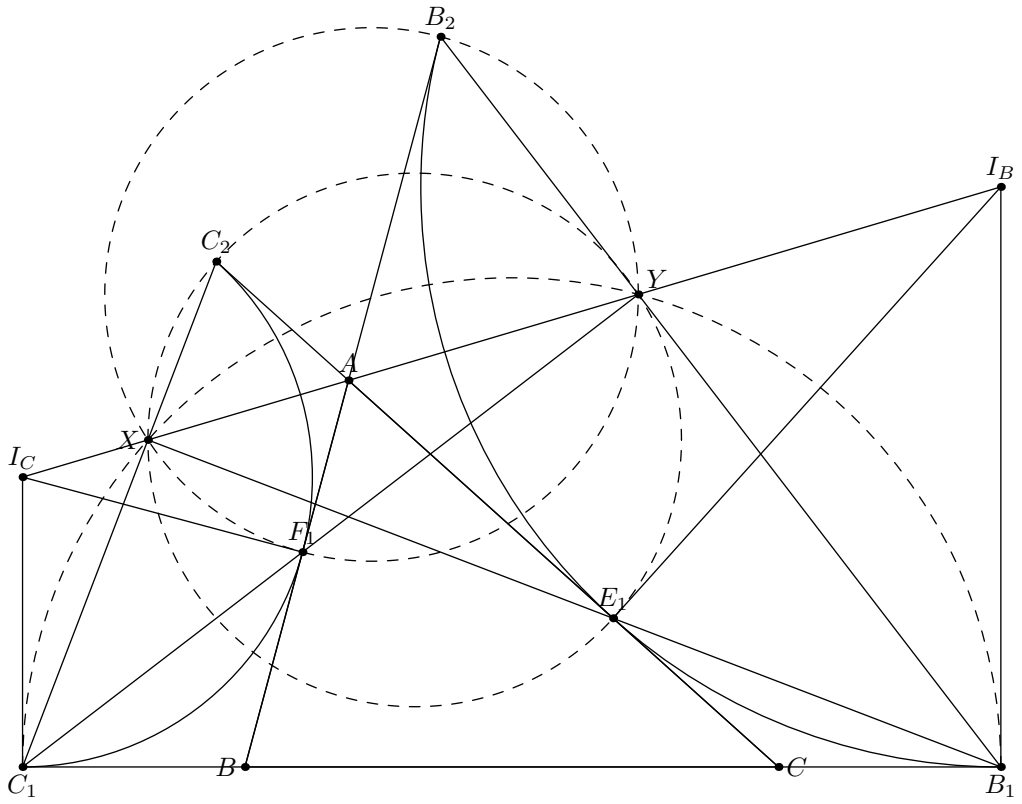


Figure 11.3: *Big Picture III*

$$Pow_{\omega_1}(I_B) = I_B B_2^2 = I_B B_1^2 = Pow_{\Omega}(I_B)$$

so $I_C - Y' - I_B$ as it is the power axis of Ω and ω_1.

Similarly, considering $\omega_2 = (C_2 X' E_1)$ we get $I_C - X' - I_B$.

Combining those facts with *Lemma 98* we obtain $I_C - X' - A - Y' - I_B$ so $X = X'$ and $Y = Y'$ which ends the proof.

□

Problem 68. Given is triangle ABC. The angle bisectors of angles $\angle ABC$ and $\angle BCA$ intersect the sides CA and AB at the points B_1 and C_1, and intersect each other at the point I. The line B_1C_1 intersects the circumcircle of triangle ABC at the points M and N . Prove that the circumradius of triangle MIN is twice as long as the circumradius of triangle ABC.

Solution. Let I_B, I_C be the excenters opposite B, C respecively. Then, *Lemma 90*, $(AIBI_C)$. By PoP chasing:

$$I_C C_1 \cdot C_1 I = AC_1 = C_1 B = MC_1 \cdot C_1 N$$

by (ABC).

Then, $(I_C MIN)$. Similarly, $I_B \in (MIN)$ so $(I_C MINI_B)$.

From *Lemma 98* and *Lemma 12*, we get that $(I_C MINI_B)$ is the reflection of $(I_A I_B I_C)$ over $I_B I_C$ and as $(I_A I_B I_C)$ is two times larger that its nine-point circle, we get the thesis.

□

Hence we can formulate the following lemma:

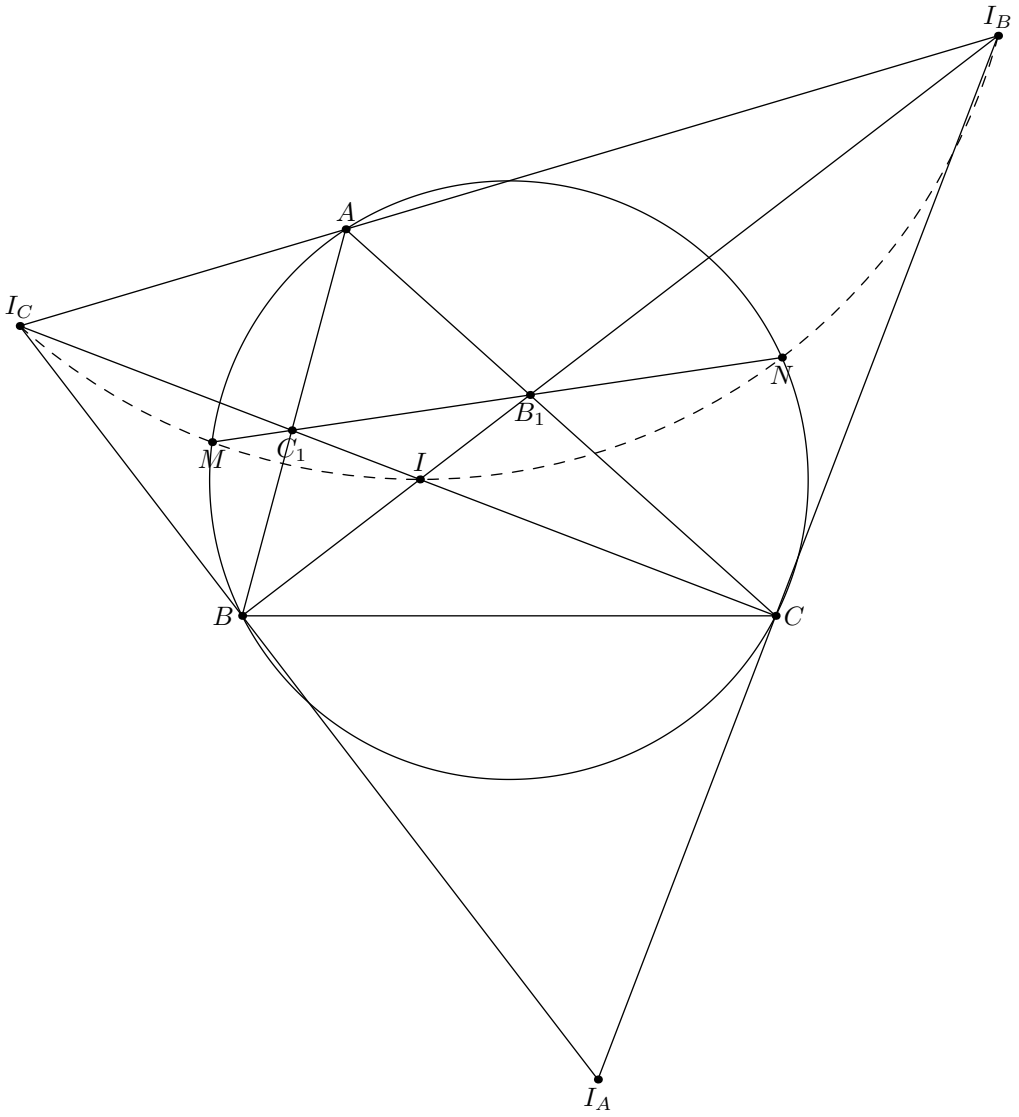


Figure 11.4: *Problem 68.*

Lemma 100

Let angle bisectors from B and C intersect sides AC, AB at points X, Y respectively and let line XY intersect (ABC) at points P, Q. Then:

- (I_CPIQI_B)
- connection to *Lemma 12* – wrt $\triangle I_AI_BI_C$

  Here (ABC) is the nine-point circle and (I_CPIQI_B) is the circle (BHC) from that lemma.

Problem 69. Let $\triangle ABC$ be a triangle with orthocenter H, circumcenter O. Points D, E, F are the feet of altitude from A, B, C respectively. Lines AH, EF interesect at X and lines BH, DF intersect at Y. Let N be the midpoint of OH. Prove that $XY \perp CN$.

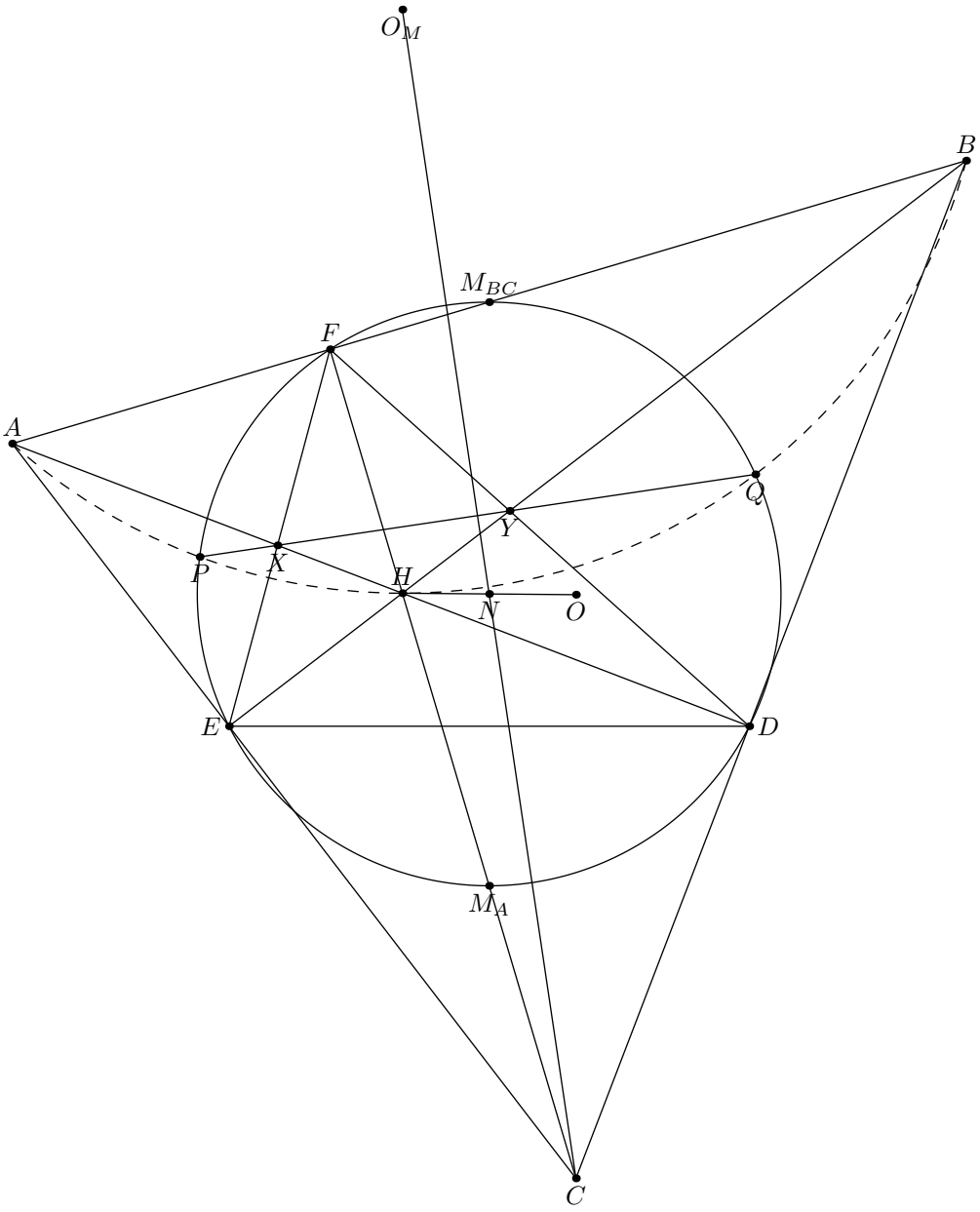


Figure 11.5: *Problem 69.*

Solution. Let's redraw this figure upside down. Clearly, (DEF) is the nine point circle and N is its center.

Let line XY intersect (DEF) at P, Q. Then from *Lemma 100*, $(APHQB)$. Denote its center by O_M. Then, we get $O_M - N - C$ from *Lemma 12*. Moreover, $O_M N \perp PQ$ as $PQ = (DEF) \cap (APHQB)$.

Hence, $XY \perp CN$ and we are done.

□

Problem 70. USAMO 2017/3

Let ABC be a scalene triangle with circumcircle Ω and incenter I. Ray AI meets BC at D and Ω again at M; the circle with diameter DM cuts Ω again at K. Lines MK and BC meet at S, and N is the midpoint of IS. The circumcircles of $\triangle KID$ and $\triangle MAN$ intersect at points L_1 and L. Prove that Ω passes through the midpoint of either IL_1 or IL.

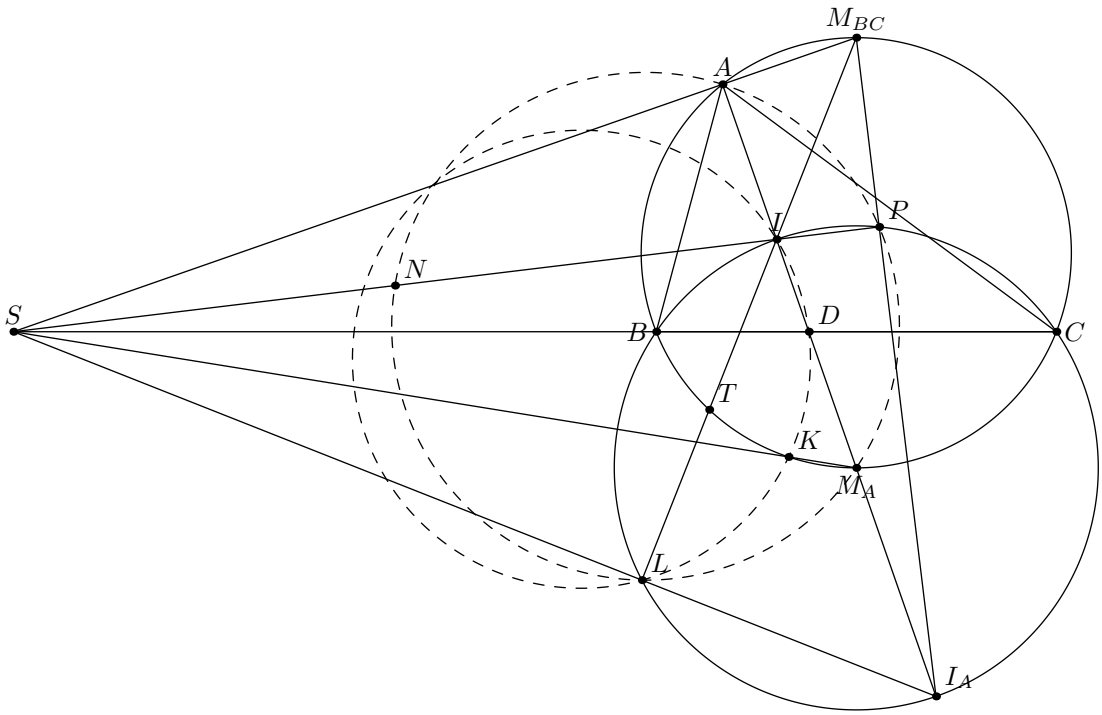


Figure 11.6: *Problem 70*

Solution. Slight change of notation (because I don't like it), let $M_A := M$ and M be the midpoint of BC. Also, let M_BC be the antipode of M_A wrt (ABC), I_A be the A-excenter and $P = M_{BC}I_A \cap (BIC)$.

Now consider the following radical axis (notice that $M_BC - D - K$ we have all those circles for obvious right-angled reasons):

$$(DMM_AK) \cap (ABC) = KM_A$$
$$(DMM_AK) \cap (AM_{BC}MD) = DM = BC$$
$$(AM_{BC}MD) \cap (ABC) = AM_{BC}$$
$$(AM_{BC}PI) \cap (ABC) = AM_{BC}$$
$$(AM_{BC}PI) \cap (BIC) = IP$$
$$(BIC) \cap (ABC) = BC$$

so using the first 3 we have by the 3 radical axis theorem $S - A - M_{BC}$ and then using the latter 3: $S - I - P$.

Now point I is the orthocenter of $\triangle SM_{BC}I_A$, let L be the foot of M_{BC} onto SI_A then $M_{BC} - I - L$ and $\angle ILI_A = 90^o$ so $L \in (BICI_A)$. We'll prove that it is the desired point that lies on both (KID) and (M_AAN).

Points $(NAPM_AL)$ form the nine-point circle of $\triangle SM_{BC}I_A$ and $(KDIL)$ can be proven by PoP:

$$M_{BC}I \cdot M_{BC}L = M_{BC}C^2 = M_{BC}M \cdot M_{BC}M_A = M_{BC}D \cdot M_{BC}K$$

Finally, let T be the midpoint of IL, it lies on (ABC) by *Lemma 109* and we are done. □

Now you might be wondering why this is in the excircle section. It's very reasonable as we haven't used absolutely any excircle knowledge. Well it's because some of the points that appear in the problem are quite intersting.

Point T is the A-mixtlinear point as you'll see in the next section.

Point K is such that AK is the symmedian by *Lemma 62*.

Point L, the most intersting, is the Miquel point of (BCI_BI_C) so $L \in (I_AI_BI_C)$.

Should we make a separate Lemma out of it ? Sure

Lemma 101

L is the miquel point of (BCI_BI_C)

Moreover, $(AIDLS)$ where D is the foot of I onto BC.

L is also an important character in *Lemma 109*.

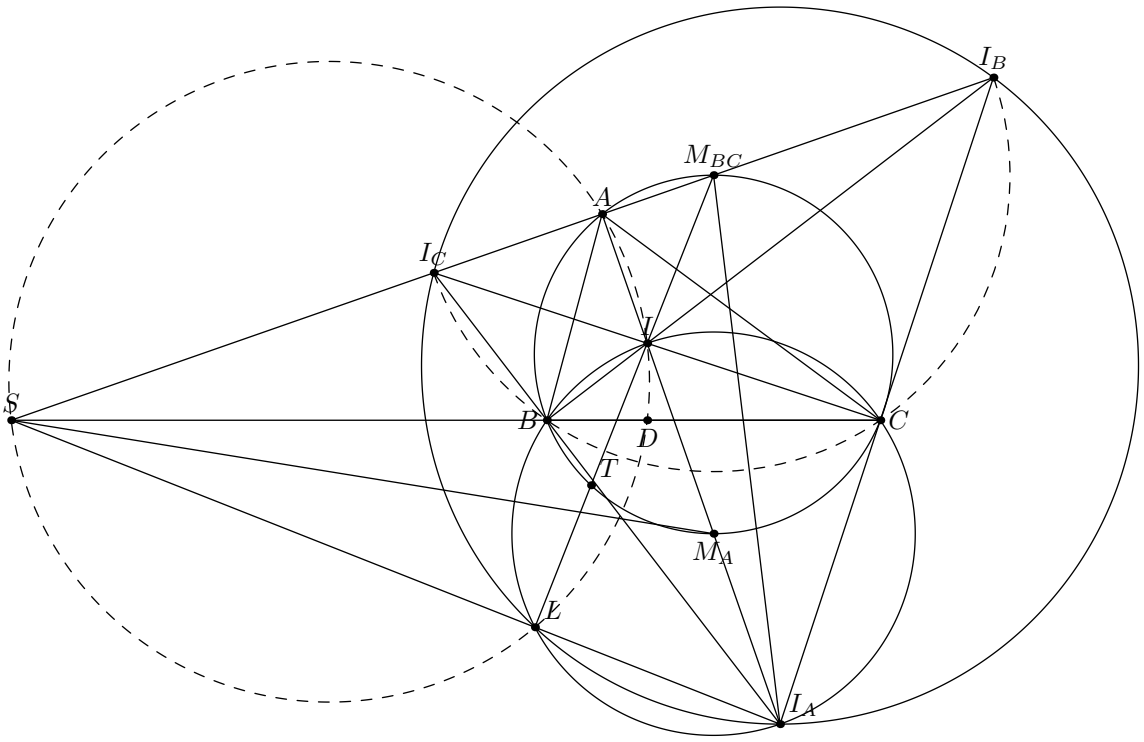


Figure 11.7: *Lemma 101.*

Problem 71. MBL 2024 Qualifying Quiz

A circle ω is tangent to the sides BC, CA, AB of triangle ABC in points D, E, F, respectively. The segment DP is the diameter of ω. Line AP intersects ω in points P and Q. The point M is

the centre of side BC. Lines MQ and EF intersect in point R. Prove that the circumcircle of triangle PQR passes through the centre of ω.

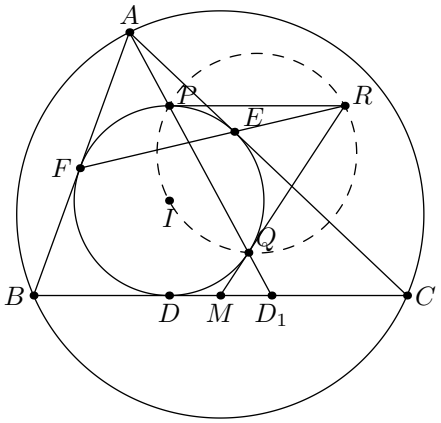


Figure 11.8: *Problem 71*

Solution. From *Lemma 95*, line AP intersects BC at D_1 - the excircle touchpoint. By angle chasing:

$$\angle DIM = \angle DPD_1 = \frac{1}{2}\angle DIQ = \angle MIQ$$

so line MQ is tangent to the incircle and, hence, $\angle IQR = 90^o$.

Line EF is the polar of A wrt the incircle, so by *Lemma 125*, we have that $A \in$ polar of R. Therefore, AQ is the polar of R and RP in tangent to the incircle. From $\angle IPR = 90^o = \angle IQR$ we have $(IQRP)$ and so we're done. □

11.1 Problems

Problem 72. Let ABC be a triangle. The incircle of ABC touches the sides AB and AC at the points Z and Y, respectively. Let G be the point where the lines BY and CZ meet, and let R and S be points such that the two quadrilaterals $BCYR$ and $BCSZ$ are parallelograms. Prove that $GR = GS$.

Problem 73. USAMO 2001 P2

Let ABC be a triangle and let ω be its incircle. Denote by D_1 and E_1 the points where ω is tangent to sides BC and AC, respectively. Denote by D_2 and E_2 the points on sides BC and AC, respectively, such that $CD_2 = BD_1$ and $CE_2 = AE_1$, and denote by P the point of intersection of segments AD_2 and BE_2. Circle ω intersects segment AD_2 at two points, the closer of which to the vertex A is denoted by Q. Prove that $AQ = D_2P$.

11.2 Reading

[Ram14] — Prasanna Ramakrishnan, *All About Excircles!*

Chapter 12

Mixtlinear

The *A-mixtilinear incircle* is tangent to AB, AC, and internally tangent to (ABC). The *A-mixtilinear touchpoint* T_A is the tangency point of the mixtilinear incircle and (ABC).

Let points B_1, C_1 be the touchpoints of the A mixtilinear incircle with sides AB, AC respectively. Let M_B, M_C be midpoints of arcs AC, AB not containing B, C, respectively.

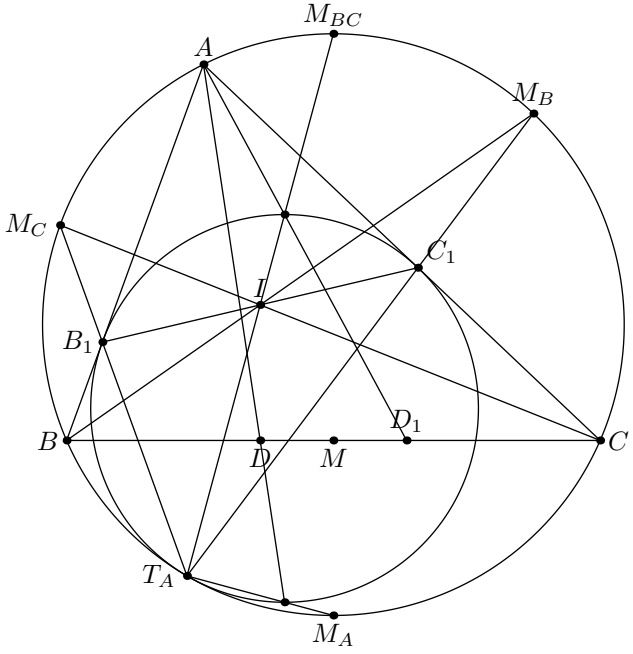


Figure 12.1: *Mixtlinear incircle*

Lemma 102

Lines AT_A and AD_1 are isogonal in $\triangle ABC$.

Lemma 103: Mixtlinear properties

- $M_{BC} - I - T_A$
- $B_1 - I - C_1$
- I is the midpoint of B_1C_1

- $AI \perp B_1C_1$
- $T_A - C_1 - M_B$
- $T_A - B_1 - M_C$

Problem 74. EGMO 2023 P6

Let ABC be a triangle with circumcircle Ω. Let S_b and S_c respectively denote the midpoints of the arcs AC and AB that do not contain the third vertex. Let N_a denote the midpoint of arc BAC (the arc BC including A). Let I be the incenter of ABC. Let ω_b be the circle that is tangent to AB and internally tangent to Ω at S_b, and let ω_c be the circle that is tangent to AC and internally tangent to Ω at S_c. Show that the line IN_a, and the lines through the intersections of ω_b and ω_c, meet on Ω.

Lemma 104

Lines AD and T_AM_A meet on the "bottom point" of the A-mixtilinear incircle and AD_1 and T_AM_{BC} meet at the "top point" of the A-mixtilinear incircle.

Proof. A homothety sending the incircle to the A-mixtillinear circle at A means that the top and bottom points of the A-mixtilinear incircle are on AD_1 (from *Lemma 95*) and AD respectively. Similarly the homothety sending the A-mixtillinear incircle to (ABC) at T_A means that the top and bottom points of the A-mixtilinear incircle are on T_AM_{BC} and T_AM_A respectively. □

12.1 Configurations

Lemma 105: Mixtlinear configuration I

Let $K = BI \cap EF$, $L = CI \cap EF$ and M_1 be the midpoint of KL.
Then

- $M_{BC} - M_1 - I - T_A$
- this line is an I-symmedian in $\triangle BIC$

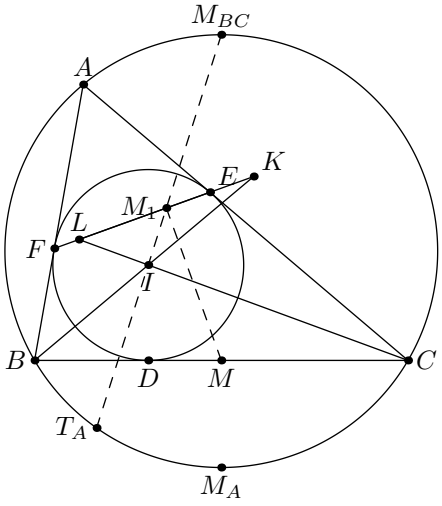


Figure 12.2: *Mixtlinear configuration I*

Proof. It follows from *Lemma 91* that quadrilateral $BCKL$ is cyclic and M is its circumcenter. Then M_1 is the foot of M onto KL.

We know that $\angle M_{BC}CM_A = \angle M_{BC}BM_A = 90^o$ and M_A is the circumcenter of $\triangle BIC$, so by *Lemma 59*, T_AM_{BC} is the I-symmedian of $\triangle BIC$.

Then $\angle BIT_A = \angle MIC$. And, as $\triangle BIC \simeq \triangle LIK$, $\angle MIC = \angle M_1IK$, so $T_A - I - M_1$. □

Problem 75. Nicaragua IMO TST 2019, P7

Let ABC be a triangle with $AC \neq AB$, ω its inscribed circle and I the center of ω. Let E and F be the points of tangency of ω with the sides CA and AB, respectively. Lines BI and CI meet EF at K and L, respectively. Points P and Q are located on the side BC so that $\angle LQC = 90^o = \angle KPB$. Show that the circumcircles of the triangles PBF and QCE meet on the circumcircle of triangle ABC.

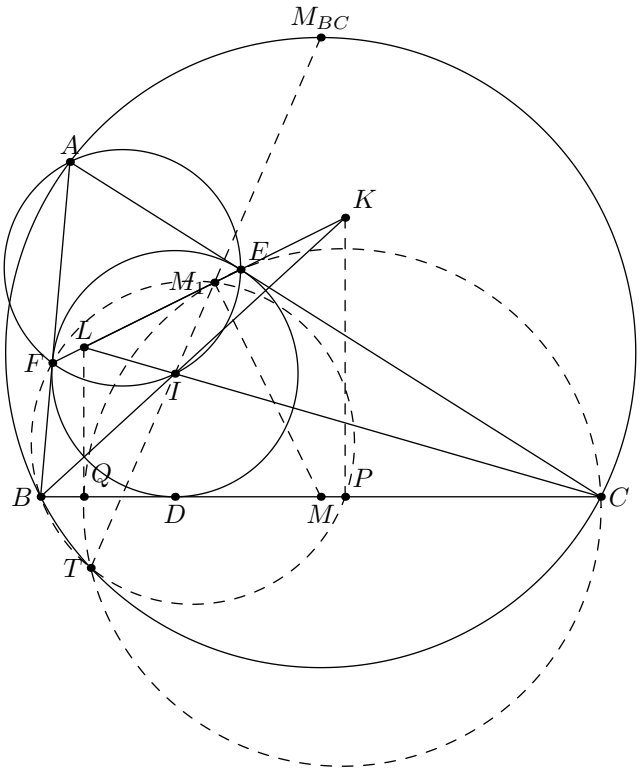


Figure 12.3: *Problem 75*

Solution. Let M, M_1 be the midpoints of segments BC, KL respectively and recall *Lemma 105*.

Notice that $\angle MM_1K = 90^o = \angle KPM$ so $(KPMM_1)$ is cyclic. Denote $\angle ABC = 2 \cdot \beta$, then $\angle KMP = \angle KMC = 2 \cdot \angle KBC = 2 \cdot \beta$ and $\angle MM_1P = \angle MKP = 90^o - \angle KMP = 90^o - 2 \cdot \beta$. Further, $\angle FBP = 2 \cdot \beta$ and $\angle FM_1P = 90^o + \angle MM_1P = 180^o - 2 \cdot \beta$, so $M_1 \in (PBF)$.

Similiarly, $M_1 \in (QCE)$. Denote $\angle BAC = 2 \cdot \alpha$ and $T = M_1M_{BC} \cap (ABC)$ where M_{BC} is the midpoint of arc BC containing A.

From angle chasing:

$$\angle AFM_1 = 90^o - \alpha = \angle BCM_{BC} = \angle BTM_{BC} = \angle BTM_1,$$

so $T \in (PBFM_1)$. Similiarly, $T \in (QCEM_1)$ and T is the desired point of intersection.

□

Let $N = IT \cap BC$, $S = BC \cap B_1C_1$, $L = M_AN \cap (ABC)$ and R be the midpoint of SN.

Lemma 106: Mixtlinear configuration II

- TL and TM are isogonal conjugates wrt $\triangle BTC$
- AT and TM_{BC} are isogonal conjugates wrt $\triangle M_C T M_B$
- $CM_C \parallel M_B M_{BC}$, $BM_B \parallel M_C M_{BC}$
- $(BTIB_1)$; CI and BM_C are tangent to it; it's harmonic
- $(CTIC_1)$; BI and CM_B are tangent to it; it's harmonic
- (AM_BTM_C) and it is harmonic
- $S - T - M_A$
- $S - L - M_{BC}$
- RT is the radiacal axis of (ABC) and A-mixtlinear
- (AB_1C_1L)
- $(SLNT)$; it's the T- Apollonius circle of $\triangle BTC$
- N is the orthocenter of $\triangle AM_AM_{BC}$

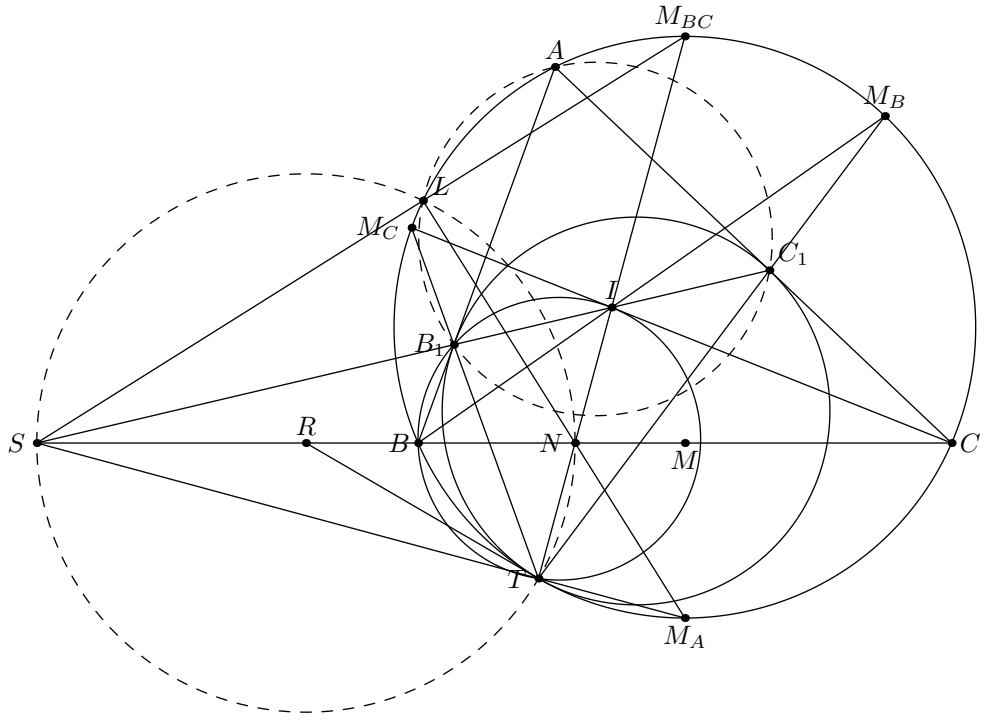


Figure 12.4: *Mixtlinear configuration II*

12.2 Reading

[Jaf18] — Jafet98, *On mixtilinear incircles*

[Che15]— Evan Chen, *A Guessing Game: Mixtilinear Incircles*

Chapter 13

Sharky-Devil point

The A-Sharkydevil Point, K, is the intersection of $(AEIF)$ and (ABC), $K \neq A$. Furthermore, let P be the foot of D onto EF.

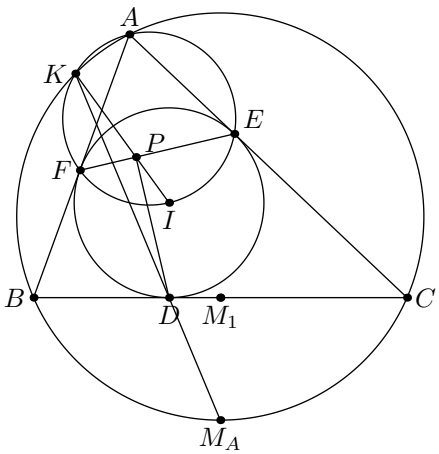


Figure 13.1: *Lemma 107 & Lemma 108*

Lemma 107

Points M_A, D and K are collinear.

Proof. Consider an inversion with center M_A and ray M_AB - $I(M_A, M_AB)$. Notice that point $A' = AM_A \cap BC$ is the inversion image of point A because $I((ABC)) = BC$ and from *Lemma 90* point I is constant under this inversion.

Then, the circle $(AEIF)$ is transformed into a circle with diameter IA' and it follows instantly that point D also lies on that circle. Then $I((AEIF)) = (A'ID)$.

Combining that with the definitions of points $K = (ABC) \cap (AEIF)$ and $D = BC \cap (A'ID)$ we conclude that $D = I(K)$. Therefore, point K, its inversion image D and the center of inversion M_A must be collinear. □

Lemma 108

Points I, P and K are collinear.

Proof. Consider a spiral similarity at K, taking $\triangle KFE$ to $\triangle KBC$. Then it takes I to M_A.

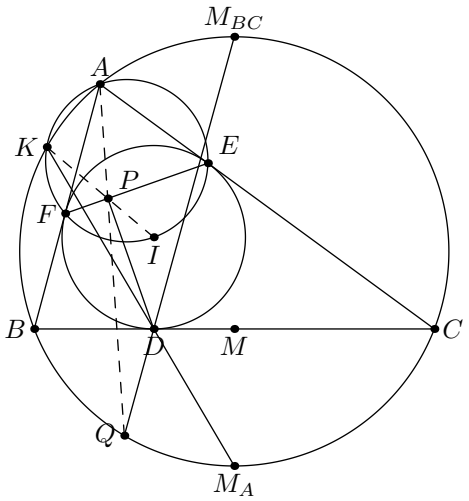


Figure 13.2: *Problem 76*

So $\triangle KFB \simeq \triangle KIM_A$. Now notice $DP \parallel AM_A$ so $\triangle KFB \simeq \triangle KPD$.
Then that spiral similarity must take P to D, and as $K - D - M_A$ so must $K - P - I$. $\square$

Problem 76. Polish MO Camp 2022 First Mathematical Match P9

Incircle of triangle ABC with incenter I touches sides BC, AC, AB at D, E, F respectively. Let P be the foot of D onto EF. Rays AP and IP intersect with (ABC) at Q and K. Denote the midpoint of BC by M. Prove that point D is the incenter of $\triangle KQM$.

13.1 Nice configurations

Let T be the A-mixtlinear incircle touchpoint, $S = AK \cap BC$, $L = IT \cap (BIC)$ (note that we have $M_{BC} - I - T$ from *Lemma 103*), $R = KM_{BC} \cap DI$ and $Z_0 = AM_{BC} \cap BC$.

Lemma 109: Incircle configuration I

The following facts hold:

- Line SI is tangent to $(AEIF)$ and (BIC)
- Point T is the midpoint of IL
- $S - T - M_A$
- line IL is the polar of S wrt (BIC)
- $(KIDTS)$
- line IM_A is tangent to $(KIDTS)$
- K is the miquel point of $BFEC$
- $K - I - A'$
- (Z_0AIHL)

Lemma 110

Circle (KDR) is tangent to line BC and (ABC).

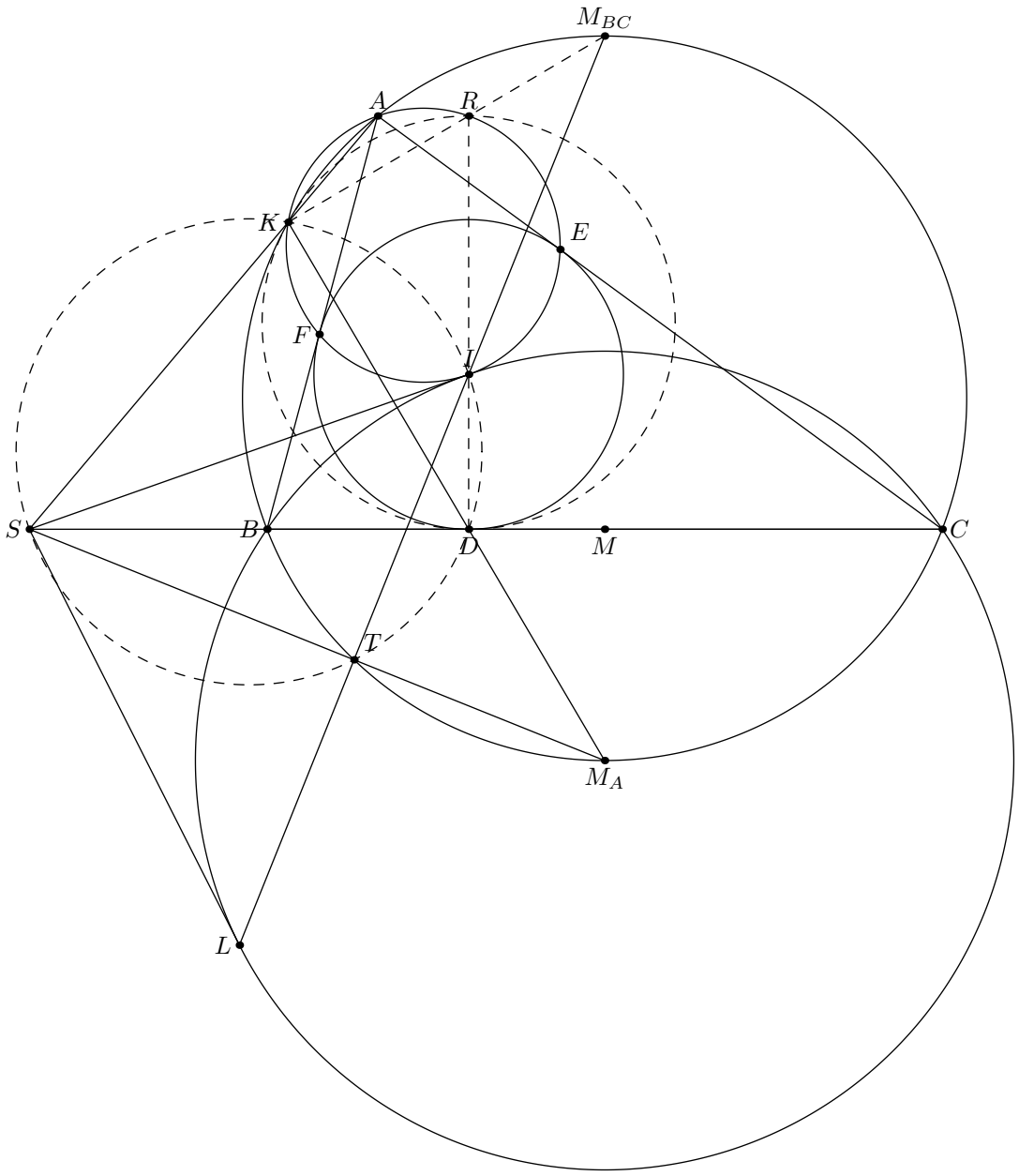


Figure 13.3: *Incircle configuration I*

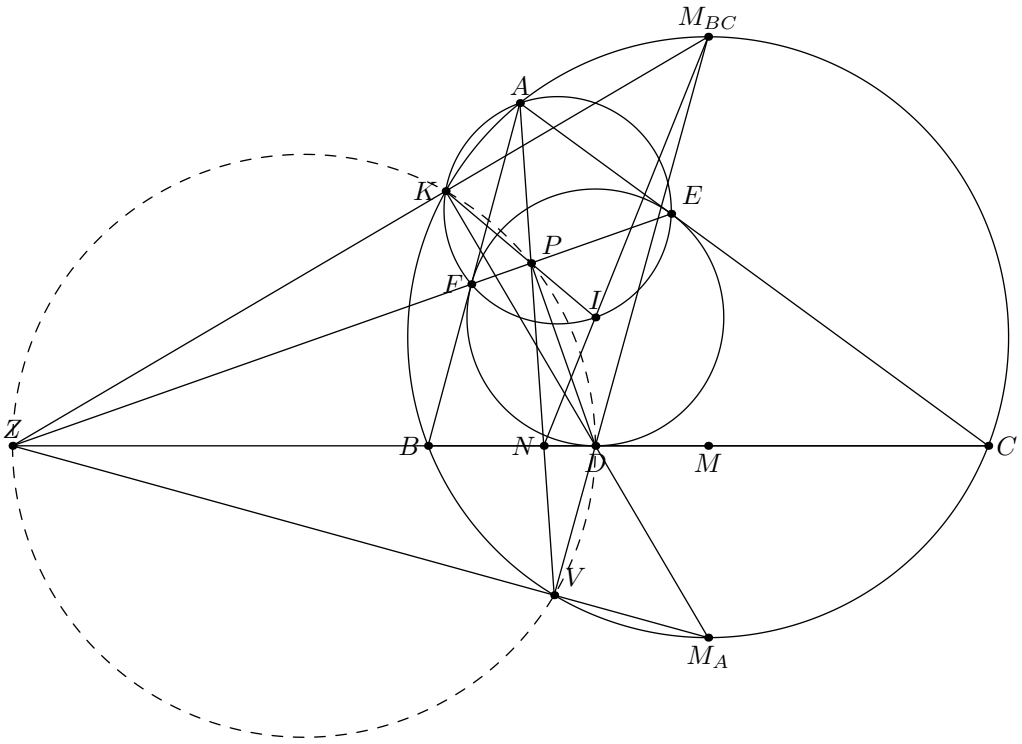


Figure 13.4: *Incircle configuration II*

Lemma 111: Incircle configuration II

Let $N = BC \cap IM_{BC}$ (recall we've already met N in *Lemma 106*).
The following facts hold:

- $(DPKZV)$ is orthogonal to both the incrcle and the circumcircle
- $A - P - N - V$

Problem 77. ELMO SL 2019 G3
Let $\triangle ABC$ be an acute triangle with incenter I and circumcenter O. The incircle touches sides BC, CA, and AB at D, E, and F respectively, and A' is the reflection of A over O. The circumcircles of ABC and $A'EF$ meet at G, and the circumcircles of AMG and $A'EF$ meet at a point $H \neq G$, where M is the midpoint of EF. Prove that if GH and EF meet at T, then $DT \perp EF$.

Proof. A' is obviously the antipode.
Let $S = A'G \cap EF$ and $K = AS \cap (ABC)$. It's easy to see that S has equal powers wrt (ABC) and the incircle - so K must be the *Sharkydevil point*.
Furthermore, $\angle AMS = 90^o = \angle AGS$ so $S \in (AMG)$.
Let $o_1 = (A'EFG)$, $o_2 = (AMGS)$ and i be the incircle.

$$i \cap o_1 = EF$$

$$o_1 \cap o_2 = GH$$

let

$$i \cap o_2 = XY$$

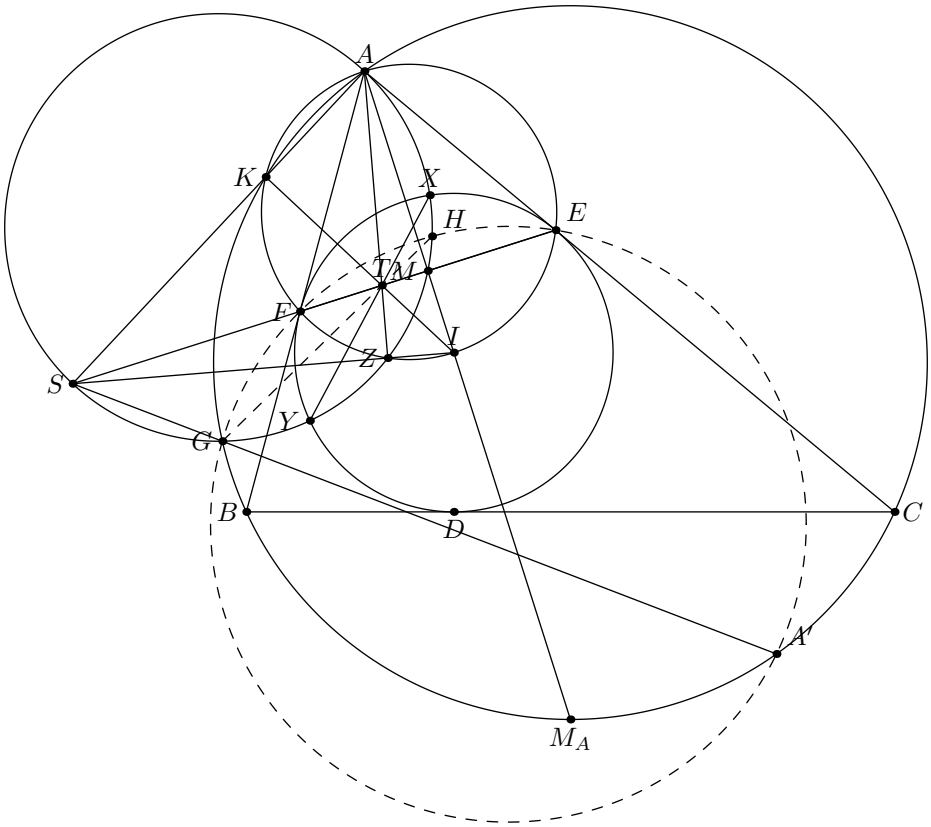


Figure 13.5: *Problem 77.*

then from *Lemma 137* - as $T = EF \cap GH$ - $T \in XY$

Let Z be the second intersection of $(AEIF)$ and o_2 - considering *Lemma 137* for $(AEIF)$, i and o_2 we get $T \in AZ$

As AI is the diameter of $(AEIF)$ and AS is the diameter of o_2 - we have $S - Z - I$. Then T is the orthocenter of $\triangle AIS$ - and $I - T - K$ uncovering T to be the foot of D onto EF from *Lemma 108* and we are done.

□

Lemma 112: Incircle configuration III

Let $T' = AT \cap (DEF)$ with T' closer to A. Furthermore, let $Y = AK \cap EF$, $M_1 = mid(EF)$, $U = A'Y \cap (ABC)$. Then:

- $T' - P - D$
- T' is the reflection of the orthocenter of DEF over EF
- $T' \in (KIDTS)$
- (YKM_1I)
- (AM_1UY)
- $(A'EFU)$
- $P \in (A'EFU) \cap (AM_1UY)$

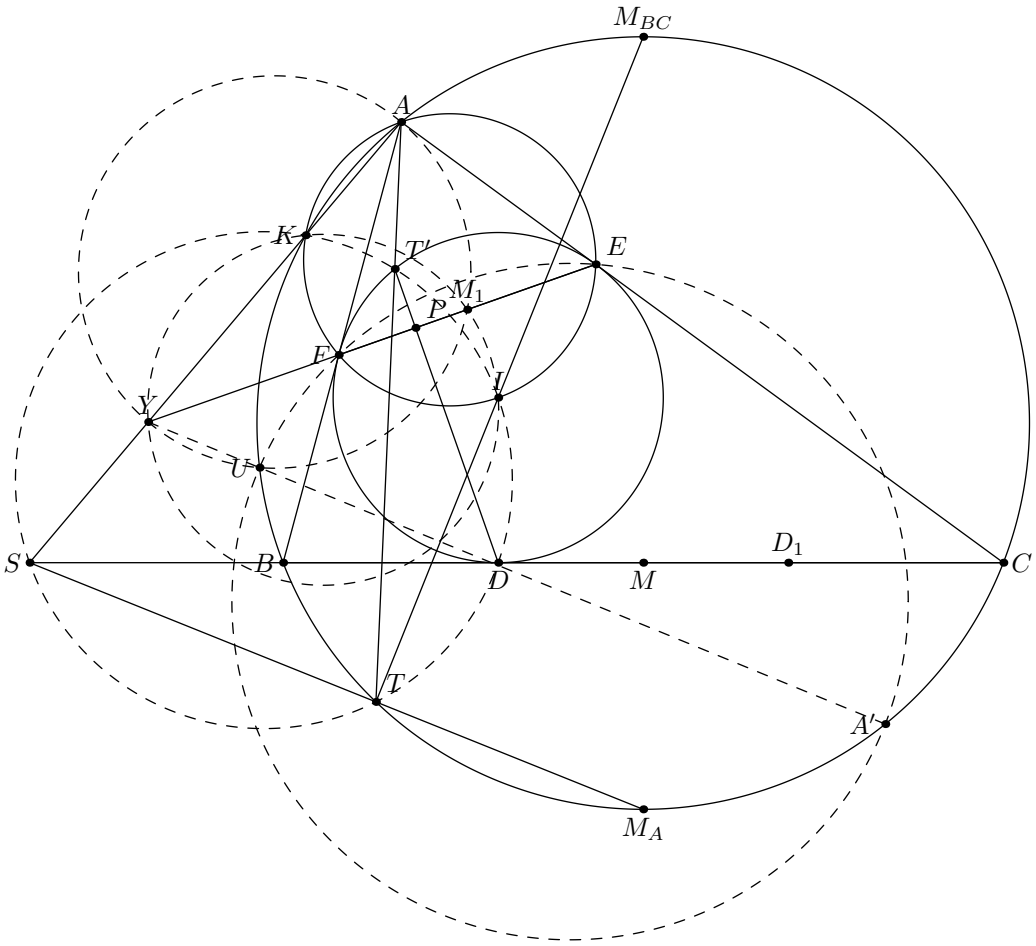


Figure 13.6: *Incircle configuration III*

Problem 78. 2024 Turkey TST P1

In triangle ABC, the incenter is I and the circumcenter is O. Let AI intersects (ABC) second time at P . The line passes through I and perpendicular to AI intersects BC at X. The feet of the perpendicular from X to IO is Y. Prove that A, P, X, Y cyclic.

Solution. Rename P as M_A. We identify point X as the same as S from *Lemma 109* and define points K, T, D, M_{BC} the same as in the lemma. Let $M = AM_A \cap DY$.

Now from *Lemma 119* we have:

$$(M_A, M_{BC}; O, D) \underset{I}{=} (I, T; Y, D)$$

so then from *Lemma 137* tagents to $(SYKIDT)$ at I and T meet at M.

From angle chasing:

$$\angle TYM = \angle TKD = \angle TAM$$

$$\angle MYO = \angle DKI = \angle MAO$$

Then points $(MTYAO)$ lie on a circle and:

$$\angle M_AAY = \angle MAY = 180^o - \angle MTY$$

$$\angle MTY = \angle MTD + \angle DTY = \angle TSD + \angle DST = \angle M_ASY$$

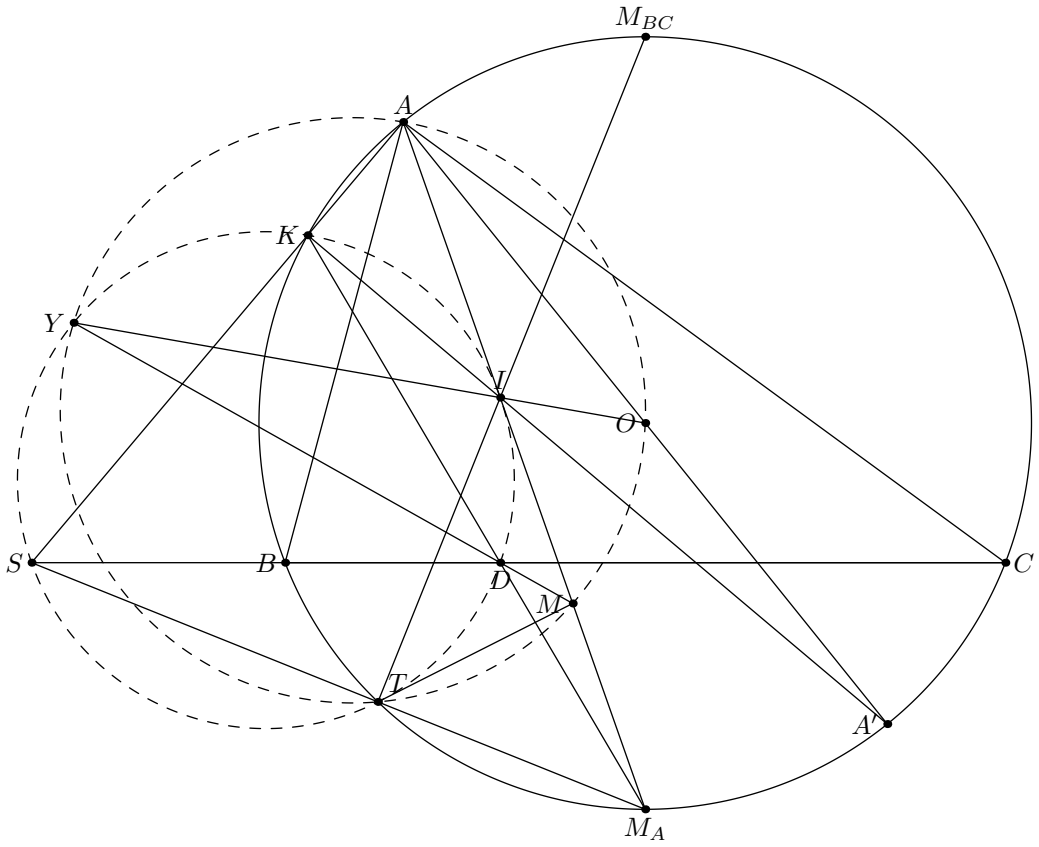


Figure 13.7: *Problem 78.*

which ends the proof.

□

13.2 Further configurations

Motivated by the following problems we can formulate a new interesting configuration.

Problem 79. IMO SL 2002 G7

The incircle Ω of the acute-angled triangle ABC is tangent to BC at K. Let AD be an altitude of triangle ABC and let M be the midpoint of AD. If N is the other common point of Ω and KM, prove that Ω and the circumcircle of triangle BCN are tangent at N.

Problem 80. IMSC 2023

Let k be the inscribed circle of non-isosceles triangle ABC, which center is I. Circle k touches sides BC, CA, AB in points P, Q, R respectively. Line QR intersects BC in points M. Let a circle which contains points B and C touch k in point N. Circumscribed circle of MNP intersects line AP in point L, different from P. Prove that points I, L and M are collinear.

Solution. Let the line tangent to k and (BCN) at N intersect BC at R_1, and let I_A be the A-extender. Then:

$$R_1N^2 = R_1P^2 = R_1B \cdot R_1C$$

so $R_1 \in k \cap (BIC)$. Let k and (BIC) intersect at X, Y.

Notice that k and (BIC) are orthogonal, $\angle IXI_A = IYI_A = 90^o$ so XY is the polar of I_A wrt k.

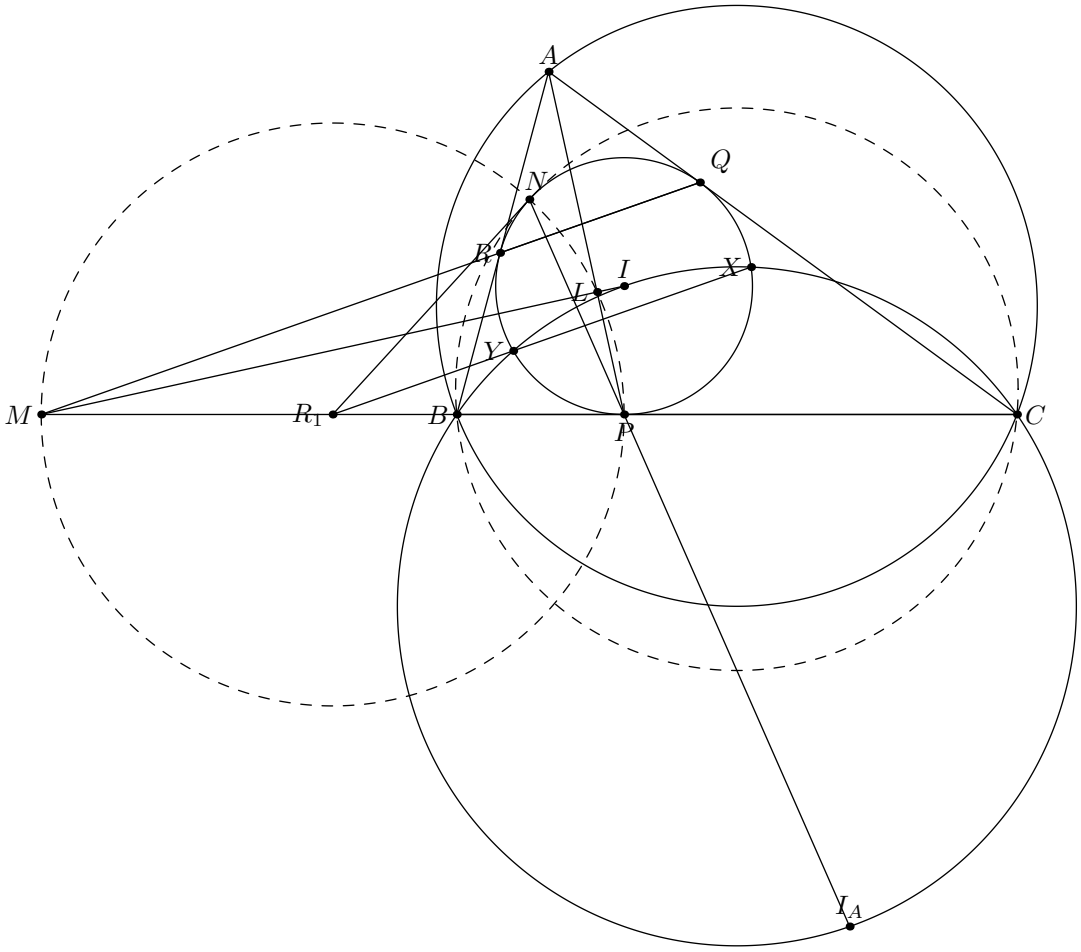


Figure 13.8: *Problem 80*

Then from *Lemma 125*, $I_A \in$ polar of $R_1 = NP$ and from *Lemma* **??**, R_1 is the midpoint of MP.

Which means that $R_1M = R_1P = R_1N$ and R_1 is the midpoint of (MNP). Then $\angle MLP = 90^2$.

Furthermore, $M \in$ polar of $A = QR$ wrt k so AP is the polar of M which must mean that $MI \perp AP$. But the means that $L = AP \cap MI$ and we are done.

□

We can summarize the configuarations of those problems in the following way.

Let N be the intersection of line I_AD with the incircle and let M' be the midpoint of DI_A, M_1 the midpoint of the altitude from A and R the midpoint of DZ.

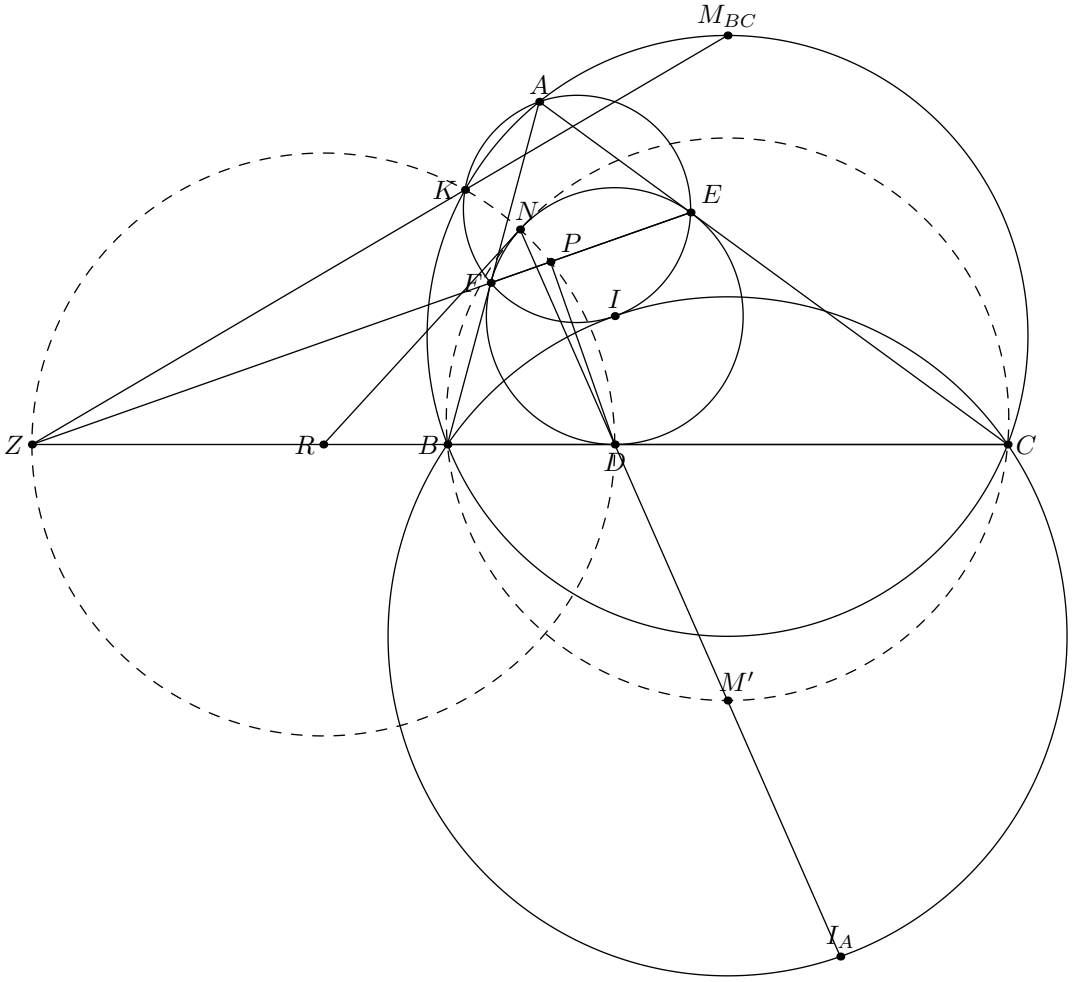


Figure 13.9: *Incircle configuration IV*

Lemma 113: Incircle configuration IV

Then:

- $M_1 - N - D - M' - I_A$
- (BNC) is tangent to the incircle
- $M' \in (BNC)$

- $N \in (DPKZ)$
- NI_A is the polar of R wrt the incircle
- $R \in (BIC) \cap$ incircle
- R lies on the orthic axis of ABC

Now recall, *Problem 25*, it turns out that there is a stunning connection between it and *Problem 79*.

You can read about it in the second dot of Reading – *Section 13.4*.

Problem 81. Let the incircle of $\triangle ABC$ be tangent to sides BC, AC, AB at D, E, F respectively and have the incenter at I. Let I_B, I_C be the excenters opposite vertices B, C, respectively. Let P, Q be midpoints of segments I_BE, I_CF, respectively. The circumcircle of APC intersects AB at R, $R \neq A$ and the circumcircle of ABQ intersects AC at $S \neq A$. Show that lines PR, QS, AI coincide.

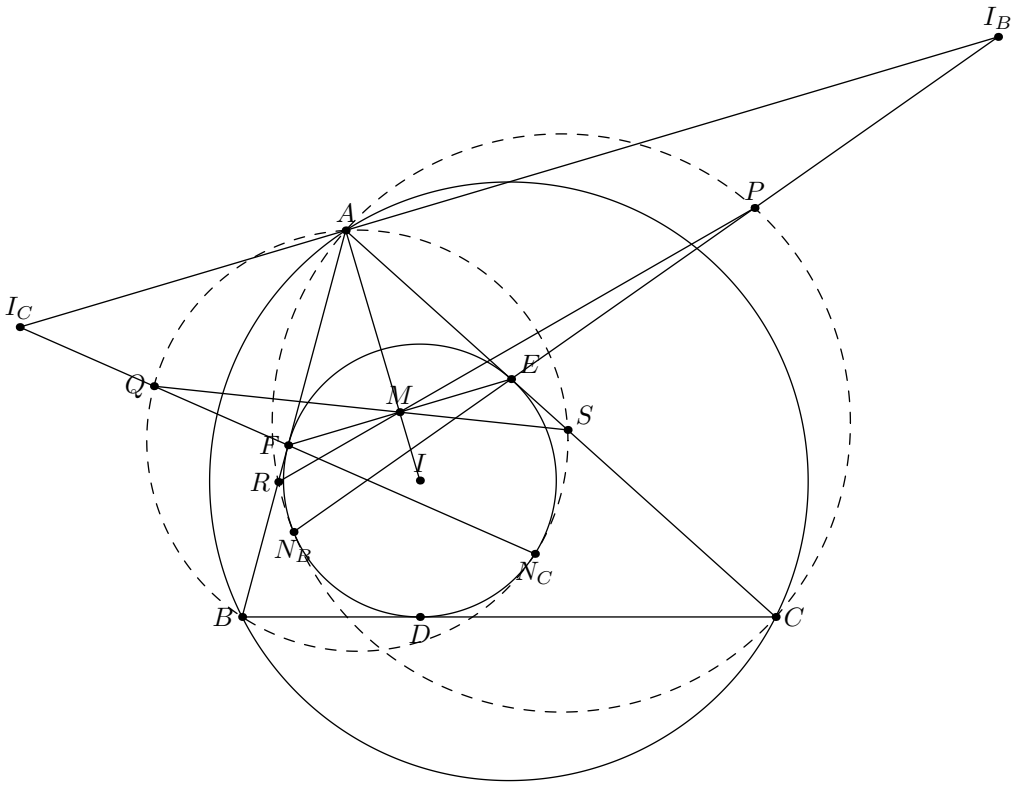


Figure 13.10: *Problem 81.*

Solution. We immediately recognise the configuration from *Lemma 113*. Therefore, $(ACPR)$ is tangent to the incircle, denote that point of tangency by N_B, and $N_B - E - P - I_B$. Analogously, $(ABQS)$ is tangent to the incircle, denote that point by N_C, and $N_C - F - Q - I_C$.

Denote the incircle by ω. Then notice that ω is the *A-mixtlinear incircle* of $\triangle ACR$. Lines AI and RP are angle bisectors in this triangle, so their intersection, say M, is the incenter of $\triangle ACR$. Recalling *Lemma 103*, we know that M is the midpoint of the segment connecting the points of tangency of the mixtlinear to the sides of the triangle, in this case EF.

Analogously, we show that AI and QS intersect at the midpoint of EF, which finishes the proof. □

13.3 Problems

Problem 82. North Korea TST 2013/1

The incircle of a non-isosceles triangle ABC with the center I touches the sides BC, CA, AB at A_1, B_1, C_1 respectively. The line AI meets the circumcircle of ABC at A_2. The line B_1C_1 meets the line BC at A_3 and the line A_2A_3 meets the circumcircle of ABC at A_4 ($\neq A_2$). Define B_4, C_4 similarly. Prove that the lines AA_4, BB_4, CC_4 are concurrent.

Problem 83. ELMO SL 2010 G4

Let ABC be a triangle with circumcircle ω, incenter I, and A-excenter I_A. Let the incircle and the A-excircle hit BC at D and E, respectively, and let M be the midpoint of arc BC without A. Consider the circle tangent to BC at D and arc BAC at T. If TI intersects ω again at S, prove that SI_A and ME meet on ω.

Problem 84. GQMO 2020 P1

Let ABC be a triangle with incentre I. The incircle of the triangle ABC touches the sides AC and AB at points E and F respectively. Let ℓ_B and ℓ_C be the tangents to the circumcircle of BIC at B and C respectively. Show that there is a circle tangent to EF, ℓ_B and ℓ_C with centre on the line BC.

13.4 Reading

Incenter-Related Configurations -Part (I) And all parts I - VI (linked on the website).

The Fables of a G4 and a G7

Part IV

Projective Geometry

Chapter 14

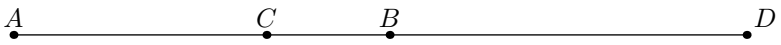
Cross-ratios

14.1 Definition

Let A, B, C, D be four distinct collinear points. The *cross-ratio* of the quadruple $(A, B; C, D)$, denoted by $(A, B; C, D)$, is defined using directed segment lengths as:

$$(A, B; C, D) = \frac{\vec{AC}}{\vec{BC}} \div \frac{\vec{AD}}{\vec{BD}} = \frac{\vec{AC} \cdot \vec{BD}}{\vec{BC} \cdot \vec{AD}}$$

where $\vec{XY}$ denotes the directed distance from X to Y along the line.



Remark. Because directed segment lengths are signed, the cross-ratio $(A, B; C, D)$ is negative if and only if one of the pairs $\{A, B\}$ and $\{C, D\}$ separates the other on the line. If C and D both lie inside or both lie outside segment AB, the cross-ratio is positive.

Lemma 114: Cross-Ratio of a Pencil of Lines

Let a, b, c, d be four lines concurrent at a vertex O. Let a transversal line ℓ intersect lines a, b, c, d at points A, B, C, D, respectively. Then the cross-ratio $(A, B; C, D)$ depends only on the directed angles between the lines and is given by:

$$(A, B; C, D) = \frac{\sin\angle(a, c)}{\sin\angle(b, c)} \div \frac{\sin\angle(a, d)}{\sin\angle(b, d)} = \frac{\sin\angle(a, c)\sin\angle(b, d)}{\sin\angle(b, c)\sin\angle(a, d)}$$

We denote this pencil cross-ratio as $O(A, B; C, D)$ or $O(a, b; c, d)$.

Proof. Applying the Law of Sines to triangles $\triangle OAC$ and $\triangle OBC$ sharing altitude $h = d(O, \ell)$ yields:

$$\frac{AC}{BC} = \frac{[\triangle OAC]}{[\triangle OBC]} = \frac{\frac{1}{2}OA \cdot OC \sin\angle(a, c)}{\frac{1}{2}OB \cdot OC \sin\angle(b, c)} = \frac{OA \cdot \sin\angle(a, c)}{OB \cdot \sin\angle(b, c)}$$

Similarly, for triangles $\triangle OAD$ and $\triangle OBD$:

$$\frac{AD}{BD} = \frac{OA \cdot \sin\angle(a, d)}{OB \cdot \sin\angle(b, d)}$$

Dividing the two ratios causes the lengths OA and OB to cancel completely:

$$(A, B; C, D) = \frac{AC/BC}{AD/BD} = \frac{\sin\angle(a, c)\sin\angle(b, d)}{\sin\angle(b, c)\sin\angle(a, d)}$$

Since the right-hand side depends solely on the angles subtended at O, the cross-ratio is identical for any transversal line ℓ. □

Theorem 115: Invariance under Perspectivity

Let A, B, C, D be four collinear points on line ℓ, and let A', B', C', D' be their projections through a point P onto another line ℓ'. Then:

$$(A, B; C, D) \overset{P}{=} (A', B'; C', D')$$

Furthermore, for any four points A, B, C, D lying on a circle (or conic) Γ and any point $P \in \Gamma$, the pencil cross-ratio $P(A, B; C, D)$ is constant regardless of the choice of $P \in \Gamma$.

Proof. The line projections preserve the pencil cross-ratio $P(a, b; c, d)$, which equals both $(A, B; C, D)$ and $(A', B'; C', D')$. For points on a circle Γ, inscribed angle equality implies that $\angle(PA, PC) = \angle(QA, QC)$ for any $P, Q \in \Gamma$, making the pencil cross-ratio $P(A, B; C, D)$ invariant. □

Lemma 116: Symmetries and Permutations of Cross-Ratio

If $(A, B; C, D) = \lambda$, then permutations of the four points yield the following values:

$$\begin{aligned}
&(A, B; C, D) = (B, A; D, C) = (C, D; A, B) = (D, C; B, A) = \lambda \\
&(B, A; C, D) = (A, B; D, C) = \frac{1}{\lambda} \\
&(A, C; B, D) = (B, D; A, C) = 1 - \lambda \\
&(A, D; C, B) = 1 - \frac{1}{\lambda} = \frac{\lambda - 1}{\lambda}
\end{aligned}$$

14.2 Harmonic

The most interesting case, and where you are most likely to encounter the cross ration, is the harmonic case.

Lemma 117: Harmonic Cross Ratio

Four collinear points A, B, C, D form a *harmonic quadruple* (or *harmonic bundle*), written as $(A, B; C, D) = -1$, if:

$$\frac{\vec{AC}}{\vec{BC}} = -\frac{\vec{AD}}{\vec{BD}}$$

We say that D is the *harmonic conjugate* of C with respect to segment AB.

This may not seem intuitive but we will identify some common harmonic quads in the lemmas below. Hopefully, that will help you develop some intuition for finding the harmonic cross ratio in classical configurations.

Lemma 118: Metric Relations in Harmonic Quadruples

Suppose $(A, B; C, D) = -1$. Then the following distance relations hold:

1. **Harmonic Length Relation:**
   $\frac{1}{2}AB \cdot CD = BC \cdot AD = BD \cdot AC$

2. **Newton's Relation:** Let M be the midpoint of segment AB. Then $(A, B; C, D) = -1$ if and only if: $AM^2 = MC \cdot MD$

Proof. Identify $AC \cdot BD = BC \cdot AC$ comes directly from the cross ratio condition.
Now let's prove

$$AB \cdot CD = BC \cdot AD + BD \cdot AC$$
$$(AC + CB) \cdot (CB + BD) = BC \cdot AD + BD \cdot AC$$
$$AC \cdot CB + AC \cdot BD + CB^2 + CB \cdot BD = BC \cdot AD + BD \cdot AC$$
$$CB \cdot (AC + CB + BD) = BC \cdot AD$$

and we are done.
Now suppose M is the midpoint of AB, we will show

$$AM^2 = MC \cdot MD$$
$$AM^2 = (AC - MA) \cdot (BD + MB)$$
$$AM \cdot (2 \cdot AM - AC + BD) = AC \cdot BD$$
$$\frac{1}{2} AB \cdot CD = AC \cdot BD$$

which we showed above.
Now for the other direction of M satisfies this condition then it must be □

Lemma 119

Let AB be a segment and M its midpoint. Then:

$$(A, B; M, X_{AB}) = -1$$

Lemma 120

Let ABC be a triangle and D, E, F points on sides BC, AC, AB respectively. Denote $X = BC \cap EF$. Then, if lines AD, BE, CF coincide, we have:

$$(B, C; D, X) = -1$$

Proof. Apply Ceva's Theorem to $\triangle ABC$ with concurrent cevians AD, BE, CF:

$$\frac{BD}{DC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = 1$$

Apply Menelaus's Theorem to $\triangle ABC$ with transversal line $X - F - E$:

$$\frac{BX}{XC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = -1$$

Dividing Ceva's relation by Menelaus's relation yields $\frac{BD/DC}{BX/XC} = -1$, which is $(B, C; D, X) = -1$. □

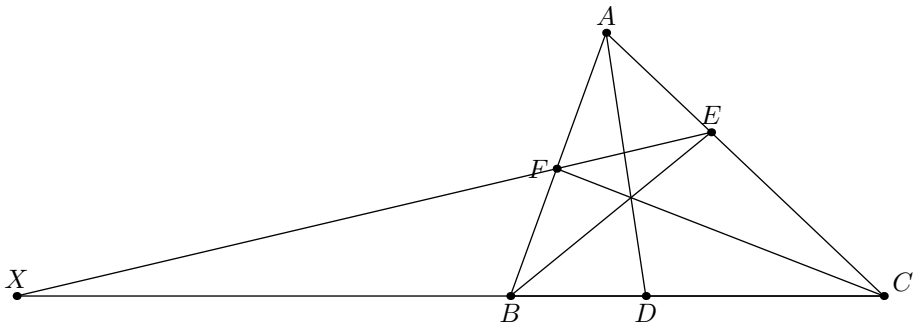


Figure 14.1: *Lemma 120*

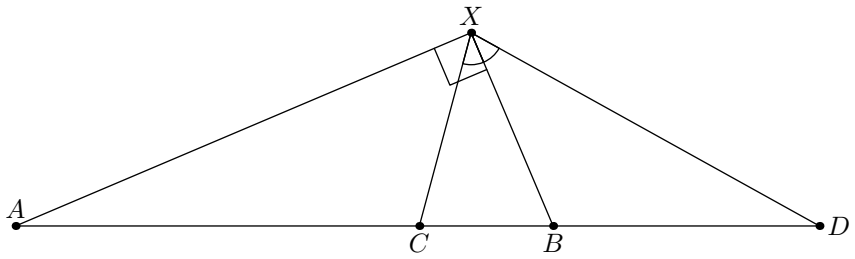


Figure 14.2: *Lemma 121*

Lemma 121

If points A, B, C and D lie on a line ℓ, and point X lies outside of ℓ, then any two of the following conditions implies the third:

- $(A, B; C, D) = -1$
- XB is the angle bisector of $\angle CXD$
- $XA \perp XB$

Problem 85. Let ABC be a triangle, and let D, E, F be the points of tangency of the incircle of triangle ABC with the sides BC, CA and AB respectively. Let X be in the interior of ABC such that the incircle of XBC touches XB, XC and BC in Z, Y and D respectively. Prove that $EFZY$ is cyclic.

14.3 Moving points

A *projective transformation* (or *homography*) between two projective lines $\ell_1 \to \ell_2$ is a map f that preserves cross-ratios:

$$(f(A), f(B); f(C), f(D)) = (A, B; C, D) \quad \forall A, B, C, D \in \ell_1$$

Lemma 122: Fundamental Theorem of Moving Points

Let f, g be two projective transformations.

1. If f has at least 3 distinct fixed points P_1, P_2, P_3 (i.e., $f(P_i) = P_i$), then f is the identity map ($f(P) = P$ for all P).

2. If $f(P_i) = g(P_i)$ for 3 distinct points P_1, P_2, P_3, then $f \equiv g$ everywhere.

14.4 Problems

Problem 86. IMO 1995 SL G3

ABC is a triangle. The incircle touches BC, CA, AB at D, E, F respectively. X is a point inside the triangle such that the incircle of XBC touches BC at D. It touches CX at Y and XB at Z. Show that EFZY is cyclic.

Problem 87. ELMO 2015 P3

Let ω be a circle and C a point outside it; distinct points A and B are selected on ω so that CA and CB are tangent to ω. Let X be the reflection of A across the point B, and denote by γ the circumcircle of triangle BXC. Suppose γ and ω meet at $D \neq B$ and line CD intersects ω at $E \neq D$. Prove that line EX is tangent to the circle γ.

Problem 88. Sharygin 2022 P19

Let I be the incenter of triangle ABC, and K be the common point of BC with the external bisector of angle A. The line KI meets the external bisectors of angles B and C at points X and Y . Prove that $\angle BAX = \angle CAY$.

Problem 89. IMO SL 2005 G6

The median AM of a triangle ABC intersects its incircle ω at K and L. The lines through K and L parallel to BC intersect ω again at X and Y. The lines AX and AY intersect BC at P and Q. Prove that $BP = CQ$.

Problem 90. IMO SL 2004 G8

A cyclic quadrilateral $ABCD$ is given. The lines AD and BC intersect at E, with C between B and E; the diagonals AC and BD intersect at F. Let M be the midpoint of the side CD, and let $N \neq M$ be a point on the circumcircle of $\triangle ABM$ such that $\frac{AN}{BN} = \frac{AM}{BM}$. Prove that E, F, N are collinear.

14.5 Reading

[Poh07] — **Cosmin Pohoata, *Harmonic Division and its Applications***

[Bur12] — **Dominik Burek, *Dwustosunek i biegunowe*** (In Polish)

[Rem10] — **Alexander Remorov, *Projective Geometry***

[Ron21c] — **Victor Rong, *Projective Geometry***

[Zac19] — **Zack Chroman, Gopal K. Goel, Anant Mudgal, *The Method of Animation***

[Zve19] — **Vladyslav Zveryk, *The Method of Moving Points***

[Wil22] — **Wilhelm Carmevik, Neo Dahlfors, Alvin Palmgren Hjalmar Rusck, *The Method of Moving Points***

Chapter 15

Apollonian circle

Lemma 123: Apollonian Circle

Let AB be a segment and k a positive real number. Then the locus of points P such that $\frac{AP}{BP} = k$ is a circle called the *Apollonian circle*.

In case $k = 1$, it is the perpenticular bisector of AB.

Lemma 124: A-Apollonian circle

The A-Apollonian circle ω_A of $\triangle ABC$ is defined as the Apollonian circle of segment BC with ratio $k = \frac{AB}{AC} = \frac{c}{b}$, passing through vertex A. It has the following properties:

- diameter of ω_A is segment A_1A_2, where A_1 and A_2 are the feet of the internal and external angle bisectors of $\angle A$ on side line BC.
- ω_A is orthogonal to (ABC).
- ω_A intersects (ABC) at A and K, where K is the second intersection of the A-symmedian line AK with (ABC).
- The A-Humpty point P_A lies on the A-Apollonian circle ω_A.

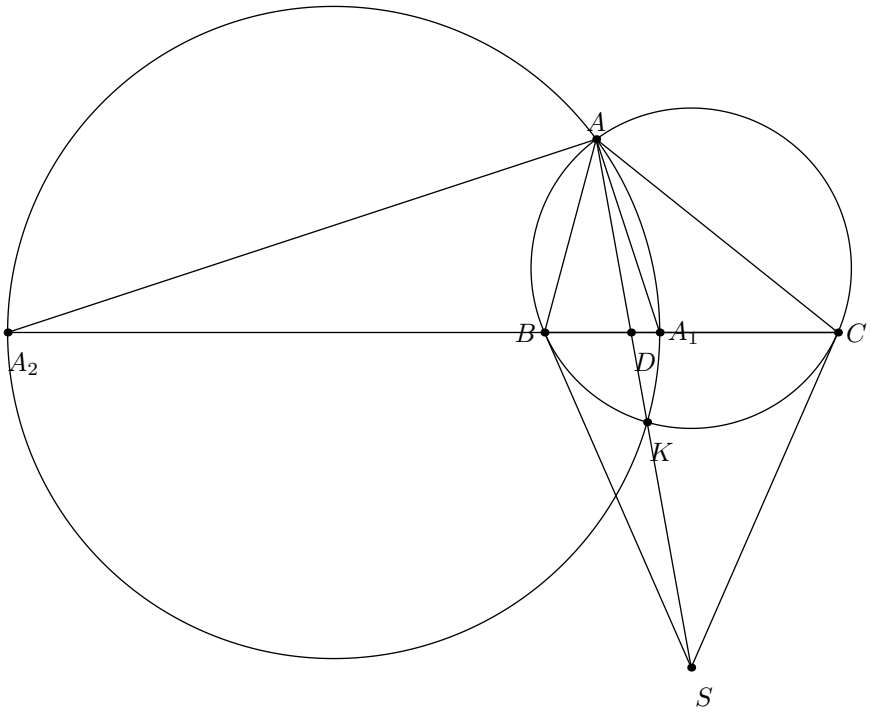


Figure 15.1: *Lemma 124*

Chapter 16

Polars

Let ω be a circle with center O and radius r, and let $P \neq O$ be a point in the plane. Let P' be the inverse of P with respect to ω, satisfying $OP \cdot OP' = r^2$. The *polar* of P with respect to ω is the line ℓ_P passing through P' perpendicular to line OP. Conversely, P is called the *pole* of line ℓ_P.

1. **Exterior Point Construction:** If P lies outside ω, let T_1 and T_2 be the points of tangency of the two tangents drawn from P to ω. Then line T_1T_2 is the polar of P.
2. **Harmonic Construction:** A line through P intersects circle ω at A and B. If Q is chosen on line AB such that $(A, B; P, Q) = -1$, then Q lies on the polar of P.
3. **Interior Point Construction:** If P lies inside ω, draw two distinct chords AB and CD passing through P. Let the tangents to ω at A and B meet at X, and tangents at C and D meet at Y. Then line XY is the polar of P.

16.1 Polar theorems

Lemma 125: La Hire

Let ω be a circle. A point X lies on the polar of point Y if and only if Y lies on the polar of X. That is:

$$X \in \text{polar}(Y) \iff Y \in \text{polar}(X).$$

Lemma 126: Collinear Poles and Concurrent Polars

3 poles colliner = 3 polars coincide

A triangle $\triangle PQR$ is said to be *self-polar* with respect to a circle ω centered at O if each vertex is the pole of the opposite side. That is, PQ is the polar of R, QR is the polar of P, and RP is the polar of Q.

Lemma 127: Brocard

Let $ABCD$ be a cyclic quadrilateral inscribed in a circle ω with center O. Let the diagonal and opposite side intersections be:

$$P = AC \cap BD, \quad Q = AB \cap CD, \quad R = AD \cap BC.$$

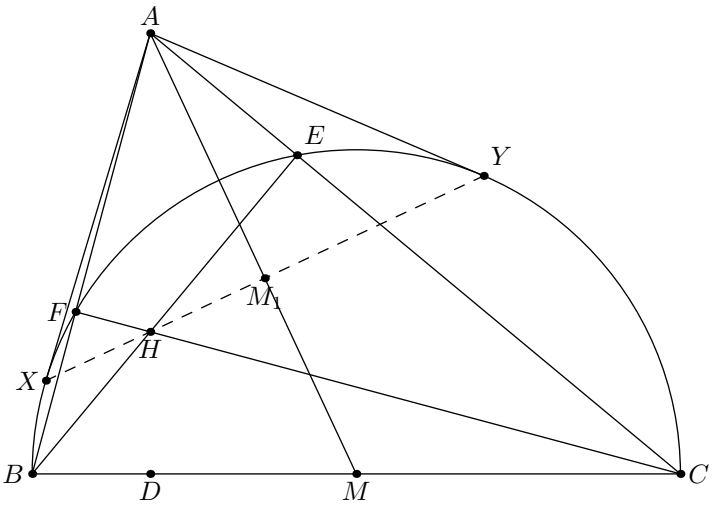
Then:

1. Triangle $\triangle PQR$ is self-polar with respect to ω.
2. O is the orthocenter of $\triangle PQR$.

Another lemma. You may think that it's just Brocard again but really it is in fact a Brocard in a different font.

Lemma 128

Let $\triangle ABC$, H its orthocenter and M the midpoint of BC. Then the line perpendicular to AM from H is the *polar* of A wrt $(BEFC)$.



16.2 Pole - polar duality

Pole-polar duality allows us to translate complex configurations into dual forms:

Primal Configuration	$\Longleftrightarrow$	**Dual Configuration**
Point P	$\Longleftrightarrow$	Line ℓ_P (polar of P)
Line ℓ	$\Longleftrightarrow$	Point L (pole of ℓ)
Point P lies on line ℓ	$\Longleftrightarrow$	Line ℓ_P passes through point L
Collinear points P, Q, R	$\Longleftrightarrow$	Concurrent lines ℓ_P, ℓ_Q, ℓ_R
Intersection $P = \ell_1 \cap \ell_2$	$\Longleftrightarrow$	Line $p = L_1L_2$ through poles
Circle ω	$\Longleftrightarrow$	Dual conic / set of tangent lines

16.3 Problems

Problem 91. Circles ω_1 and ω_2 with centers at O_1, O_2 intesect at X, Y. The common tangent touches ω_1 at A and ω_2 at B. Tangents to ω_1, ω_2 at X intersect line O_1O_2 at K, L respectively. Denote $AK \cap \omega_1 = N \neq A$ and $BL \cap \omega_2 = M \neq B$. Prove that lines AM, BN and O_1O_2 coincide.

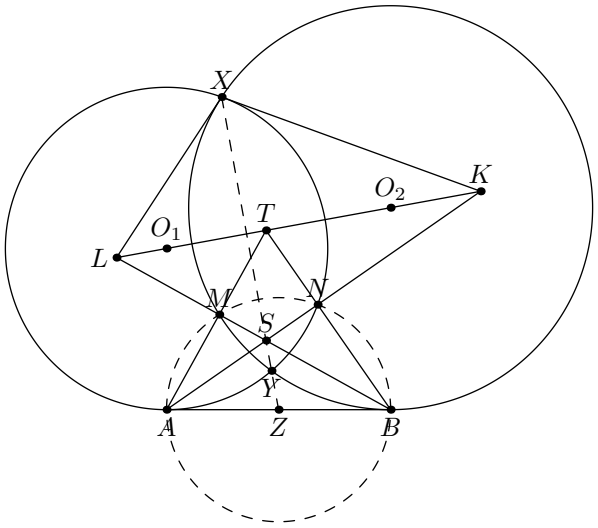


Figure 16.1: *Problem 91*

Solution. Let $Z = XY \cap AB$.

Wrt ω_2:

$$XY = polar(L)$$

from symmetry, so from *Lemma 125*, $L \in polar(Z)$ so

$$BL = polar(Z)$$

and ZM is tangent to ω_2.

Analogusly,

$$AK = polar(Z)$$

wrt ω_1 and ZN is tangent to it.

Considering *Lemma 80*, we have $ZN = ZA = ZB = ZM$ and so $(AMNB)$ with center at Z.

Applying the 3 radiacal axis theorem to ω_1, ω_2, $(AMNB)$ we get that AK, BL, XY coincide, denote that point by S.

Wrt $(AMNB)$:

$$AN = polar(O_1)$$

$$BM = polar(O_2)$$

so

$$O_1O_2 = polar(S)$$

Now let $T = AM \cap BN$, from *Lemma 127*, $S \in polar(T)$ so

$$T \in polar(S) = O_1O_2$$

which we were supposed to show. □

Problem 92. Let Ω, ω be two internally tangent circles, ω inside Ω. Point I is the circumcenter of ω. Chord AB of circle Ω is tangent to ω. Denote the midpoints of arc AB containing and not containing the point of tangency to ω by M and N, respectively. Tangents from N to ω are tangent to it at points C and D where $ABCD$ is a convex quadrilateral. Let $X = AD \cap BC$. Prove that points X, I, M are collinear.

Solution. Divide the solution into the following two parts:

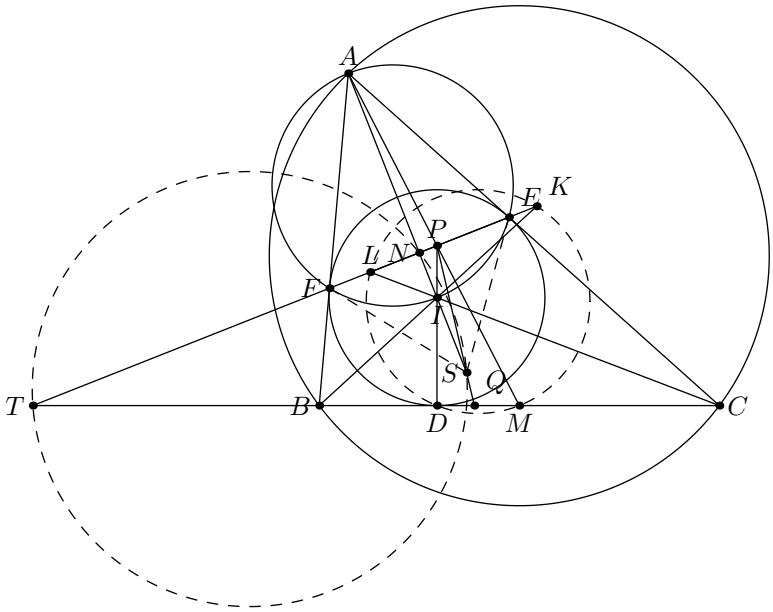


Figure 16.2: *Problem 93.*

- Points A, B, C, D lie on a circle with circumcenter N

  Let γ be a circle with center N and ray AN^2. Now consider an intersion I around it. Clearly, it transforms Ω into line AB so circle ω must be orthogonal to γ.

  Then the tangency points from N into ω must lie on γ so quadrilateral $ABCD$ is indeed cyclic with center at N.

- Polars of points I, X, M with respect to γ coincide

  Line AB is the polar of M because $\angle NBM = \angle NAM = 90^o$.

  Line CD is the polar of I because lines IC and ID are tangent to γ.

  Furthermore, let $Y = AB \cap CD$ and $Z = AC \cap BD$. Then from *Lemma 127* we have that YZ is the polar of X.

  So point Y is the intersection of lines AB, CD and YZ, the polars of points M, I and X, respectively. Combining that fact with *Lemma 126* we obtain the desired result.

□

Problem 93. Taiwan TST Quiz for 56th IMO

In scalene triangle ABC with incenter I, the incircle is tangent to sides CA and AB at points E and F. The tangents to the circumcircle of $\triangle AEF$ at E and F meet at S. Lines EF and BC intersect at T. Prove that the circle with diameter ST is orthogonal to the nine-point circle of $\triangle BIC$.

Problem 94. Given is an acute $\triangle ABC$. Line perpendicular to BC passing through A intersects circle with diameter BC at points K, M such that $AK < AM$. Line perpendicular to AC passing through B intersects circle with diameter AC at points L, N such that $BL < BN$. Prove that lines AB, KN, ML coincide.

16.4 Reading

[Olš20] — Radek Olšák, *Polar Duality in Olympiad Geometry*

Chapter 17

Inversions

Let Ω be a circle centered at O with radius $r > 0$. An *inversion* about Ω is a transformation I_Ω of the extended Euclidean plane $\mathbb{R}^2 \cup \{\infty\}$ defined as follows:

1. Every point $P \neq O, \infty$ is mapped to a point P' on ray OP such that:
$$OP \cdot OP' = r^2.$$
2. $I_\Omega(O) = \infty$ and $I_\Omega(\infty) = O$.
3. Points on circle Ω are fixed points ($P' = P$).

Lemma 129: Properties of Inversion

Let I_Ω be an inversion centered at O with radius r.

- Inversion is an involution ($I_\Omega \circ I_\Omega = \mathrm{id}$), meaning $(P')' = P$.
- For any two points $A, B \neq O$ with inverse images A', B':
$$A'B' = r^2 \cdot \frac{AB}{OA \cdot OB}.$$
- $\triangle OAB \sim \triangle OB'A'$, which implies directed angle equality $\angle OAB = -\angle OB'A'$.
- Cline Mapping Rules (circle/line = cline):
  - A line passing through O maps to itself.
  - A line not passing through O maps to a circle passing through O.
  - A circle passing through O maps to a line not passing through O.
  - A circle not passing through O maps to another circle not passing through O.
- A circle γ is orthogonal to the circle of inversion Ω if and only if γ inverts to itself ($\gamma' = \gamma$).
- Tangent clines invert to tangent clines.

Most likely, you will be using one specific inversion:

Lemma 130: $\sqrt{bc}$ Inversion

In a reference triangle $\triangle ABC$ with side lengths a, b, c, the $\sqrt{bc}$ *inversion at* A is the

composite transformation:

$$I_{\sqrt{bc}} = \text{Ref}_{AI} \circ I_{(A,\sqrt{bc})},$$

consisting of an inversion centered at A with radius $r = \sqrt{bc} = \sqrt{AB \cdot AC}$, followed by a reflection across the interior angle bisector AI of $\angle BAC$.

This inversion has the following properties:

- since $AB \cdot AC = bc$ and AI bisects $\angle BAC$, vertex B swaps with C.
- line BC swaps with the circumcircle (ABC).
- since $AI \cdot AI_A = bc$, the incenter I swaps with the A-excenter I_A.
- the B-excenter I_B swaps with the C-excenter I_C.
- the A-altitude foot D swaps with the A-antipode A' on (ABC), because $AD \cdot AA' = h_a \cdot 2R = bc$.
- the A-median AM_A swaps with the A-symmedian AK_A.
- the A-mixtilinear incircle ω_A swaps with the A-excircle.
- circle (BOC) swaps with (BHC).

Remark (The $\sqrt{\frac{bc}{2}}$ Inversion Extension)**.** By scaling the radius to $r = \sqrt{\frac{bc}{2}}$ (equivalent to composing $I_{\sqrt{bc}}$ with a homothety $\text{h}(A, \frac{1}{2})$), the *Nine-Point Circle* n_9 swaps directly with circle (BOC), and circumcenter O swaps with altitude foot D.

Lemma 131: Orthic Inversion

In triangle $\triangle ABC$ with orthocenter H and altitude feet D, E, F, the *Orthic Inversion* is the inversion centered at H with power:

$$r^2 = AH \cdot HD = BH \cdot HE = CH \cdot HF$$

Under this inversion:

- Vertices Swap with Altitude Feet:

$$A \to D, \quad B \to E, \quad C \to F.$$

- (ABC) swaps with the Nine-Point Circle n_9.
- intersection points K, L of altitude line EF with (ABC) satisfy $AK = AL = \sqrt{AH \cdot AD}$ and remain fixed under inversion centered at A.

17.1 Reading

[Han15] — Gunmay Handa, *Inversion on the Fly*

[Roy24] — Aharshi Roy, $\sqrt{bc}$ *inversion FTW!*

Part V

Quadrilaterals

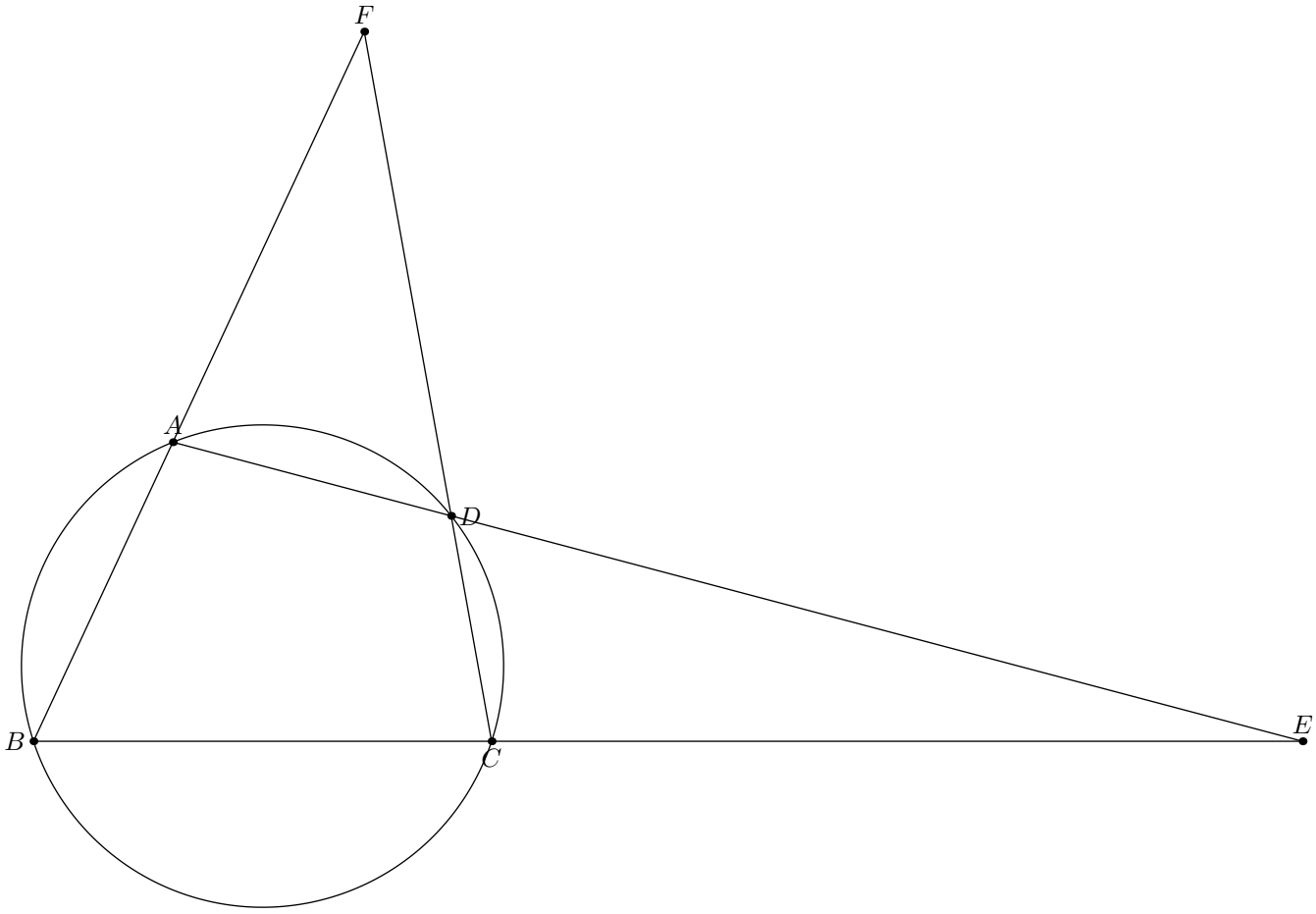


Figure 17.1: *Complete cyclic quadrilateral*

Chapter 18

The Big Picture

18.1 Miquel point

Lemma 132: Miquel's Theorem for Triangles

Let D, E, F be points on lines BC, CA, AB of $\triangle ABC$, respectively. Then the circumcircles (AEF), (BFD), and (CDE) concur at a single point M, known as the *Miquel point* of $\triangle ABC$ with respect to D, E, F.

Lemma 133: Miquel's Theorem for Quadrilaterals

For every quadrilateral $ABCD$, there exists a *Miquel point* M - the center of spiral similarity of $AB \to DC$ and $AD \to BC$.

Let $E = AB \cap CD$ and $F = AD \cap BC$. Then the four circumcircles (EAC), (EBD), (FAB), and (FCD) concur at point M.

18.2 Big Diagram

Let $ABCD$ be a cyclic quadrilateral with circumcenter O and let $E = AB \cap CD$, $F = AD \cap BC$, $P = AC \cap BD$. Moreover, let M be the *Miquel point* of $(ABCD)$.

Lemma 134: Cyclic Quadrilateral - The Big Picture

- $E - M - F$
- $O - P - M$
- $OM \perp EF$
- $(AOCM)$, $(BODM)$
- MO bisects $\angle CMA$ and $\angle BMD$
- P and M are inverse images of each other wrt $(ABCD)$
- $\triangle PQR$ is *self-polar*
- O is the orthocenter of $\triangle PQR$

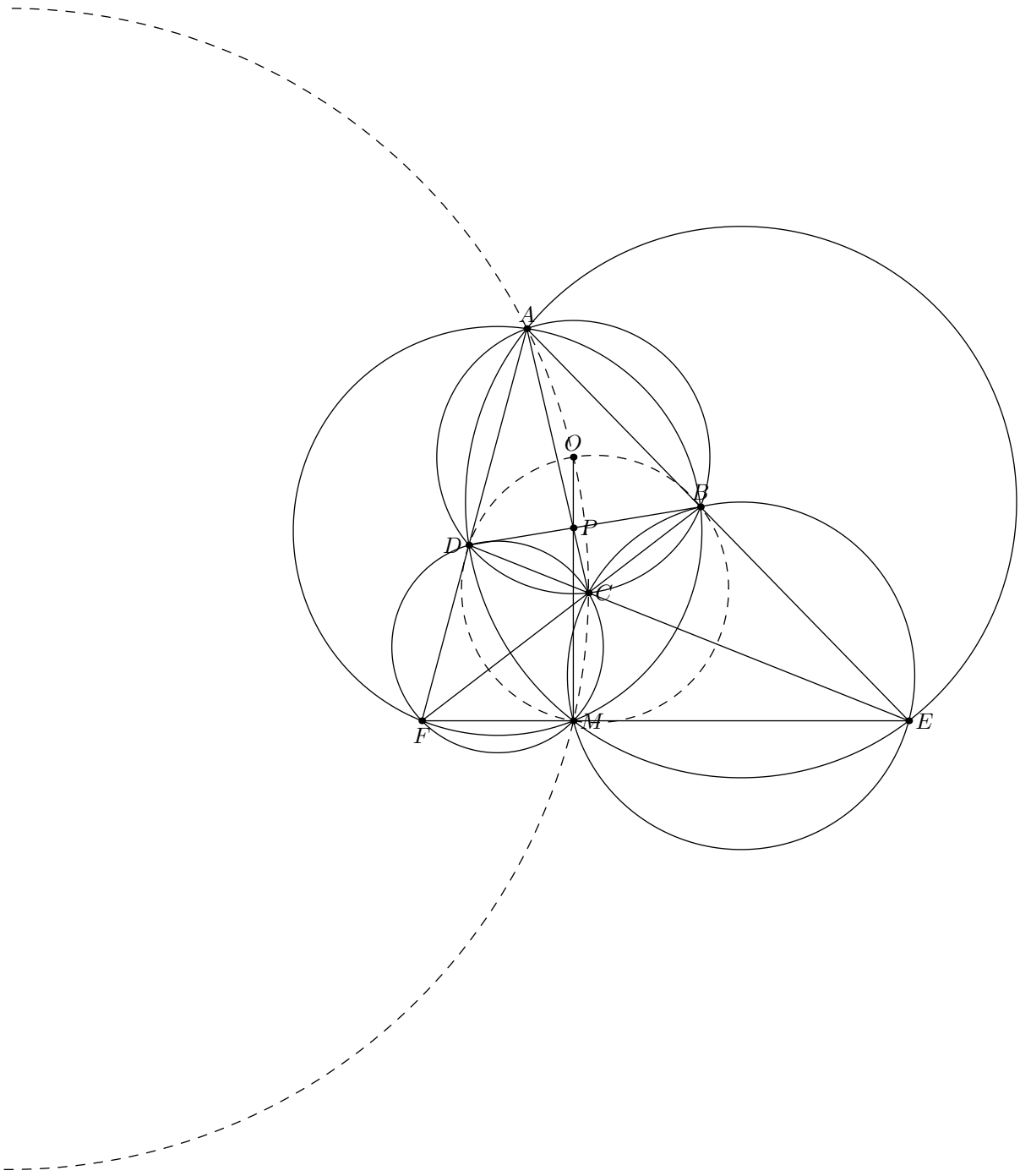


Figure 18.1: *Lemma 134*

Problem 95. GOTEEM 2020/5

Let ABC be a triangle and let B_1 and C_1 be variable points on sides BA and CA, respectively, such that $BB_1 = CC_1$. Let $B_2 \neq B_1$ denote the point on (ACB_1) such that BC_1 is parallel to B_1B_2, and let $C_2 \neq C_1$ denote the point on (ABC_1) such that CB_1 is parallel to C_1C_2. Prove that as B_1 and C_1 vary, the circumcircle of AB_2C_2 passes through a fixed point other than A.

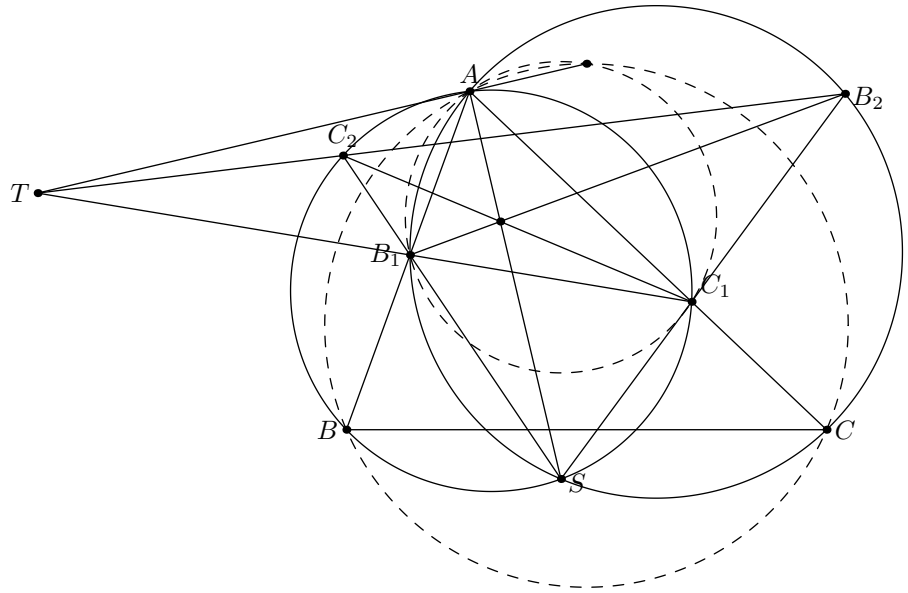


Figure 18.2: *Problem 95.*

Solution. Let S be the second intersection of □

Problem 96. USA TST 2007

Triangle ABC is inscribed in circle ω. The tangent lines to ω at B and C meet at T. Point S lies on ray BC such that $AS \perp AT$. Points B_1 and C_1 lies on ray ST (with C_1 in between B_1 and S) such that $B_1T = BT = C_1T$. Prove that triangles ABC and AB_1C_1 are similar to each other.

18.3 Problems

Problem 97. IMO 2020 SL G6

Let I and I_A be the incenter and the A-excenter of an acute-angled triangle ABC with $AB < AC$. Let the incircle meet BC at D. The line AD meets BIA and CIA at E and F, respectively. Prove that the circumcircles of triangles AID and $IAEF$ are tangent to each other.

18.4 Reading

[Zha09] — Yufei Zhao, *Cyclic Quadrilaterals — The Big Picture*

[Ron21a] — Victor Rong, *Complete Quadrilaterals and the Miquel Point*

[Kal] — Helia Kalanaki, *Complete Quadrilaterals*

Chapter 19

Newton - Gauss - Bodenmiller

Lemma 135: Newton-Gauss line

The *Newton-Gauss line* is the line joining the midpoints of the three diagonals of a complete quadrilateral.

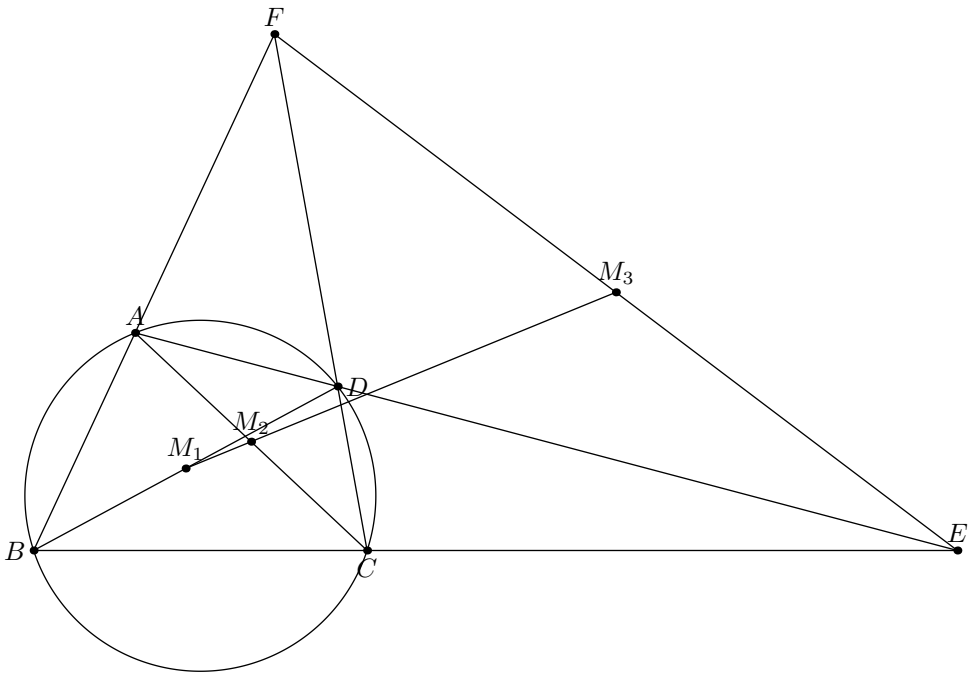


Figure 19.1: *Newton-Gauss line*

For the next theorem we will need a help from another lemma:

Lemma 136

Let ABC be a triangle and let M, N be points on sides CA and AB respectively.Then the orthocenter H of triangle ABC lies on the radical axis of the circles with diameters BM and CN.

Proof. Let ω_1, ω_2 be circles with diameters BM, CN and B_1, C_1 the feet of altitudes from B, C respectively. Notice that $B_1 \in \omega_1$ and $C_1 \in \omega_2$. Then:

$$Pow_{\omega_1}(H) = BH \cdot HB_1 = CH \cdot HC_1 = Pow_{\omega_2}(H)$$

and hence we're done. □

Lemma 137: Gauss-Bodenmiller Theorem

The circles with diameters AC, BD, EF are coaxial.

Their radical axis, called *Aubert line*, is a line passing through each of the four orthocenters of the triangles ADF, BCF, DCE, ABE.

Furthermore, it is perpenticular to the *Newton-Gauss line*.

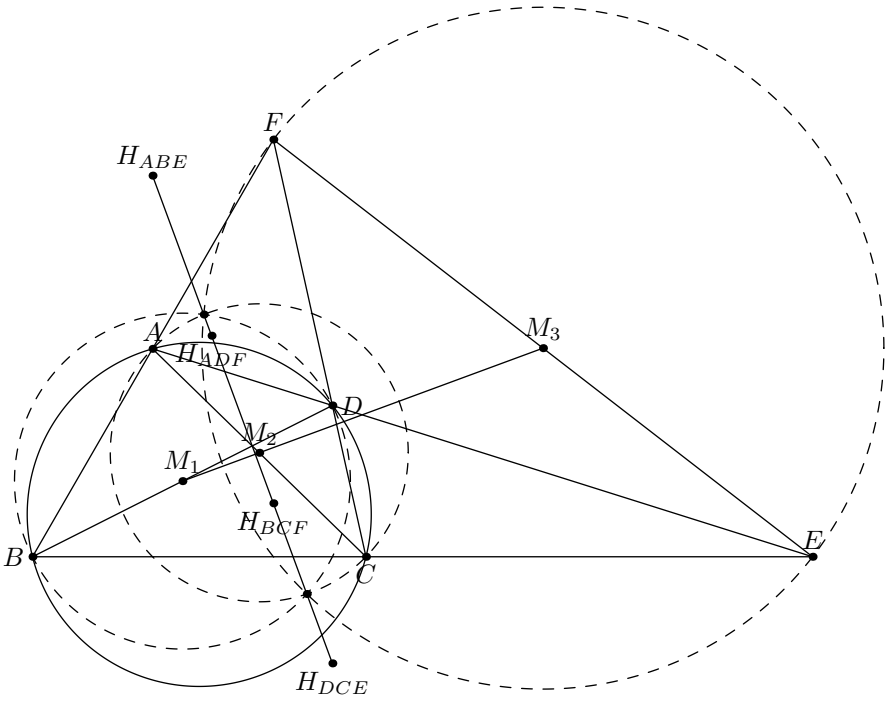


Figure 19.2: *Aubert line*

Proof. Note that segments AC, BD are cevians in triangle BCF so from *Lemma 136* we know that H_{BCF} lies on the radical centre of the circles with diameters AC and BD. Similarly, points H_{ADF}, H_{DCE}, H_{ABE} also lie on the common radical axis of the 3 circles. □

I think this is the single most demanding diagram in this book.

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